

11 Plaice in the North Sea and Skagerrak

ple.27.420 – *Pleuronectes platessa* in Subarea 4 and Subdivision 20

The stock was benchmarked in 2022 and the Stock Annex was updated. Therefore, only a comprehensive description of the stock assessment results and deviations from the Stock Annex are presented within this Section of the report.

11.1 General

11.1.1 Stock structure

Plaice in the Skagerrak (Subdivision 20) is considered to have two components: an Eastern and a Western. The latter occurs in a mix with plaice migrating in from the North Sea (Ulrich *et al.*, 2013) and the predominance of catches occurs on summer feeding aggregations in the Western Skagerrak. In a benchmark (WKPLE; ICES, 2015) it was decided that plaice in the Skagerrak would be assessed together with the North Sea stock.

In addition, as in previous years, 50% of the mature animals from 7.d in quarter 1 are included in the North Sea plaice assessment, since North Sea plaice migrates into the area in that season (WKFLAT; ICES, 2010).

11.1.2 Ecosystem considerations

Available information on ecosystem aspects can be found in the Stock Annex. In addition, the ICES Working Group on the Ecosystem Effects of Fishing Activities (WGECO; ICES, 2014) met in April 2014 and addressed a specific question in relation to North Sea plaice, in response to a request from WGNSSK in 2013:

“According to WGNSSK estimates, the North Sea is currently ongoing a plaice outburst without precedent. However, plaice is not included in multispecies models, so the consequences of this outburst on the North Sea ecosystem are unclear and would potentially require additional focus”.

WGECO addressed the trends shown in the stock assessment of plaice, which show how increasing fishing pressure on the stock has progressively moved SSB away from the desired state (in the 1980s and 1990s), and then how management has rectified this situation in recent years, which has brought the North Sea plaice stock in a situation unlike any other over the whole 58-year period for which data are available. The group investigated a possible relationship of these trends with abundance of benthic biomass, which is a predominant food source for plaice. Q1 IBTS data showed a two-fold increase in demersal benthivore biomass over the last 29-year period of the survey, and that species composition of the demersal benthivore guild has changed as well. The data showed that predation loading by plaice on benthic invertebrates increased by a factor of 13.8 in just eleven years (2000–2011).

The increase in the consumption of benthic invertebrate prey by the whole demersal benthivore guild, and particularly by plaice, raises the question as to whether the abundance of benthic invertebrate prey might be becoming limiting. If the biomass of demersal benthivore fish is approaching its carrying capacity, then growth rates in the dominant species in the guild might

start to decline (which is in this case plaice growth rates). Computed growth coefficients for the 1956 to 2002 cohorts showed a strong declining linear trend over the whole period (albeit with clear systematic variation in the residuals), and this has been related to increasing water temperature in the North Sea. However, fitting a 4th order polynomial function to the data suggested a marked decline in cohort growth towards the end of the time-series. This is perhaps indicative of plaice becoming food limited, possibly suggesting that B_{MSY} targets for the stock might be marginally too high to be supported by available benthic invertebrate food supplies. However, this evidence is by no means conclusive as polynomial functions are known to show a tendency for marked swings at the extremes of the data range.

More in-depth analysis in WGECO 2018 using the recent years' data showed that the co-occurrence of reduced size at age and increasing stock abundance has led to a negative relationship in period 2006–2016. This correlative indication of density-dependent growth reduction, is further strengthened by a coinciding reduction in physical condition across a range of sizes, hinting that food scarcity may indeed be the mechanism behind the patterns (WGECO; ICES, 2018a).

11.1.3 Fisheries

A basic description of the fisheries is available in the Stock Annex. In recent years, pulse trawling, aiming at reduction of fuel consumption and reduction of bottom disturbance, has been adopted in fisheries. In 2011, approximately 30 derogation licenses for pulse trawls were taken into operation, which increased to 42 in 2012. An additional 42 derogation licenses have been extended in spring 2014. In 2016 and 2018, ICES published advice on ecological and environmental effects of pulse trawling, compared to traditional beam trawls (ICES, 2016; ICES, 2018b). It was concluded that pulse trawling has fewer environmental and ecological effects than beam trawls. Pulse trawls have been increasingly used in the North Sea flatfish fisheries since 2009. Over this period, the fishing mortality has reduced, and stock biomass has increased, mostly due to an overall decrease in effort. The shift in fishing method has resulted in a change in distribution of the fishery. Pulse trawling has increased in areas such as off the Thames estuary and the Belgian coast but decreased in others. This change is related to lighter gear, which can be used on softer grounds than the beam trawls (ICES, 2018b).

Following the EU decision in February 2019 to revise the technical measures regulations, pulse trawling was prohibited since 30 June 2021 which was expected to cause large changes in the Dutch beam trawl fleet in the coming years.

11.1.4 ICES Advice for 2025

The information in this Section is taken from the ICES advice sheet:

ICES advises that when the MSY approach is applied, catches in 2025 should be no more than 176 988 tonnes.

11.1.5 Management

An EU multiannual management plan (MAP) has been agreed by the EU for this stock (EU, 2018). There is no agreement with Norway and UK regarding this plan, and it is not used as the basis of the advice for this shared stock. ICES was requested by the EU and UK to provide advice based on the MSY approach and to include F_{MSY} ranges in the catch scenarios.

11.2 Data available

During the benchmark of the eastern English Channel (7.d) plaice stock (WKFLAT) it was decided that 50% of Q1 mature fish catches taken in the eastern English Channel are actually plaice from the North Sea stock migrating in and out of the area. Before 2015, 50% of the Q1 eastern English Channel (7.d) plaice landings were included in the assessment of the North Sea plaice stock. Since 2015, 50% of the mature fish in both landings and discards in Q1 were added to the North Sea stock and the time series was updated, such that in previous years also 50% of the mature catches from Q1 were added. See the Stock Annex for plaice in Division 7.d for further details.

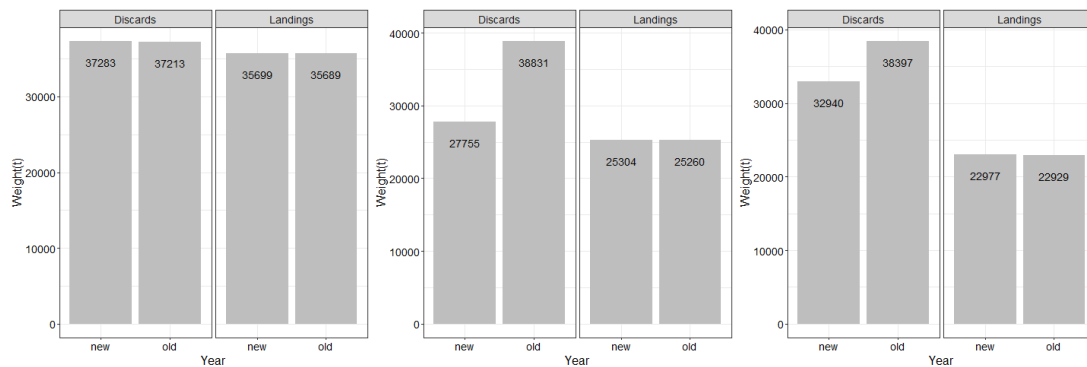
During the benchmark on plaice (WKPLE; ICES, 2015), it was decided that plaice from the Skagerrak would be added to the North Sea stock. Since then, the assessment has been a combined assessment with Skagerrak plaice.

11.2.1 Changes in historical data

The time series of plaice catches from Division 7.d., including landings since 1980 and discards since 2006, was updated in 2025 (this report) following a benchmark WKBNSCS (ICES, 2025).

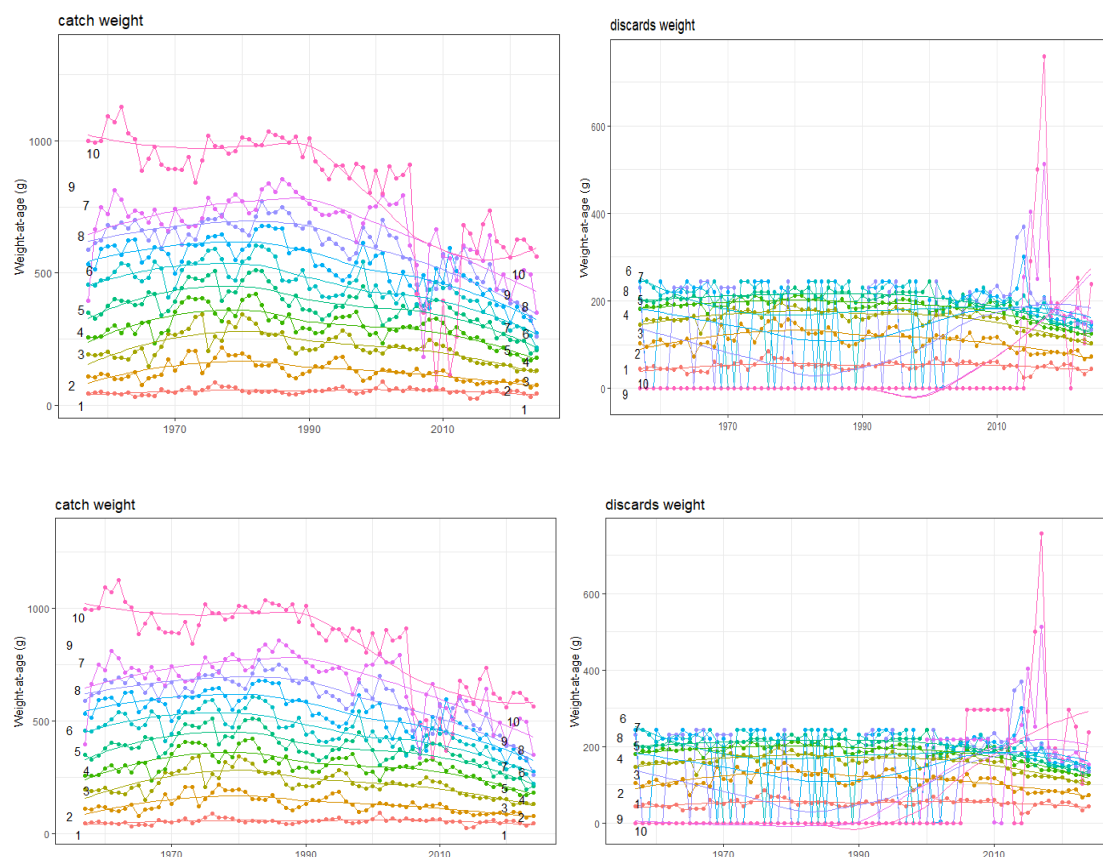
Catch data from 2021- 2023 were updated and re-raised in InterCatch in 2025 due to 1) update of UK-England and Wales catches in 2021-2023 (summarized in table below) and 2) update of Netherlands discards in 2022-2023. The discards raising and allocation strategy was consistent with the strategy described in Section 11.2.2. Additionally, an error (i.e., a very high discards of Dutch OTB fleet due to mismatch of metier types) in matching discards strata was found in 2022 data and corrected. As a result, the new raised discards of 2022 reduced 28.5%. The comparison of raised old and new landing/discards are illustrated in the table and figure below.

stock	Year	CatchCategory	CATON.old	CATON.new	CATON.diff	pcDiff
ple.27.420	2021	B	17.569	17.699	0.13	0.74
ple.27.420	2021	D	17.863	113.249	95.386	533.99
ple.27.420	2021	L	2287.021	2296.921	9.9	0.43
ple.27.420	2022	B	1.364	1.389	0.025	1.83
ple.27.420	2022	D	8.942	50.691	41.749	466.89
ple.27.420	2022	L	1744.897	1788.44	43.543	2.5
ple.27.420	2023	B	4.037	5.549	1.512	37.45
ple.27.420	2023	D	29.743	120.445	90.702	304.95
ple.27.420	2023	L	2383.461	2431.391	47.93	2.01



Plaice in Subarea 4 and Subdivision 20. Comparison of the updated InterCatch raised catch data to the old data for 2021-2023.

Some zero mean-weight for age 9 and 10+ in several historical years (around 2010) were observed. These observations have non-zero discards but zero weight. These erroneous mean weights values (non-zero discards but zero weight) were imputed using the average of the discards weight for the corresponding ages (age 9: 0.219 kg; age 10+: 0.296).



Plaice in Subarea 4 and Subdivision 20. Upper row: catch and discards weight-at-age for some unrealistic weight for age 9 and 10+. Lower row: corrected catch and discards weight-at-age.

11.2.2 InterCatch processing

Since 2012, national research institutes submitted landings and discard estimates by métier and quarter in InterCatch. Figures 11.2.1 and 11.2.2 show the landings and discards coverage by country and by métier in Subarea 4 and Subdivision 20. For Subarea 4, 80% of the total discards

in 2024 were obtained from sampling. For Subdivision 20, 84% of the total discards were obtained from sampling. BMS landings, where reported, were included as discards in the assessment since 2016. Approximately 53% and 82% of the landings in weight were sampled in Subarea 4 and Subdivision 20 respectively, to obtain information on age-composition.

Of the métiers for which discards are monitored in sampling programmes, the largest part of these discards is covered in the TBB_DEF_70–99_mm fleet. In most discards monitoring programmes, age composition information is also collected. To raise the number of discards for landings that had no discards and to raise the landings and discards for which no age distribution was known, the same grouping strategy was used (see table below). The TBB and OTB fleets that covered most of the catches each had their own group (TBB<100, TBB ≥100, OTB/OTM<100, and OTB/OTM≥100). Other major groups include seines, shrimp, and gillnets. All discards raising and age allocations were done per quarter. If discards/age structures were present for data for the whole year only, these were added to all quarters. If there were no discards/age structures in a specific quarter and métier, a similar métier type (from the same quarter) or all other quarters (from the same métier) were used. Allocations to calculate the age compositions were done separately for discards and landings.

Plaice in Subarea 4 and Subdivision 20. Summary of the imported/Raised and Sampled/Estimated data by area.

CatchCategory	RaisedOrImported	SampledOrEstimated	Area	CATON	perc
Landings	Imported_Data	Sampled_Distribution	27.4	7631	53
Landings	Imported_Data	Estimated_Distribution	27.4	6858	47
Discards	Imported_Data	Sampled_Distribution	27.4	17098	80
Discards	Raised_Discards	Estimated_Distribution	27.4	3754	18
Discards	Imported_Data	Estimated_Distribution	27.4	519.1	2
BMS landing	Imported_Data	Estimated_Distribution	27.4	12.07	100
Landings	Imported_Data	Sampled_Distribution	27.3.a.20	4003	82
Landings	Imported_Data	Estimated_Distribution	27.3.a.20	869.8	18
Discards	Imported_Data	Sampled_Distribution	27.3.a.20	2102	84
Discards	Raised_Discards	Estimated_Distribution	27.3.a.20	407.7	16
Discards	Imported_Data	Estimated_Distribution	27.3.a.20	< 1	< 1
BMS landing	Imported_Data	Estimated_Distribution	27.3.a.20	0	NA

Plaice in Subarea 4 and Subdivision 20. Grouping strategies to raise discards.

Group for discards raising and age allocation*	quarter + area	process of strata
TBB<100(excluding CRU_16-31)	Each quarter + all area	Germany-Q1 (discards ~9.3 of discards/landing ratio) excluded in raising, because the samples are not representative.
TBB>=100	year + all area	No discards available, using OTB_DEF>100 and raise by year

Group for discards raising and age allocation*	quarter + area	process of strata
TBB/OTB_CRU_16-31	Each quarter + all area	Denmark provided one strata, include. Raise discards also using OTB/TBB <100 (excluding SWE. ENG-Q3, NLD-Q4 in OTB<100 group, exclude GER-Q1 in TBB<100 group)
OTB/OTM-CRU/DEF/SPF<100(excluding CRU_16-31)	Each quarter + all area	Raise by quarter, exclude SCO annual discards, exclude Sweden; Q3 removed UK(England) due to too high discards rate (~54.6 discards/landing ratio); Q4 removed NLD OTB_DEF_70-99 strata (~23.1) too high discards
OTB/OTM-CRU/DEF/SPF>=100	Each quarter + all area	Raise by quarter, exclude SCO annual discards. Q4 all available strata give 0 discards rate, so we raise using Q3 and Q4 to raise Q4.
SSC/SDN<100	year + all area	No discards provided, so we raise by year and using OTB<100 (excluding SWE, ENG-Q3, NLD-Q4)
SSC/SDN>=100	year + all area	Raise by year, all discards are provided by DEN.
GNS/GTS/GTR	year + all area	No discards provided. We raise using OTB/TBB >=100
Others	year + all area	Raise using all available strata

Plaice in Subarea 4 and Subdivision 20. Grouping strategies to allocate age structures.

Group for discards raising and age allocation*	quarter + area	process of strata
TBB<100(excluding CRU_16-31)	Each quarter + 4/320	
TBB≥100	Each quarter + 4/320	Landing allocated by year due to lack of sampled strata (3 strata by DEN in Q2 and Q4). Discards allocated by year using OTB_DEF>100. Annual SCO OTB>100 are included.
TBB/OTB_CRU_16-31	Each quarter + all area	Landing: There is one strata from DEN (Q3). We raise include OTB/TBB<100 as well. Discards: one strata from GER(Q2), we raise include OTB/TBB<100 as well.
OTB/OTM-CRU/DEF/SPF<100(excluding CRU_16-31)	Each quarter + all area	SCO annual age compositions are excluded (they are only used if I need to raise by year). Landing: All age samples are from DEN. Discards: SWE age samples are included.
OTB/OTM-CRU/DEF/SPF≥100	Each quarter + all area	Discards allocated by year due to lack of sampled strata; SCO age samples are included.
SSC/SDN<100	year + all area	Raised by year. Both landing and discards are raised using OTB <100.
SSC/SDN≥100	year + all area	Raised by year. For landing/discards: there are many strata from DEN, so we raise using DEN samples.

Group for discards raising and age allocation*	quarter + area	process of strata
GNS/GTS/GTR	year + all area	Landing age composition raised based on DEN GNS/GTR samples. Discards raised using OTB/TBB $\geq$ 100 fleet.
Others	year + all area	Raise using all available strata.

* all_0_0 are treated as ≥ 100 . Mesh size 90-119 is classified as < 100 .

11.2.3 Landings

According to ICES data, in 2024, BMS landings were 12.07 tonnes and majority were reported from UK. Meanwhile the official reported BMS landings were 78.94 tonnes from all countries. For the assessment in this report, BMS was treated as discards.

Total ICES estimated landings (including 7.d and Subdivision 20) of North Sea plaice in 2024 was 19 499 tonnes. Of these 14 489 tonnes came from the Subarea 4, 4872 tonnes came from Subdivision 20, and 138 tonnes came from 7.d. The landings in Subarea 4 decreased 26% compared to 2023. The landings in Subdivision 20 increased 43% compared to 2023. Total landings (in tonnes) are presented in Table 11.2.1 and landings in numbers at age in Table 11.2.2 and Figure 11.2.4. Since 2010, the majority of landings were age 3–6.

11.2.4 Discards

The discards time series used in the assessment includes Dutch, Danish, German and UK discards observations for 2000–2024, as described in the Stock Annex. From Belgium, discards data have been available as well but were only used in the assessment since 2012 when it became available in InterCatch. See Section 11.2.1 for more information on the use of InterCatch for raising discards rates across métiers and countries. The Dutch discards data for 2009 and 2010 were derived from a combination of the observer programme that has been running since 2000, and a new self-sampling programme. The estimates from both programmes were combined to come up with an overall estimate of discarding by the Dutch beam trawl fleet. Since 2011, estimates were derived exclusively from the self-sampling data. There is an on-going project within WMR to validate these estimates by examining matched (same vessel and haul) trips where both observer estimates, and self-sampling estimates are derived.

To reconstruct the number of plaice discards at age before 2000, catch numbers at age data was reconstructed in 2005 based on a model-based analysis of growth, selectivity of the 80-mm beam trawl gear, and the availability of undersized plaice on the fishing grounds. Discards numbers at age are presented in Table 11.2.3 and Figure 11.2.4. Figure 11.2.3 presents a time series of landings, catches and discards from these different sources. The total discards weight has been gradually decreasing since the first year of observed discards in 2000. The discards ratios are illustrated in Figure 11.2.7. Since 2010, the majority of discards were age 1–3.

11.2.5 Catch

The catches of 2024 in Subarea 4 reached 26% of the 136 413 tonnes catch TAC for 2024. The catches of 2024 in Subdivision 20 reached 40% of the 18 250 tonnes catch TAC for 2024. The total catch-at-age as used in the assessment including all landings and all discards are presented in Table 11.2.4. These include catch of NS plaice in the 1st quarter from 7.d and catch from the

Subdivision 20. Table 11.2.5 presents details of ICES catch estimates of NS plaice stock in Subarea 4, Subdivision 20 and in the 1st quarter from Division 7.d. Landings-at-age, discards-at-age and catch-at-age plots are presented in Figures 11.2.4 and 11.2.5. The log-catch curves are illustrated in Figures 11.2.6.

11.2.6 Weight-at-age

Stock weights-at-age are presented in Table 11.2.6. Stock weight-at-age has varied considerably over time, especially for the older ages. Landings, discards and catch weights-at-age are presented in Table 11.2.7, 11.2.8 and 11.2.9 respectively. Catch weights-at-age are derived from the discards and landings weights-at-age according to the relative contributions of each to the overall catch for each age. Figure 11.2.8 presents the stock, discards, landings and catch weights-at-age. Notably, there has been a long-term decline in the observed stock weights-at-age.

11.2.7 Maturity and natural mortality

During the benchmark in 2017, a time-varying maturity ogive was estimated using Dutch commercial landings 1957–2015, but the new ogives had marginal effect on the estimated SSB. Therefore, the previously used, time-invariant maturity ogive (Table 11.2.10) was chosen.

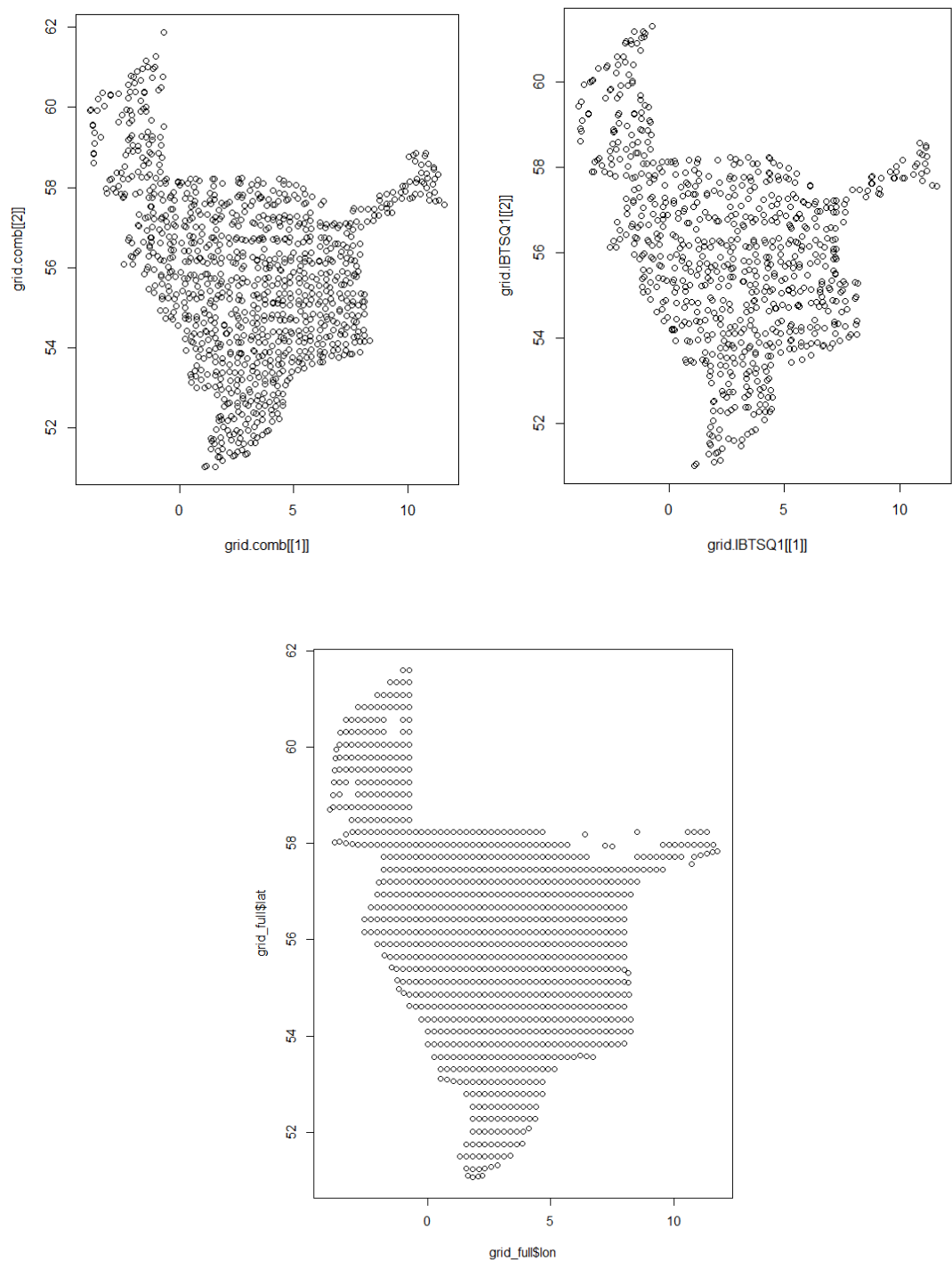
An age-dependent natural mortality was adopted in the 2022 benchmark (WKNSCS; ICES, 2023a). A weight-dependent mortality was first estimated via the Peterson-Wroblewski method based on the stock weight-at-age until 2020. The final natural mortality was then averaged across years for each age (Table 11.2.10). This age-dependent, time-invariant M is expected to be fixed for the following assessment years.

11.2.8 Survey data

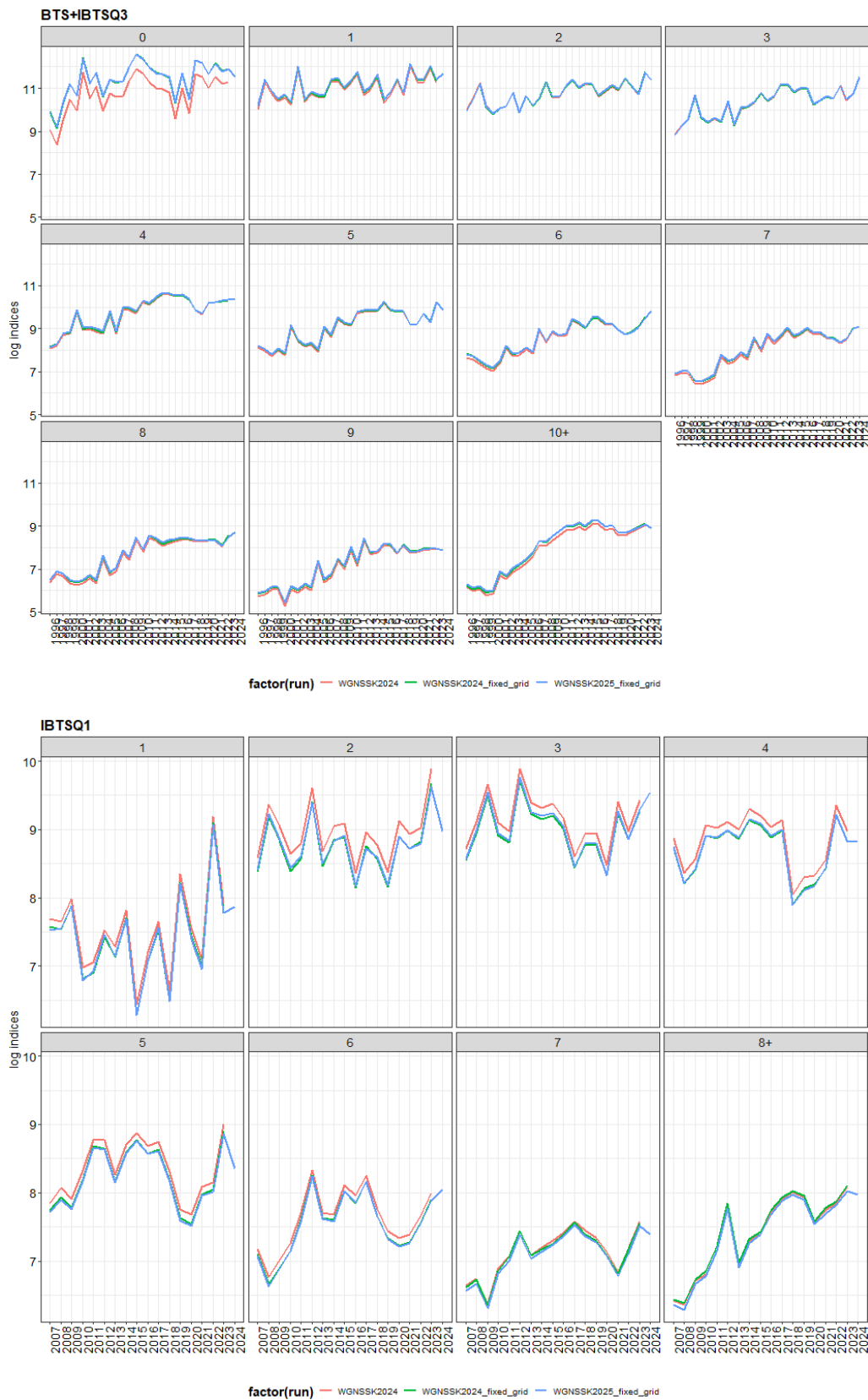
The following five survey indices are used in the plaice assessment:

- Beam Trawl Survey and IBTS Q3 combined (BTS-IBTS Q3); (1996–2024); Age 1–10+
- Beam Trawl Survey RV Isis (BTS-Isis) for the older part of the time series; (1985–1995); Age 1–8
- Sole Net Survey 1 (SNS1); (1970–1999); Age 1–6
- Sole Net Survey 2 (SNS2); (2000–2024); Age 1–6
- IBTS Q1 plaice index; 2007–2024; Age 1–8+

The most important surveys for demersal fish species in the greater North Sea area are the BTS (3rd Quarter) and the IBTS (1st and 3rd Quarter). The BTS covers divisions 4.b, 4.c and the English Channel, while the IBTS also covers Division 4.a and the Skagerrak (Subdivision 20). The spatial distributions of plaice biomass per haul for these 3 surveys in 2024 are illustrated in Figure 11.2.9. Both BTS-IBTS Q3 and IBTS Q1 indices were estimated using smoother based delta-GAM method (Berg *et al.*, 2014). The retrospective analyses of the two delta-GAM indices are shown in Figure 11.2.10, indicating very robust estimates across the last 5 years. In 2025, due to an update of the statistical programming software used (R), the delta-GAM indices prediction grid was changed from a time-varying irregular grid into a time-invariant regular grid (see figure below). Sensitivity runs have been conducted to test the impact of the changed grid on the indices, by comparing 1) the 2024 WG run using the old grid; 2) the 2024 run using the new grid; 3) the 2025 WG run (adding 2024 data year) using the new grid. Results show negligible difference (see figure below). This change helps to reduce bias associated with variability in the age distribution. Consequently, the assessment has been slightly revised, and the reference points have been updated in 2025 (see section 11.5, Annex8).



Plaice in Subarea 4 and Subdivision 20. Top left: original BTS+IBTSQ3 grid in WGNSSK 2024, generated every year; top right: IBTSQ1 grid in WGNSSK 2024, generated every year; bottom: the new time-invariant grid used in WGNSSK 2025.



Plaice in Subarea 4 and Subdivision 20. Sensitivity runs of survey indices using the original and the new grid.

An annual spatial abundance map could be estimated per age from the delta-GAM model for each of these two indices (Figure 11.2.11). The Q3 survey indicates that nursery areas for younger plaice are found in the Belgium-Netherlands-Germany-Denmark coastal area. As they get older,

they move north-west towards the centre of North Sea and Scotland coastal area. On the other hand, the IBTS Q1 survey shows slightly different age distributions. This is likely due to the spawning and nursery season in Q1.

Table 11.2.11 and Figure 11.2.12 show the survey index values. Two moderately strong year-class 2013 and 2016 were observed. Very strong 2018, 2021 and 2023 year-classes were observed. Additionally, all surveys show an increasing trend for older fishes (age ≥ 5) during 2000 and 2015.

The internal consistency of the survey indices (Figure 11.2.12) appears relatively high for BTS-IBTS Q3, but low for the SNS and IBTS Q1 surveys. The log-catch curves of ages 1–6 for the surveys are illustrated in Figure 11.2.14. In general, SNS has a low selectivity for older ages. Compared to BTS, IBTS has a higher selectivity for older ages. Overall, all surveys show relatively consistent catch selectivity pattern over the time series (which is the assumption for the stock assessment), except for IBTS–Q1 where the time series is too short to validate.

11.3 Assessment

11.3.1 Model parameters and diagnostics

SAM assessment model (Nielsen and Berg, 2014; Albertsen *et al.*, 2018) was adopted since the 2022 benchmark. The table below gives an overview of parameters used in the SAM assessment model:

\$minAge

1

\$maxAge

10

\$maxAgePlusGroup

Is last age group considered a plus group for each fleet (1 yes, or 0 no).

1 0 1 1 0 0

\$keyLogFsta

Coupling of the fishing mortality states (normally only first row is used).

0 1 2 3 4 5 6 7 8 8

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1

\$corFlag

Correlation of fishing mortality across ages (0 independent, 1 compound symmetry, 2 AR(1), 3 separable AR(1)).

2

\$keyLogFpar

Coupling of the survey catchability parameters (normally first row is not used, as that is covered by fishing mortality).

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1

0 1 2 3 4 5 6 7 -1 -1

8 9 10 11 12 12 12 12 12 12

13 14 15 15 15 15 16 17 -1 -1

18 19 20 21 22 23 -1 -1 -1 -1

24 25 26 27 28 29 -1 -1 -1 -1

\$keyVarF

Coupling of process variance parameters for log(F)-process (normally only first row is used)

0 1 2 2 2 2 2 3 3 3

\$keyVarLogN

Coupling of process variance parameters for log(N)-process

0 1 1 1 1 1 1 1 1 1

\$keyVarObs

Coupling of the variance parameters for the observations.

0 1 1 1 1 1 1 2 2 2

3 4 4 4 4 4 5 5 -1 -1

6 7 7 7 7 7 7 8 9 10

11 12 12 12 12 12 13 13 -1 -1

14 15 16 17 17 17 -1 -1 -1 -1

18 19 20 21 22 22 -1 -1 -1 -1

\$obsCorStruct

Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstructured). | Possible values are:
"ID" "AR" "US"

"ID" "ID" "ID" "AR" "AR" "AR"

\$keyCorObs

Coupling of correlation parameters can only be specified if the AR(1) structure is chosen above.

NA's indicate where correlation parameters can be specified (-1 where they cannot).

#1-2 2-3 3-4 4-5 5-6 6-7 7-8 8-9 9-10

NA NA NA NA NA NA NA NA NA

NA NA NA NA NA NA NA -1 -1

NA NA NA NA NA NA NA NA NA

0 0 0 0 0 0 0 -1 -1

1 1 1 1 1 -1 -1 -1 -1

2 2 2 2 2 -1 -1 -1 -1

\$fbarRange

lowest and highest age included in Fbar

2 6

Model diagnostics including standardized catch and survey residuals and retrospective plots are illustrated in figures 11.3.1–11.3.3. There are age and cohort patterns in both catch (plus age) and BTS-IBTSQ3 survey (age 5+ in last 10 years) residuals, implying some lack of fitting from the model. Further investigations will be conducted to improve the assessment. The retrospective analyses do not show any pattern, and the Mohn’s rho values are all within the acceptance limits and peels fall within confidence intervals.

11.3.2 Assessment results

Figure 11.2.3 illustrates the trends in observed catch, landing and discards. Reported landings gradually increased up to the late 1980s and then rapidly declined until 1995, in line with the decrease in TAC. The landings show a general decline from 1989 onwards, increasing slowly but steadily since 2007, and decreasing again since 2016. Discards were particularly high in 1997 and 1998 (reconstructed), and in 2001 and 2003 (observed), resulting from strong year-classes.

Figure 11.3.4 and Table 11.3.1 present the model estimated $F(2-6)$, SSB, and recruitment. The estimated SSB in 2025 is 1 230 426 tonnes and it is well above $MSY B_{trigger}$. SSB has markedly increased since 2008, following a substantial reduction in fishing mortality (F) since 1999. The estimated F in 2024 is 0.040, and it is below F_{MSY} since mid-2000. The estimated recruitment in 2024 is 7 407 716 thousand.

The estimated model parameters are presented in Table 11.3.2. The estimated fishing mortality and stock numbers are shown in Tables 11.3.3–11.3.4 and Figure 11.3.5, respectively.

The stock dynamics are partly affected by the occurrence of strong year-classes. Between 2000 and 2010, recruitment has been increasing. After 2010, recruitment has been fluctuating around its geometric mean. A high 2019 recruitment has been detected in all surveys (BTS-IBTS Q3, SNS and DYFS), but not shown in catches. Recruitment in 2022 turned out to be high again. The increased stock size in 2000-2014 is also partly the direct consequence of reduced fishing mortality. Additionally, the age composition in SSB (Figure 11.3.6) implies that the weight of older aged plaice (age ≥ 8) has been increasing since 2010. Information from surveys (BTS, IBTS-Q3, SNS and DFS) implies that older fishes are likely migrating to the northwestern part of the North Sea (WGBEAM; ICES 2020), where the targeted fishing effort is low (Figure 11.3.7). The leave-one-out analysis (Figure 11.3.8) indicates that BTS-IBTS Q3 is dominating the assessment, which is consistent with the fact that this survey has the highest area and age coverage as well as the lowest uncertainty in survey indices.

11.4 Short-term forecasts

Due to some adjustments of survey prediction grid and update of catch data (2021-2023), the reference points were recalculated during the 2025 working group. Short-term forecast was conducted under the new reference points (Section 11.5).

A stochastic short-term forecast was conducted under SAM structure. This includes simulating from the uncertainties of stock number, F , N process error in survival. In forecasting, the process error for F is switched off. The settings of the forecast are listed in the table below.

Variable	Assumption	Arguments in SAM forecast function
Initial stock size	Starting populations are simulated from the estimated distribution at the start of the base year (including co-variances).	Year.base=last_data_year; deterministic=FALSE nr_sims=5001

Variable	Assumption	Arguments in SAM forecast function
Survival	Forecast according to N processes	deterministic=FALSE
Maturity	Fixed values, time-invariant	-
Natural mortality	Fixed age-dependent Peterson M, determined during WKNSCS2022	-
F and M before spawning	0	-
Weight-at-age in the catch	Average over the last 3 years	ave.years=last 3 years
Weight-at-age in the stock	Average over the last 3 years	ave.years= last 3 years
Exploitation pattern	Simulated according to the average of the last 3 years	overwriteSelYears=last 3 years processNoiseF=FALSE
Median Fbar value in intermediate year	Last data year (Fsq)	fval = c(...,Flast,...)
Proportion of landing	Average over the last 3 years	ave.years=last 3 years; splitLD = TRUE
Recruitment in intermediate year	Sampling from recent 10 years time interval, include the last data year	rec.years=recent 10 years
Recruitment in TAC year	Sampling from recent 10 years time interval, include the last data year	rec.years=recent 10 years

The coefficient of variation (CV) of the recruitment estimates up to 2024 are low (<0.015 on log scale). Therefore, 2024 recruitment is included in the recruitment resampling in the forecast.

Although at historical low level, the mean weight in recent years do not show any strong decreasing trend, therefore, catch and stock weight for forecast was chosen as the average of the last 3 years.

Consistent to the reduced fishing efforts, the F_{bar} is still on a decreasing trend. Thus, F in the intermediate year was chosen as F of last data year. The trend of the exploitation pattern in recent years is unclear. Therefore, average of last 3 years was chosen for the intermediate year exploitation pattern as well as for the discards ratio.

Given the choice of recruitment and F in the intermediate year, the estimated median values for the catches and stock size are listed below.

Assumption	$F_{(2-6)} 2025$	SSB 2026	Recruitment 2025-2026	Landings 2025	Discards 2025
$F_{2025} = F_{2024}$	0.040	1 313 280 t	4 477 139 thousands	25 250 t	27 582 t

A series of F options were assumed for the TAC year. Resulting management options for 2026 are given in Table 11.5.1. The forecasting under F_{MSY} option is illustrated in Figure 11.3.9.

The change in advice (-7.3%) is due to a combination of the upwards revision of the size of the large 2023 year-class and a change in the F_{MSY} reference point from 0.152 to 0.123. Figures making a comparison with last year's advice are illustrated in Figures 11.3.10-11.3.13.

Mature individuals from the plaice stock in Division 7.e (ple.27.7e) and stock in Subarea 4 and Subdivision 20 (ple.27.420) migrate to Division 7.d in the first quarter. Thus, a proportion of the catch of these two stocks occurs in Division 7.d. For the ple.27.7e, ple.27.7d, and ple.27.420 stocks, the catch advice is given at stock level but the catch corresponding to each area is provided in the catch scenario table (ICES, 2024a). Of the plaice caught in Division 7.d, 12.86% is assumed to originate from the North Sea and 3.86% from the western English Channel (average ratio calculated over the years 2003–2024); both of these predicted catches are added to the predicted values for the ple.27.7d stock and applied to projected catch, projected landings, and projected discards. Following ICES advice basis, catches in Division 7.d will consist of 1151 tonnes from the ple.27.7d stock, plus 178 tonnes from the ple.27.420 stock and 53 tonnes from the ple.27.7e stock. All catch scenarios for ple.27.7e and ple.27.420 stocks assume that fishing opportunities for ple.27.7d stock in 2025 correspond to the “ICES basis for the advice scenario”; for the ple.27.7d stock, however, this amount depends on the scenario.

11.5 Biological reference points

The reference points were updated during the 2025 working group (WD 2, Annex 8).

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{trigger}$	544 290	B_{pa} ; in tonnes.	This report
	F_{MSY}	0.123	Stochastic simulations (EqSim) based on the recruitment period 1957–2024.	This report
Precautionary approach	B_{lim}	391 695	Lowest SSB level that in the past has resulted in good recruitment; in tonnes	This report
	B_{pa}	544 290	$B_{lim} \times \exp(1.645 \times \sigma)$, $\sigma = 0.20$; in tonnes.	This report
	F_{pa}	0.141	The F that provides a 95% probability for SSB to be above B_{lim} ($F_{P.05}$ with advice rule).	This report
	MAP MSY $B_{trigger}$	544 290	MSY $B_{trigger}$; in tonnes.	This report
EU Management plan*	MAP B_{lim}	391 695	B_{lim} ; in tonnes.	This report
	MAP F_{MSY}	0.123	F_{MSY}	This report
	MAP range F_{lower}	0.094–0.123	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY.	This report
	MAP range F_{upper}	0.123–0.141	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY. Capped at F_{pa} .	This report

* EU multiannual plan (MAP) for the North Sea (EU, 2018).

11.6 Quality of the assessment

The current assessment assumes no discard survival.

The abundance index of SNS surveys targeting juvenile plaice has been declining because of the earlier migration of younger fish to offshore areas, which may result in an underestimation of the recruiting year classes

11.7 Status of the stock

SSB in 2025 is estimated around 1 230 313 tonnes which is well above $MSY B_{trigger}$, B_{pa} , and B_{lim} . Fishing mortality in 2024 is estimated to be at a value of 0.040, which is below F_{MSY} of 0.123 (Figure 11.4.1).

11.8 Management considerations

Plaice is mainly taken by beam trawlers in a mixed fishery with sole in the southern and central part of the North Sea. There are a number of EC regulations that affect the fisheries on plaice and sole in the North Sea, e.g., as a basis for setting the TAC, limiting effort, minimum landing size and minimum mesh size.

11.8.1 Multiannual plan North Sea

An EU multiannual management plan (MAP) has been agreed by the EU for this stock (EU, 2018). There is no agreement with Norway regarding this plan, and it is not used as the basis of the advice for this shared stock. ICES was requested by the EC and UK to provide advice based on the MSY approach and to include F_{MSY} ranges in the catch scenarios.

11.8.2 Effort regulations (North Sea)

Regulated effort restrictions in the EU were introduced in 2003 (annexes to the annual TAC regulations) for the protection of the North Sea cod stock. In addition, a long-term plan for the recovery of cod stocks was adopted in 2008 (EC regulation 1342/2008). In 2009, the effort management programme switched from a days-at-sea to a kW-day system (EC regulation 43/2009), in which different amounts of kW-days are allocated within each area by member state to different groups of vessels depending on gear and mesh size. Effort ceilings are updated annually. A minor part of the fleets exploiting sole, i.e. otter trawls (OTB) with a mesh size equal to or larger than 100 mm included in Figure 11.2.1, have since 2009 been affected by the regulation. The beam trawl fleet (BT2) was affected by this regulation only once in 2009 but not afterwards.

The overall fleet capacity and deployed effort of the North Sea beam trawl fleet has been substantially reduced since 1995, likely due to a number of reasons, including the above-mentioned effort limitations for the recovery of the cod stock. 25 vessels were decommissioned in 2014. In addition, the current sole and plaice long-term management plan specifically reduces effort as a management measure. However, the evaluation of amendments to the plan in 2012 showed that the plan is consistent with the precautionary approach and the principle of maximum sustainable yield (MSY) also without reductions of effort.

Between 2014 and 2018, the majority of the Dutch beam trawlers have switched to pulse trawls, resulting in reallocation of fishing effort to fishing grounds in the southern part of Division 4.c targeting sole. The discards ratio for plaice has been increasing in the same period, from 40% in

2014 to 63% in 2023. Pulse trawling has been prohibited since 30 June 2021, following the EU decision in February 2019 to revise the technical measures regulations. In the meantime, the Dutch fleets targeting plaice are experiencing substantial decommission partly due to the high fuel prices.

11.8.3 Technical measures

Technical measures applicable to the mixed flatfish beam-trawl fishery in the southern North Sea affect both sole and plaice. The minimum mesh size of 80 mm selects sole at the minimum landing size. However, this mesh size generates high discards of plaice with a larger minimum landing size than sole. For the overall fleet the discards ratio has been slightly decreasing since 2003 and increasing up again since 2016. In 2020, discards ratio was approximately 48% by weight. Mesh enlargement would reduce the catch of undersized plaice but would also result in loss of marketable sole. Furthermore, the size selectivity of the fleet may lead to a shift in the age and size at maturation. For example, in recent years plaice and sole have become mature at younger ages and at smaller sizes than in the past (Grift *et al.*, 2003). The introduction of the Omega (mesh size) meter in 2010 has led to a slight increase in the effective mesh size in the fishery.

Technical management measures have caused a shift towards two categories of vessels: 2000 HP (the maximum engine power allowed) and 300 HP. The 300 HP vessels are allowed to fish within the 12-nautical mile coastal zone and in the Plaice Box. The Plaice Box is a partially closed area along the continental coast that was implemented in phases, starting in 1989. The area has been closed to most categories of vessels >300 HP all year round since 1995. The most recent EU-funded evaluation by Beare *et al.* (2010) reported the Plaice Box as having very little impact on the plaice stock.

11.9 Issues for future benchmarks

11.9.1 Data

- Migration between plaice in Subarea 4 and Subdivision 20 and plaice in Division 7d is known to occur. However, current estimates are based on historical studies and may not reflect current dynamics. New studies may be needed to better quantify recent plaice migration.
- Plaice have heterogeneous age distributions in the North Sea: younger ages are distributed more closely to coastal area while older ages are distributed towards north-west of the North Sea. In recent years, strong younger age signals appeared in IBTS Q3 survey around Scotland coast. The accuracy and uncertainty of these signals need to be investigated, e.g., age readings, gear selectivity (Scottish gear has a different selectivity).
- The abundance indices of SNS and DYFS surveys targeting age 0-1 plaice have been declining due to the earlier migration of younger fishes to offshore areas. A combined SNS-DYFS-BTS-IBTSQ3 survey index is desired to cover the entire stock area for all ages.
- Catch data in some historical years are inconsistent: large old ages discards in some years, and zero discards weight with large discards. Additionally, the inclusion of Subdivision 20 was not done in InterCatch. The catch data need to be re-estimated within InterCatch.

11.9.2 Assessment

- Residual age and year patterns in BTS-IBTS Q3 survey and catches need to be solved.
- Reduce “error” in discards estimation by including non-zero survival in assessment.

- Add survey uncertainty in the assessment.

11.9.3 Short-term forecast

- The recruitment signal of age 0 in SNS and DFS surveys are no longer used in the short-term forecast. A new method needs to be considered to include SNS and DYFS age 0 indices in assessment model to predict recruitment in the intermediate year, e.g., combined indices of all surveys including age 0.

11.10 References

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11.11 Tables and Figures

Table 11.2.1. Plaice in Subarea 4 and Subdivision 20. Official landings, ICES landings and TAC in tonnes. Division 7.d Q1 not included.

North Sea												Skagerrak		
Year	Belgium	Denmark	France	Germany	Netherlands	Norway	Sweden	UK	Others	Total	ICES estimate	TAC NS	Total	TAC SK
1982	7103	24 532	1046	3628	55 715	16	6	20 889	0	112 935	154 546	140 000	8083	
1983	8916	19114	1185	2397	53 608	17	22	17 407	0	102 666	144 030	164 000	7130	
1984	10 220	23361	1145	2485	61 478	17	14	17183	0	115 903	156 149	182 000	7920	
1985	9965	28 236	1010	2197	90 950	23	18	15 912	0	148 311	159 838	200 000	10 096	
1986	7232	26 332	751	1809	74 447	21	16	17 294	0	127 902	165 347	180 000	11 378	
1987	8554	21 597	1580	1794	76 612	12	7	20 638	0	130 794	153 670	150 000	12 502	
1988	11 527	20 259	1773	2566	77 724	21	2	24 497	43	138 412	154 475	175 000	10 820	
1989	10 939	23 481	2037	5341	84 173	321	12	26 104	0	152 408	169 818	185 000	5997	
1990	13 940	26 474	1339	8747	78 204	1756	169	25 632	0	156 261	156 240	180 000	10 048	
1991	14 328	24 355	508	7926	67 945	560	103	27 839	0	143 564	148 003	175 000	6679	
1992	12 006	20 890	537	6818	51 064	836	53	31 277	0	123 481	125 190	175 000	9554	11 200

North Sea												Skagerrak		
Year	Belgium	Denmark	France	Germany	Netherlands	Norway	Sweden	UK	Others	Total	ICES estimate	TAC NS	Total	TAC SK
1993	10 814	16 452	602	6895	48 552	827	7	31 128	0	115 277	117 113	175 000	9854	11 200
1994	7951	17 056	407	5697	50 289	524	6	27 749	0	109 679	11 0392	165 000	9551	11 200
1995	7093	13 358	442	6329	44 263	527	3	24 395	0	96 410	98 356	115 000	9379	11 200
1996	5765	11 776	379	4780	35 419	917	5	20 992	0	80 033	81 673	81 000	8003	11 200
1997	5223	13 940	254	4159	34 143	1620	10	22 134	0	81 483	83 048	91 000	7813	11 200
1998	5592	10 087	489	2773	30 541	965	2	19 915	1	70 365	71 534	87 000	6449	11 200
1999	6160	13 468	624	3144	37 513	643	4	17 061	0	78 617	80 662	102 000	7049	11 200
2000	7260	13 408	547	4310	35 030	866	3	20 710	0	82 134	81 150	97 000	7001	11 200
2001	6369	13 797	429	4739	33 290	1805	3	19 147	0	79 579	81 847	78 000	9351	9400
2002	4859	12 552	552	3927	29 081	1897	2	16 740	1	69 611	70 217	77 000	7080	6400
2003	4580	13 742	318	3800	27 372	1699	2	13 892	0	65 405	66 489	73 250	7065	1400
2004	4344	12 121	241	3649	23 659	1737	1	15 312	0	61 064	61 436	61 000	8172	9500
2005	3414	11 385	192	3380	22 266	1660	0	10 438	0	52 735	55 700	59 000	6146	7600
2006	3524	11 909	175	3599	22 764	1430	0	12 431	0	55 832	57 943	57 441	8444	7600

North Sea												Skagerrak		
Year	Belgium	Denmark	France	Germany	Netherlands	Norway	Sweden	UK	Others	Total	ICES estimate	TAC NS	Total	TAC SK
2007	3894	8132	227	2643	21 464	1224	4	11 558	0	49 146	49 744	50 261	7682	8500
2008	3396	8232	188	3138	20 312	1051	20	11 409	0	47 746	48 875	49 000	7912	9300
2009	3537	8156	272	2931	23 152	1116	1	13 143	0	52 308	54 973	55 500	6299	9300
2010	3699	9369	385	3787	26 707	1090	5	14 789	0	59 831	60 674	63 825	8810	9300
2011	4588	11 636	375	3812	29 274	1224	3	15 500	0	66 412	67 386	73 400	8183	7900
2012	5066	12 243	281	3873	32 193	1025	5	16 916	0	71 602	73 830	84 410	7651	7900
2013	6508	13 650	250	4921	33 488	843	3	19 336	0	78 999	78 905	97 070	6815	9142
2014	7152	11 737	276	4224	27 720	584	5	17 380	0	69 078	70 847	111 631	8324	10 056
2015	5750	14 425	226	5171	30 860	169	7	17 395	0	74 003	74 963	128 376	9804	10 056
2016	6664	16 547	169	4371	32 228	94	9	18 800	0	78 882	81 059	131 714	9772	11 766
2017	5316	12 517	151	2621	31 049	67	5	15 130	0	66 856	65 442	129 917	8643	17 639
2018	4887	9670	112	2616	24 742	69	3	9743	0	51 842	50 783	112 643	6009	15 343
2019	3912	6585	62	2074	21 185	57	3	7456	0	41 334	39 970	125 435	7597	16 782
2020	2560	5639	25	1403	18 588	36	5	5604	0	33 860	32 736	146 852	7256	19 647

North Sea												Skagerrak		
Year	Belgium	Denmark	France	Germany	Netherlands	Norway	Sweden	UK	Others	Total	ICES estimate	TAC NS	Total	TAC SK
2021	2318	4907	9	1261	15 943	24	4	4878	0	29 344	28 925	143 419	6539	19 188
2022	1302	3530	12	934	11 167	18	3	4475	0	21 442	21 311	125 692	3840	16 816
2023*	1664	3306	9	1499	7431	40	7	4723	0	18 680	19 566	132 922	3395	17 783
2024*	1449	2493	6	767	5498	38	5	3943	0	14 199	14 489	136 413	4836	18 250

* Official values are preliminary.

Table 11.2.2. Plaice in Subarea 4 and Subdivision 20. Landings (SOP corrected) in numbers by age (including 1st quarter of 7.d) in thousands as input.

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	0	4792	66428	49659	35282	9867	12248	10026	5523	12059
1958	0	7581	23612	65979	36274	20836	8696	8507	6497	13981
1959	0	16914	31085	26040	41988	23432	14173	6547	6739	16530
1960	0	5998	62285	51359	21462	27510	14280	9073	5121	15253
1961	0	2299	33913	68965	33209	12958	14909	9900	6090	14889
1962	0	2075	34677	64548	48387	19939	8757	8733	5081	12373
1963	0	4424	21886	78412	55414	32413	13096	6965	7183	16913
1964	0	14818	40789	65219	57838	37369	15937	6644	4010	17012
1965	0	9913	42439	53486	43919	30320	18464	8602	4237	17686
1966	0	4220	66196	52428	37336	27870	16801	10981	6585	15201
1967	0	6101	30905	115157	42204	22491	16496	8163	6862	11397
1968	0	9750	41883	39251	127220	17638	10642	10396	4039	13754
1969	3	15892	47819	38186	37657	107955	11016	6440	8669	17029
1970	74	16850	49861	54712	39642	34174	76862	6149	4078	14459
1971	20	30568	49876	34580	26919	23659	17471	30711	6626	17468
1972	2296	37561	63958	54402	23695	17479	14787	11211	19111	16094
1973	1332	33342	62095	76769	44397	14518	9335	10347	6392	25194
1974	2305	23972	57595	43677	42588	20391	8300	6554	5773	22790
1975	1042	29878	65465	33211	27004	22509	12613	6292	4362	20923
1976	2892	34498	79621	98846	14129	10156	9352	6553	3022	12871
1977	3225	57062	43359	66120	83841	9157	5922	5030	4068	9206
1978	1102	58412	60114	52398	48310	34240	5728	3232	2333	7201
1979	1316	57933	118662	48879	47805	39864	24187	4154	2802	9272
1980	996	66094	136273	79034	25549	18321	14018	8622	1898	5497
1981	259	103352	125925	59563	36671	12751	9805	8296	5006	6092
1982	3373	48353	212184	71166	29190	16976	7704	5552	4539	8776
1983	1214	119693	115330	100471	29591	12959	8239	4225	3013	8309

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1984	108	63507	280478	62835	41492	15416	6842	5594	2730	6551
1985	120	72805	146836	201625	37939	17106	7442	3781	2814	5831
1986	1669	66934	165985	106460	101683	27971	9840	4704	2834	7083
1987	1	85152	118414	120780	81304	44589	13540	4669	2347	5610
1988	1	15200	253812	85346	59949	31492	19348	6199	3435	6402
1989	1254	46810	108271	238241	58767	21668	11605	8025	2321	5807
1990	1546	33765	104795	119827	169463	29946	9054	4690	3803	4207
1991	1425	43064	87196	122232	76074	78728	15410	5390	3215	5635
1992	3386	43768	86357	81469	88534	37542	30444	7229	3295	6977
1993	3416	53555	99804	80854	63274	35043	14745	11500	3705	5884
1994	1375	44554	105863	86992	47578	27680	17279	6661	5449	5458
1995	7779	36760	82648	84776	47912	24573	14746	5285	2495	3897
1996	1103	43345	68153	52959	37284	19159	12400	5882	2800	4990
1997	897	43121	88686	49361	31750	18673	9519	5037	3054	4401
1998	197	30593	74438	62337	22793	9152	5704	2870	1984	3360
1999	549	8689	158087	47390	31778	14078	4039	2625	1597	3234
2000	2603	15656	40818	171990	25935	12586	2980	1136	954	2121
2001	4523	37094	58677	57194	101522	11493	4740	1213	650	2364
2002	1228	16235	60446	55421	44217	43445	6574	2274	791	1635
2003	700	45057	50743	54869	34673	20301	18131	1767	670	884
2004	553	12161	119172	39012	23732	13283	5143	4747	448	570
2005	2951	19047	29715	90836	20113	10882	5895	2749	2289	647
2006	372	20538	80190	34297	51116	8074	5587	2304	1326	1395
2007	1436	21434	42105	55943	20362	21832	3092	2011	607	1317
2008	401	13264	52292	44976	33950	7523	7981	978	735	936
2009	1562	12648	62120	42621	24863	18530	3385	4261	587	812
2010	2115	20021	49208	69775	25169	12626	9766	1869	2516	1276
2011	410	13379	45670	62162	51615	14834	6656	6992	1266	2758

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2012	163	6219	60711	62096	45022	32055	7576	3391	3484	1924
2013	550	10527	63674	77208	42368	29534	15354	3958	2467	3796
2014	7	5526	41043	78027	52621	22065	13011	8405	2477	3449
2015	0	3851	42710	67011	60995	32335	12811	8898	4061	4840
2016	0	4183	39280	85224	61018	39884	19111	7717	5315	5126
2017	27	5290	24921	58084	57725	30872	16848	7582	3070	3213
2018	17	7859	24839	34137	43492	31008	16014	8980	5380	4127
2019	1	6566	43892	32320	18826	18141	11464	6933	3952	3931
2020	80	5634	19002	44795	19094	10231	11667	7598	4807	6438
2021	35	6400	21054	25254	32890	8718	5313	5751	3812	4379
2022	0	3374	14891	28294	13448	10980	5015	2490	1960	3257
2023	3	4517	8651	15177	26866	8159	7893	2835	1482	2291
2024	0	1641	19589	14191	10301	10181	3695	2293	1575	2287

Table 11.2.3. Plaice in Subarea 4 and Subdivision 20. Discards in numbers by age (including 1st quarter of 7.d) in thousands as input.

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	32356	45596	9220	909	961	25	0	0	0	0
1958	66199	73552	23655	2572	2137	65	0	0	0	0
1959	116086	127771	46402	11407	4737	106	0	0	0	0
1960	73939	167893	44948	997	1067	519	0	0	0	0
1961	75578	144609	89014	538	1612	130	0	0	0	0
1962	51265	181321	87599	21716	799	186	0	0	0	0
1963	90913	136183	129778	9964	2112	188	0	0	0	0
1964	66035	153274	64156	33825	3011	323	0	0	0	0
1965	43708	426021	59262	3404	923	267	0	0	0	0
1966	38496	163125	349358	14399	1402	125	0	0	0	0
1967	20199	133545	87532	152496	623	260	0	0	0	0
1968	73971	72192	46339	26530	22436	58	0	0	0	0

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1969	85192	67378	16747	19334	773	2024	0	0	0	0
1970	123569	152480	27747	1287	5061	161	0	0	0	0
1971	69337	96968	42354	2675	426	81	0	0	0	0
1972	70002	55470	33899	5714	567	73	0	0	0	0
1973	132352	49815	4008	673	1289	67	0	0	0	0
1974	211139	308411	3652	285	611	109	0	0	0	0
1975	244969	280130	190536	4807	253	123	0	0	0	0
1976	183879	140921	71054	18013	174	41	0	0	0	0
1977	256628	103696	79317	33552	9317	129	0	0	0	0
1978	226872	154113	27257	10775	1244	570	0	0	0	0
1979	293166	215084	57578	18382	589	310	0	0	0	0
1980	226371	122561	932	687	193	86	0	0	0	0
1981	134142	193241	1850	373	431	55	0	0	0	0
1982	411307	204572	4624	1109	216	98	0	0	0	0
1983	261400	436331	30716	2235	804	72	0	0	0	0
1984	310675	313490	52651	24529	1492	69	0	0	0	0
1985	405385	229208	35566	2221	200	78	0	0	0	0
1986	1117345	490965	48510	26470	1451	146	0	0	0	0
1987	361519	1374202	180969	1427	1348	248	0	0	0	0
1988	348597	608109	459385	61167	882	177	0	0	0	0
1989	213291	485845	193176	85758	7224	115	0	0	0	0
1990	145314	279298	168674	28102	5011	177	0	0	0	0
1991	183126	301575	141567	40739	5528	939	0	0	0	0
1992	138755	219619	94581	34348	4307	880	0	0	0	0
1993	96371	154083	48088	11966	1635	216	0	0	0	0
1994	62122	95703	35703	1038	822	144	0	0	0	0
1995	118863	82676	15753	860	663	120	0	0	0	0
1996	111250	331065	27606	3930	451	116	0	0	0	0

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1997	128653	510918	193828	588	271	108	0	0	0	0
1998	104538	646250	191631	53354	297	33	0	0	0	0
1999	127321	208401	231769	54869	278	58	0	0	0	0
2000	103468	171213	51092	64971	1230	241	263	167	1	0
2001	30346	352452	186900	74744	54276	152	45	1	1	0
2002	310442	178402	78296	13940	2834	718	109	1	1	0
2003	67798	523336	56580	20184	4358	419	5756	1	2	0
2004	233682	183508	127876	10650	1975	450	41	1	2	0
2005	93936	332157	46454	23763	4494	6007	287	6	54	0
2006	221002	227346	117609	9813	2374	252	736	195	1	1080
2007	77710	210934	73218	13956	1597	7030	190	1644	1349	1056
2008	135516	255986	37947	5324	1778	335	8852	885	41	613
2009	148673	193760	69257	9533	2022	1108	138	3220	3	8472
2010	167426	181374	60191	22762	2699	1736	2074	284	531	810
2011	117970	155602	63372	37095	12956	2941	143	2273	0	16534
2012	91977	313906	123957	33004	9508	1591	226	7	630	3084
2013	128363	157823	126148	25023	4666	1035	219	15	7	0
2014	293662	193847	117052	55539	19354	2624	477	68	291	0
2015	83478	290298	131518	39089	12638	2336	520	209	17	404
2016	79206	145188	134386	48480	21120	7463	2069	1858	180	1
2017	129907	146839	77109	57790	15797	3802	1256	233	11	7
2018	64626	267279	102272	40366	21693	5050	1486	266	55	17
2019	134695	116059.7	120622	29738	11748.8	8483.6	3134.2	1410.1	133	29
2020	94122	191628	64435	55542	9823	3631	4398	1190	1047	39
2021	72022	109482	156376	31539	17838	3960	640	1093	574	12
2022	141115	84969	61515	49242	7439	4291	1736	1174	160	93
2023	90968	262956	44599	30465	24754	4539	2317	1138	482	133
2024	77536	80638	86380	19929	10005	9885	3511	3034	1083	57

Table 11.2.4. Plaice in Subarea 4 and Subdivision 20. Catch in numbers by age (including 1st quarter of 7.d) in thousands as input.

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	32356	50388	75648	50568	36243	9892	12248	10026	5523	12059
1958	66199	81133	47267	68551	38411	20901	8696	8507	6497	13981
1959	116086	144685	77487	37447	46725	23538	14173	6547	6739	16530
1960	73939	173891	107233	52356	22529	28029	14280	9073	5121	15253
1961	75578	146908	122927	69503	34821	13088	14909	9900	6090	14889
1962	51265	183396	122276	86264	49186	20125	8757	8733	5081	12373
1963	90913	140607	151664	88376	57526	32601	13096	6965	7183	16913
1964	66035	168092	104945	99044	60849	37692	15937	6644	4010	17012
1965	43708	435934	101701	56890	44842	30587	18464	8602	4237	17686
1966	38496	167345	415554	66827	38738	27995	16801	10981	6585	15201
1967	20199	139646	118437	267653	42827	22751	16496	8163	6862	11397
1968	73971	81942	88222	65781	149656	17696	10642	10396	4039	13754
1969	85195	83270	64566	57520	38430	109979	11016	6440	8669	17029
1970	123643	169330	77608	55999	44703	34335	76862	6149	4078	14459
1971	69357	127536	92230	37255	27345	23740	17471	30711	6626	17468
1972	72298	93031	97857	60116	24262	17552	14787	11211	19111	16094
1973	133684	83157	66103	77442	45686	14585	9335	10347	6392	25194
1974	213444	332383	61247	43962	43199	20500	8300	6554	5773	22790
1975	246011	310008	256001	38018	27257	22632	12613	6292	4362	20923
1976	186771	175419	150675	116859	14303	10197	9352	6553	3022	12871
1977	259853	160758	122676	99672	93158	9286	5922	5030	4068	9206
1978	227974	212525	87371	63173	49554	34810	5728	3232	2333	7201
1979	294482	273017	176240	67261	48394	40174	24187	4154	2802	9272
1980	227367	188655	137205	79721	25742	18407	14018	8622	1898	5497
1981	134401	296593	127775	59936	37102	12806	9805	8296	5006	6092
1982	414680	252925	216808	72275	29406	17074	7704	5552	4539	8776
1983	262614	556024	146046	102706	30395	13031	8239	4225	3013	8309

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1984	310783	376997	333129	87364	42984	15485	6842	5594	2730	6551
1985	405505	302013	182402	203846	38139	17184	7442	3781	2814	5831
1986	1119014	557899	214495	132930	103134	28117	9840	4704	2834	7083
1987	361520	1459354	299383	122207	82652	44837	13540	4669	2347	5610
1988	348598	623309	713197	146513	60831	31669	19348	6199	3435	6402
1989	214545	532655	301447	323999	65991	21783	11605	8025	2321	5807
1990	146860	313063	273469	147929	174474	30123	9054	4690	3803	4207
1991	184551	344639	228763	162971	81602	79667	15410	5390	3215	5635
1992	142141	263387	180938	115817	92841	38422	30444	7229	3295	6977
1993	99787	207638	147892	92820	64909	35259	14745	11500	3705	5884
1994	63497	140257	141566	88030	48400	27824	17279	6661	5449	5458
1995	126642	119436	98401	85636	48575	24693	14746	5285	2495	3897
1996	112353	374410	95759	56889	37735	19275	12400	5882	2800	4990
1997	129550	554039	282514	49949	32021	18781	9519	5037	3054	4401
1998	104735	676843	266069	115691	23090	9185	5704	2870	1984	3360
1999	127870	217090	389856	102259	32056	14136	4039	2625	1597	3234
2000	106071	186869	91910	236961	27165	12827	3243	1303	955	2121
2001	34869	389546	245577	131938	155798	11645	4785	1214	651	2364
2002	311670	194637	138742	69361	47051	44163	6683	2275	792	1635
2003	68498	568393	107323	75053	39031	20720	23887	1768	672	884
2004	234235	195669	247048	49662	25707	13733	5184	4748	450	570
2005	96887	351204	76169	114599	24607	16889	6182	2755	2343	647
2006	221374	247884	197799	44110	53490	8326	6323	2499	1327	2475
2007	79146	232368	115323	69899	21959	28862	3282	3655	1956	2373
2008	135917	269250	90239	50300	35728	7858	16833	1863	776	1549
2009	150235	206408	131377	52154	26885	19638	3523	7481	590	9284
2010	169541	201395	109399	92537	27868	14362	11840	2153	3047	2086
2011	118380	168981	109042	99257	64571	17775	6799	9265	1266	19292

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2012	92140	320125	184668	95100	54530	33646	7802	3398	4114	5008
2013	128913	168350	189822	102231	47034	30569	15573	3973	2474	3796
2014	293669	199373	158095	133566	71975	24689	13488	8473	2768	3449
2015	83478	294149	174228	106100	73633	34671	13331	9107	4078	5244
2016	79206	149371	173666	133704	82138	47347	21180	9575	5495	5127
2017	129934	152129	102030	115874	73522	34674	18104	7815	3081	3220
2018	64643	275138	127111	74503	65185	36058	17500	9246	5435	4144
2019	134696	122625	164514	62058	30575	26625	14598	8343	4085	3960
2020	94202	197262	83437	100337	28917	13862	16065	8788	5854	6477
2021	72057	115882	177430	56793	50728	12678	5953	6844	4386	4391
2022	141115	88343	76406	77536	20887	15271	6751	3664	2120	3350
2023	90971	267473	53250	45642	51620	12698	10210	3973	1964	2424
2024	77536	82279	105969	34120	20306	20066	7206	5327	2658	2344

Table 11.2.5. Plaice in Subarea 4 and Subdivision 20. ICES catches estimates including catches in 7d during spawning migration in the 1st quarter. ICES discards since 2016 includes BMS landings.

North Sea and Skagerrak			Eastern English Channel (Q1)	
Year	ICES Landings	ICES Discards	ICES Landings	ICES Discards
1980	150461	31080	329	
1981	148198	33031	585	
1982	162619	49127	635	
1983	151160	74483	730	
1984	164070	70816	702	
1985	169933	60549	786	
1986	176725	129953	893	
1987	166173	190524	1046	
1988	165295	156423	1258	
1989	175815	107793	1281	
1990	166288	71225	1306	

North Sea and Skagerrak			Eastern English Channel (Q1)	
Year	ICES Landings	ICES Discards	ICES Landings	ICES Discards
1991	154682	80935	1116	
1992	134744	57049	860	
1993	126967	35016	655	
1994	119943	23785	826	
1995	107736	21828	719	
1996	89676	52049	741	
1997	90862	100145	867	
1998	77983	103751	716	
1999	87711	70976	814	
2000	88139	44311	1149	
2001	91078	100309	682	
2002	77319	55099	1037	
2003	73632	79275	447	
2004	69469	57478	402	
2005	61799	56250	326	
2006	66288	64160	534	129
2007	57365	42373	546	140
2008	57231	46993	223	32
2009	61487	45902	457	166
2010	69374	46570	505	272
2011	75604	41593	562	433
2012	81510	59914	510	383
2013	85717	40025	580	410
2014	80060	52937	895	710
2015	84767	50108	623	650
2016	91959	43971	824	669
2017	74217	38966	721	884
2018	57012	47792	827	831

North Sea and Skagerrak			Eastern English Channel (Q1)	
Year	ICES Landings	ICES Discards	ICES Landings	ICES Discards
2019	48063	36713	717	840
2020	40562	38110	356	253
2021	35699	37283	251	641
2022	25304	27755	297	1091
2023	22977	32940	176	640
2024	19361	23881	138	578

Table 11.2.6. Plaice in Subarea 4 and Subdivision 20. Stock weight-at-age (kg).

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	0.038	0.102	0.157	0.242	0.325	0.485	0.719	0.682	0.844	0.918
1958	0.041	0.093	0.180	0.272	0.303	0.442	0.577	0.778	0.793	0.945
1959	0.045	0.106	0.173	0.264	0.329	0.470	0.650	0.686	0.908	0.897
1960	0.038	0.111	0.181	0.272	0.364	0.469	0.633	0.726	0.845	0.918
1961	0.037	0.098	0.185	0.306	0.337	0.483	0.579	0.691	0.779	0.911
1962	0.036	0.096	0.173	0.301	0.424	0.573	0.684	0.806	0.873	1.335
1963	0.041	0.103	0.176	0.273	0.378	0.540	0.663	0.788	0.882	0.961
1964	0.024	0.113	0.184	0.296	0.373	0.477	0.645	0.673	0.845	0.973
1965	0.031	0.068	0.198	0.294	0.333	0.43	0.516	0.601	0.722	0.578
1966	0.031	0.099	0.127	0.305	0.403	0.455	0.503	0.565	0.581	0.848
1967	0.029	0.104	0.179	0.205	0.442	0.528	0.585	0.650	0.703	0.833
1968	0.055	0.094	0.175	0.287	0.344	0.532	0.592	0.362	0.667	0.746
1969	0.047	0.158	0.188	0.266	0.344	0.390	0.565	0.621	0.679	0.635
1970	0.043	0.113	0.236	0.274	0.369	0.410	0.468	0.636	0.732	0.747
1971	0.051	0.109	0.251	0.344	0.413	0.489	0.512	0.583	0.696	0.707
1972	0.056	0.158	0.218	0.407	0.473	0.534	0.579	0.606	0.655	0.759
1973	0.037	0.134	0.237	0.308	0.468	0.521	0.566	0.583	0.617	0.690
1974	0.049	0.105	0.217	0.416	0.437	0.524	0.570	0.629	0.652	0.690
1975	0.063	0.141	0.187	0.388	0.483	0.544	0.610	0.668	0.704	0.762

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1976	0.082	0.169	0.226	0.308	0.484	0.550	0.593	0.658	0.694	0.743
1977	0.064	0.184	0.265	0.311	0.405	0.551	0.627	0.690	0.667	0.759
1978	0.064	0.151	0.319	0.373	0.411	0.467	0.547	0.630	0.704	0.773
1979	0.062	0.179	0.258	0.365	0.414	0.459	0.543	0.667	0.764	0.826
1980	0.049	0.163	0.289	0.428	0.444	0.524	0.582	0.651	0.778	1.025
1981	0.041	0.140	0.239	0.421	0.473	0.536	0.570	0.624	0.707	0.849
1982	0.048	0.128	0.250	0.351	0.490	0.589	0.631	0.679	0.726	0.828
1983	0.045	0.128	0.242	0.381	0.494	0.559	0.624	0.712	0.754	0.791
1984	0.048	0.129	0.216	0.413	0.464	0.571	0.649	0.692	0.787	0.898
1985	0.048	0.146	0.232	0.320	0.452	0.536	0.635	0.656	0.764	0.869
1986	0.043	0.126	0.245	0.311	0.440	0.533	0.692	0.779	0.888	0.971
1987	0.036	0.105	0.200	0.383	0.401	0.503	0.573	0.711	0.747	0.817
1988	0.036	0.097	0.172	0.264	0.426	0.467	0.547	0.644	0.706	0.897
1989	0.039	0.101	0.192	0.247	0.362	0.484	0.553	0.616	0.759	0.837
1990	0.043	0.108	0.176	0.261	0.343	0.422	0.555	0.647	0.701	0.760
1991	0.048	0.131	0.184	0.260	0.342	0.401	0.463	0.633	0.652	0.744
1992	0.043	0.121	0.199	0.270	0.318	0.403	0.500	0.573	0.683	0.730
1993	0.050	0.119	0.208	0.315	0.330	0.391	0.490	0.587	0.633	0.723
1994	0.053	0.141	0.214	0.290	0.360	0.404	0.462	0.533	0.653	0.702
1995	0.050	0.142	0.254	0.336	0.399	0.448	0.509	0.584	0.678	0.789
1996	0.044	0.117	0.229	0.368	0.390	0.462	0.488	0.554	0.660	0.791
1997	0.035	0.115	0.233	0.359	0.439	0.492	0.521	0.543	0.627	0.734
1998	0.038	0.081	0.207	0.333	0.474	0.577	0.581	0.648	0.656	0.642
1999	0.044	0.091	0.150	0.319	0.437	0.524	0.586	0.644	0.664	0.620
2000	0.051	0.106	0.165	0.219	0.408	0.467	0.649	0.695	0.656	0.744
2001	0.061	0.122	0.202	0.233	0.331	0.452	0.560	0.641	0.798	0.816
2002	0.048	0.118	0.213	0.301	0.319	0.403	0.446	0.612	0.685	0.781
2003	0.057	0.111	0.227	0.269	0.344	0.391	0.464	0.600	0.714	0.960

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2004	0.047	0.116	0.201	0.306	0.384	0.430	0.489	0.495	0.780	0.921
2005	0.053	0.106	0.216	0.237	0.378	0.422	0.434	0.527	0.621	0.815
2006	0.052	0.130	0.190	0.316	0.354	0.424	0.439	0.506	0.583	0.688
2007	0.047	0.093	0.235	0.238	0.337	0.394	0.458	0.412	0.526	0.512
2008	0.048	0.114	0.196	0.274	0.355	0.429	0.484	0.627	0.598	0.449
2009	0.052	0.114	0.194	0.344	0.373	0.412	0.472	0.540	0.565	0.576
2010	0.053	0.116	0.179	0.340	0.361	0.401	0.448	0.572	0.568	0.655
2011	0.039	0.100	0.187	0.209	0.355	0.483	0.438	0.422	0.530	0.580
2012	0.052	0.093	0.142	0.188	0.331	0.393	0.484	0.479	0.480	0.518
2013	0.043	0.107	0.153	0.208	0.320	0.354	0.434	0.493	0.662	0.468
2014	0.048	0.104	0.158	0.202	0.312	0.380	0.439	0.484	0.458	0.615
2015	0.024	0.065	0.120	0.207	0.279	0.323	0.379	0.435	0.465	0.457
2016	0.030	0.066	0.117	0.198	0.260	0.329	0.380	0.434	0.479	0.514
2017	0.032	0.069	0.132	0.181	0.270	0.333	0.359	0.458	0.476	0.557
2018	0.036	0.064	0.116	0.165	0.215	0.276	0.327	0.366	0.412	0.595
2019	0.022	0.063	0.117	0.173	0.240	0.261	0.352	0.391	0.415	0.443
2020	0.026	0.058	0.114	0.163	0.208	0.248	0.323	0.351	0.424	0.458
2021	0.033	0.069	0.104	0.165	0.219	0.263	0.331	0.357	0.465	0.494
2022	0.023	0.071	0.107	0.15	0.203	0.259	0.274	0.287	0.488	0.472
2023	0.029	0.051	0.108	0.145	0.183	0.254	0.263	0.276	0.487	0.492
2024	0.018	0.062	0.112	0.16	0.192	0.233	0.252	0.342	0.413	0.592

Table 11.2.7. Plaice in Subarea 4 and Subdivision 20. Landings weight-at-age (kg).

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	0.000	0.165	0.201	0.258	0.353	0.456	0.533	0.589	0.396	0.998
1958	0.000	0.198	0.221	0.259	0.337	0.453	0.513	0.615	0.665	0.992
1959	0.000	0.218	0.246	0.293	0.362	0.473	0.592	0.623	0.750	1.000
1960	0.000	0.200	0.236	0.289	0.386	0.485	0.601	0.683	0.724	1.094

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1961	0.000	0.191	0.233	0.302	0.412	0.509	0.604	0.671	0.812	1.071
1962	0.000	0.211	0.248	0.300	0.400	0.541	0.570	0.692	0.777	1.127
1963	0.000	0.253	0.286	0.319	0.399	0.533	0.624	0.667	0.715	1.028
1964	0.000	0.250	0.273	0.312	0.388	0.487	0.628	0.700	0.737	1.005
1965	0.000	0.242	0.282	0.321	0.385	0.471	0.539	0.663	0.726	0.887
1966	0.000	0.232	0.270	0.348	0.436	0.484	0.559	0.624	0.690	0.933
1967	0.000	0.232	0.279	0.322	0.425	0.547	0.597	0.662	0.738	0.978
1968	0.000	0.267	0.298	0.331	0.366	0.517	0.590	0.596	0.686	0.911
1969	0.217	0.294	0.310	0.333	0.359	0.412	0.573	0.655	0.658	0.893
1970	0.315	0.286	0.318	0.356	0.419	0.443	0.499	0.672	0.744	0.892
1971	0.256	0.318	0.356	0.403	0.448	0.514	0.542	0.607	0.699	0.891
1972	0.246	0.296	0.352	0.428	0.493	0.541	0.608	0.646	0.674	0.939
1973	0.272	0.316	0.344	0.405	0.486	0.539	0.605	0.627	0.677	0.842
1974	0.285	0.311	0.354	0.405	0.476	0.554	0.609	0.693	0.707	0.926
1975	0.249	0.300	0.330	0.420	0.495	0.587	0.636	0.703	0.783	1.019
1976	0.265	0.295	0.338	0.375	0.513	0.594	0.641	0.705	0.741	0.980
1977	0.254	0.323	0.353	0.380	0.418	0.556	0.647	0.721	0.715	0.978
1978	0.244	0.315	0.369	0.397	0.438	0.491	0.609	0.687	0.776	0.950
1979	0.235	0.311	0.349	0.388	0.429	0.474	0.550	0.675	0.796	0.960
1980	0.238	0.286	0.344	0.401	0.473	0.545	0.588	0.662	0.772	1.013
1981	0.237	0.274	0.329	0.416	0.505	0.558	0.604	0.642	0.725	1.007
1982	0.279	0.262	0.311	0.424	0.514	0.608	0.664	0.712	0.738	0.984
1983	0.200	0.250	0.300	0.383	0.515	0.604	0.677	0.771	0.815	0.984
1984	0.231	0.263	0.283	0.364	0.480	0.591	0.677	0.726	0.839	1.036
1985	0.245	0.264	0.290	0.335	0.445	0.563	0.667	0.730	0.807	1.021
1986	0.221	0.269	0.303	0.339	0.405	0.473	0.668	0.750	0.856	1.014
1987	0.000	0.249	0.299	0.345	0.378	0.472	0.574	0.728	0.835	0.993
1988	0.000	0.254	0.278	0.341	0.418	0.478	0.590	0.680	0.808	1.017

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1989	0.236	0.280	0.308	0.331	0.385	0.515	0.591	0.668	0.785	0.940
1990	0.271	0.284	0.297	0.315	0.364	0.441	0.586	0.690	0.761	1.010
1991	0.227	0.286	0.292	0.302	0.360	0.452	0.526	0.666	0.743	0.924
1992	0.251	0.263	0.290	0.312	0.330	0.415	0.530	0.607	0.719	0.891
1993	0.249	0.273	0.288	0.319	0.343	0.408	0.512	0.630	0.720	0.856
1994	0.229	0.263	0.284	0.333	0.375	0.417	0.491	0.610	0.731	0.906
1995	0.272	0.277	0.301	0.335	0.375	0.420	0.474	0.593	0.734	0.906
1996	0.240	0.279	0.304	0.346	0.415	0.465	0.490	0.553	0.712	0.858
1997	0.208	0.271	0.313	0.355	0.410	0.474	0.541	0.574	0.616	0.912
1998	0.151	0.260	0.306	0.384	0.452	0.546	0.613	0.673	0.687	0.899
1999	0.245	0.253	0.280	0.347	0.415	0.416	0.538	0.637	0.748	0.804
2000	0.228	0.267	0.283	0.312	0.378	0.461	0.597	0.689	0.752	0.888
2001	0.238	0.267	0.291	0.307	0.360	0.412	0.582	0.701	0.796	0.799
2002	0.237	0.264	0.289	0.311	0.336	0.430	0.477	0.644	0.760	0.904
2003	0.232	0.252	0.285	0.320	0.353	0.389	0.482	0.635	0.763	0.857
2004	0.214	0.246	0.281	0.328	0.391	0.429	0.508	0.560	0.797	0.872
2005	0.272	0.265	0.280	0.330	0.382	0.426	0.465	0.555	0.617	0.910
2006	0.253	0.267	0.282	0.322	0.383	0.389	0.457	0.477	0.531	0.748
2007	0.263	0.268	0.303	0.343	0.364	0.432	0.507	0.486	0.587	0.632
2008	0.249	0.269	0.309	0.341	0.400	0.446	0.531	0.720	0.640	0.638
2009	0.176	0.260	0.308	0.355	0.415	0.481	0.531	0.608	0.668	0.792
2010	0.206	0.265	0.308	0.348	0.418	0.476	0.516	0.625	0.682	0.649
2011	0.235	0.242	0.281	0.341	0.414	0.504	0.604	0.521	0.556	0.804
2012	0.236	0.258	0.305	0.351	0.380	0.436	0.518	0.558	0.558	0.680
2013	0.031	0.242	0.281	0.313	0.364	0.417	0.494	0.600	0.607	0.680
2014	0.207	0.252	0.285	0.318	0.368	0.418	0.479	0.543	0.628	0.650
2015	NA	0.251	0.284	0.321	0.359	0.409	0.473	0.487	0.582	0.600
2016	NA	0.249	0.271	0.296	0.350	0.385	0.450	0.531	0.556	0.684

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2017	0.212	0.247	0.276	0.299	0.357	0.410	0.455	0.543	0.642	0.735
2018	0.167	0.243	0.259	0.287	0.306	0.356	0.400	0.447	0.439	0.589
2019	NA	0.249	0.258	0.295	0.349	0.388	0.431	0.488	0.504	0.601
2020	0.211	0.236	0.264	0.269	0.302	0.333	0.372	0.422	0.451	0.562
2021	0.182	0.245	0.248	0.272	0.296	0.366	0.398	0.431	0.53	0.627
2022	NA	0.226	0.242	0.252	0.303	0.343	0.396	0.462	0.537	0.637
2023	0.193	0.237	0.248	0.259	0.257	0.327	0.374	0.41	0.602	0.623
2024	NA	0.228	0.257	0.257	0.278	0.301	0.394	0.43	0.486	0.572

Table 11.2.8. Plaice in Subarea 4 and Subdivision 20. Discards weight-at-age (kg).

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	0.044	0.104	0.146	0.181	0.206	0.244	0.244	0.231	0	0
1958	0.047	0.096	0.158	0.188	0.2	0.244	0	0	0	0
1959	0.051	0.107	0.155	0.186	0.197	0.231	0	0	0	0
1960	0.045	0.112	0.159	0.188	0.204	0.212	0.244	0	0	0
1961	0.044	0.1	0.16	0.194	0.204	0.22	0.22	0	0	0
1962	0.042	0.098	0.155	0.193	0.213	0.221	0.221	0.231	0	0
1963	0.048	0.105	0.156	0.188	0.205	0.231	0.221	0.231	0	0
1964	0.032	0.114	0.16	0.192	0.204	0.221	0.244	0.231	0	0
1965	0.038	0.072	0.166	0.192	0.212	0.221	0.231	0	0	0
1966	0.038	0.101	0.125	0.194	0.205	0.231	0.231	0.244	0	0
1967	0.036	0.105	0.158	0.169	0.22	0.22	0.244	0.244	0	0
1968	0.06	0.096	0.156	0.191	0.192	0.244	0.22	0	0	0
1969	0.052	0.146	0.162	0.186	0.211	0.212	0	0.231	0	0
1970	0.049	0.114	0.179	0.189	0.196	0	0.22	0.231	0	0
1971	0.057	0.11	0.183	0.2	0.212	0	0	0.231	0	0
1972	0.061	0.147	0.173	0.211	0.211	0.244	0	0	0	0
1973	0.043	0.131	0.179	0.195	0.211	0.244	0	0	0	0

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1974	0.054	0.106	0.173	0.212	0.22	0.231	0.244	0	0	0
1975	0.068	0.136	0.162	0.206	0.221	0.244	0.244	0	0	0
1976	0.085	0.153	0.176	0.195	0.22	0	0.244	0	0	0
1977	0.069	0.16	0.186	0.196	0.198	0.22	0	0	0	0
1978	0.069	0.143	0.197	0.205	0.211	0.213	0.231	0	0	0
1979	0.066	0.158	0.185	0.204	0.22	0.231	0.221	0.244	0	0
1980	0.055	0.149	0.191	0.212	0.231	0	0	0	0	0
1981	0.048	0.135	0.179	0.212	0.22	0	0	0	0	0
1982	0.054	0.126	0.182	0.203	0.231	0.244	0.244	0	0	0
1983	0.051	0.126	0.18	0.205	0.211	0.244	0	0	0	0
1984	0.053	0.127	0.172	0.211	0.205	0	0.244	0	0	0
1985	0.054	0.139	0.177	0.197	0.231	0.244	0	0	0	0
1986	0.049	0.124	0.181	0.196	0.22	0.244	0.244	0	0	0
1987	0.043	0.105	0.166	0.205	0.22	0.231	0	0	0	0
1988	0.043	0.098	0.153	0.185	0.22	0.244	0	0	0	0
1989	0.046	0.102	0.163	0.181	0.196	0	0	0	0	0
1990	0.051	0.111	0.157	0.186	0.212	0.231	0	0	0	0
1991	0.055	0.13	0.161	0.185	0.203	0.221	0.231	0.231	0	0
1992	0.05	0.122	0.167	0.188	0.204	0.212	0.231	0.244	0	0
1993	0.056	0.121	0.171	0.197	0.211	0.231	0.244	0	0	0
1994	0.06	0.14	0.175	0.194	0.213	0.244	0.244	0.221	0	0
1995	0.058	0.141	0.186	0.201	0.22	0.232	0.232	0.244	0	0
1996	0.052	0.122	0.179	0.205	0.221	0.232	0	0	0	0
1997	0.044	0.117	0.178	0.203	0.221	0.244	0	0	0	0
1998	0.047	0.086	0.17	0.199	0.22	0	0.244	0	0	0
1999	0.053	0.097	0.143	0.197	0.22	0	0	0	0	0
2000	0.059	0.11	0.151	0.174	0.244	0	0.203	0	0.219	0
2001	0.068	0.122	0.167	0.178	0.197	0.244	0	0.244	0.219	0

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2002	0.056	0.119	0.17	0.182	0.172	0.208	0.003	0	0.219	0
2003	0.064	0.113	0.174	0.185	0.198	0.204	0.221	0	0.219	0
2004	0.054	0.117	0.164	0.183	0.189	0.192	0.196	0	0.219	0
2005	0.061	0.109	0.17	0.175	0.215	0.205	0.21	0.176	0.219	0
2006	0.06	0.128	0.164	0.193	0.198	0.204	0.212	0.22	0.219	0.296
2007	0.055	0.098	0.177	0.178	0.188	0.199	0.225	0.2	0.219	0.296
2008	0.056	0.116	0.163	0.186	0.187	0.23	0.22	0.191	0.219	0.296
2009	0.06	0.116	0.164	0.199	0.202	0.212	0.21	0.22	0.219	0.296
2010	0.06	0.117	0.159	0.199	0.19	0.198	0.211	0.234	0.001	0.296
2011	0.047	0.104	0.162	0.171	0.192	0.196	0.199	0.211	0	0.296
2012	0.052	0.093	0.142	0.188	0.198	0.206	0.215	0.215	0.219	0.296
2013	0.051	0.081	0.127	0.151	0.17	0.194	0.228	0.346	0.219	0
2014	0.025	0.089	0.132	0.162	0.18	0.212	0.3	0.37	0.255	0
2015	0.026	0.078	0.122	0.149	0.164	0.185	0.173	0.218	0.404	0.291
2016	0.048	0.079	0.124	0.15	0.151	0.179	0.166	0.192	0.251	0.5
2017	0.051	0.08	0.121	0.139	0.161	0.194	0.208	0.206	0.513	0.758
2018	0.058	0.084	0.121	0.137	0.149	0.152	0.159	0.179	0.196	0.163
2019	0.044	0.083	0.118	0.135	0.146	0.148	0.158	0.172	0.182	0.194
2020	0.054	0.079	0.119	0.133	0.146	0.148	0.154	0.164	0.159	0.166
2021	0.054	0.085	0.111	0.131	0.137	0.149	0.168	0.168	0.172	0.296
2022	0.045	0.087	0.109	0.126	0.142	0.154	0.156	0.147	0.184	0.252
2023	0.033	0.066	0.111	0.126	0.13	0.148	0.148	0.148	0.163	0.103
2024	0.044	0.073	0.104	0.125	0.136	0.135	0.145	0.133	0.152	0.238

Table 11.2.9. Plaice in Subarea 4 and Subdivision 20. Catch weight-at-age (kg).

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	0.044	0.11	0.194	0.257	0.349	0.455	0.533	0.589	0.396	0.998
1958	0.047	0.106	0.189	0.256	0.329	0.452	0.513	0.615	0.665	0.992

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1959	0.051	0.12	0.192	0.26	0.345	0.472	0.592	0.623	0.75	1
1960	0.045	0.115	0.204	0.287	0.377	0.48	0.601	0.683	0.724	1.094
1961	0.044	0.101	0.18	0.301	0.402	0.506	0.604	0.671	0.812	1.071
1962	0.042	0.099	0.181	0.273	0.397	0.538	0.57	0.692	0.777	1.127
1963	0.048	0.11	0.175	0.304	0.392	0.531	0.624	0.667	0.715	1.028
1964	0.032	0.126	0.204	0.271	0.379	0.485	0.628	0.7	0.737	1.005
1965	0.038	0.076	0.214	0.313	0.381	0.469	0.539	0.663	0.726	0.887
1966	0.038	0.104	0.148	0.315	0.428	0.483	0.559	0.624	0.69	0.933
1967	0.036	0.111	0.19	0.235	0.422	0.543	0.597	0.662	0.738	0.978
1968	0.06	0.116	0.223	0.275	0.34	0.516	0.59	0.596	0.686	0.911
1969	0.052	0.174	0.272	0.284	0.356	0.408	0.573	0.655	0.658	0.893
1970	0.049	0.131	0.268	0.352	0.394	0.441	0.499	0.672	0.744	0.892
1971	0.057	0.16	0.277	0.388	0.444	0.512	0.542	0.607	0.699	0.891
1972	0.067	0.207	0.29	0.407	0.486	0.54	0.608	0.646	0.674	0.939
1973	0.045	0.205	0.334	0.403	0.478	0.538	0.605	0.627	0.677	0.842
1974	0.056	0.121	0.343	0.404	0.472	0.552	0.609	0.693	0.707	0.926
1975	0.069	0.152	0.205	0.393	0.492	0.585	0.636	0.703	0.783	1.019
1976	0.088	0.181	0.262	0.347	0.509	0.592	0.641	0.705	0.741	0.98
1977	0.071	0.218	0.245	0.318	0.396	0.551	0.647	0.721	0.715	0.978
1978	0.07	0.19	0.315	0.364	0.432	0.486	0.609	0.687	0.776	0.95
1979	0.067	0.19	0.295	0.338	0.426	0.472	0.55	0.675	0.796	0.96
1980	0.056	0.197	0.343	0.399	0.471	0.542	0.588	0.662	0.772	1.013
1981	0.048	0.183	0.327	0.415	0.502	0.556	0.604	0.642	0.725	1.007
1982	0.056	0.152	0.308	0.421	0.512	0.606	0.664	0.712	0.738	0.984
1983	0.052	0.153	0.275	0.379	0.507	0.602	0.677	0.771	0.815	0.984
1984	0.053	0.15	0.265	0.321	0.47	0.588	0.677	0.726	0.839	1.036
1985	0.054	0.169	0.268	0.333	0.444	0.562	0.667	0.73	0.807	1.021
1986	0.049	0.141	0.275	0.311	0.402	0.472	0.668	0.75	0.856	1.014

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1987	0.043	0.113	0.219	0.343	0.375	0.471	0.574	0.728	0.835	0.993
1988	0.043	0.102	0.197	0.276	0.415	0.477	0.59	0.68	0.808	1.017
1989	0.047	0.118	0.215	0.291	0.364	0.512	0.591	0.668	0.785	0.94
1990	0.053	0.13	0.211	0.29	0.36	0.44	0.586	0.69	0.761	1.01
1991	0.056	0.149	0.211	0.273	0.349	0.449	0.526	0.666	0.743	0.924
1992	0.055	0.145	0.226	0.275	0.324	0.41	0.53	0.607	0.719	0.891
1993	0.063	0.16	0.25	0.303	0.34	0.407	0.512	0.63	0.72	0.856
1994	0.064	0.179	0.257	0.331	0.372	0.416	0.491	0.61	0.731	0.906
1995	0.071	0.183	0.283	0.334	0.373	0.419	0.474	0.593	0.734	0.906
1996	0.054	0.14	0.268	0.336	0.413	0.464	0.49	0.553	0.712	0.858
1997	0.045	0.129	0.22	0.353	0.408	0.473	0.541	0.574	0.616	0.912
1998	0.047	0.094	0.208	0.299	0.449	0.544	0.613	0.673	0.687	0.899
1999	0.054	0.103	0.199	0.267	0.413	0.414	0.538	0.637	0.748	0.804
2000	0.063	0.123	0.21	0.274	0.372	0.452	0.565	0.601	0.751	0.888
2001	0.09	0.136	0.197	0.234	0.303	0.41	0.577	0.701	0.795	0.799
2002	0.057	0.131	0.222	0.285	0.326	0.426	0.469	0.644	0.759	0.904
2003	0.066	0.124	0.226	0.284	0.336	0.385	0.419	0.635	0.761	0.857
2004	0.054	0.125	0.22	0.297	0.375	0.421	0.506	0.56	0.794	0.872
2005	0.067	0.117	0.213	0.298	0.352	0.347	0.453	0.554	0.608	0.91
2006	0.06	0.14	0.212	0.293	0.375	0.383	0.428	0.457	0.531	0.551
2007	0.059	0.114	0.223	0.31	0.351	0.375	0.491	0.357	0.333	0.482
2008	0.057	0.124	0.248	0.325	0.389	0.437	0.367	0.469	0.618	0.503
2009	0.061	0.125	0.232	0.326	0.399	0.466	0.518	0.441	0.666	0.339
2010	0.062	0.132	0.226	0.311	0.396	0.442	0.463	0.573	0.563	0.512
2011	0.048	0.115	0.212	0.277	0.369	0.453	0.595	0.445	0.556	0.369
2012	0.052	0.096	0.196	0.294	0.348	0.425	0.509	0.557	0.506	0.444
2013	0.051	0.091	0.179	0.273	0.345	0.409	0.49	0.599	0.606	0.68
2014	0.025	0.094	0.172	0.253	0.317	0.396	0.473	0.542	0.589	0.65

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2015	0.026	0.08	0.162	0.258	0.326	0.394	0.461	0.481	0.581	0.576
2016	0.048	0.084	0.157	0.243	0.299	0.353	0.422	0.465	0.546	0.684
2017	0.051	0.086	0.159	0.219	0.315	0.386	0.438	0.533	0.642	0.735
2018	0.058	0.089	0.148	0.206	0.254	0.327	0.38	0.439	0.437	0.62
2019	0.044	0.092	0.155	0.218	0.271	0.312	0.372	0.435	0.494	0.598
2020	0.054	0.083	0.152	0.194	0.249	0.285	0.312	0.387	0.399	0.56
2021	0.054	0.094	0.127	0.194	0.24	0.298	0.373	0.389	0.483	0.626
2022	0.045	0.092	0.135	0.172	0.246	0.29	0.334	0.361	0.51	0.626
2023	0.033	0.069	0.133	0.17	0.196	0.263	0.323	0.335	0.494	0.594
2024	0.044	0.076	0.132	0.18	0.208	0.219	0.273	0.261	0.35	0.564

Table 11.2.10. Plaice in Subarea 4 and Subdivision 20. Natural mortality-at-age and maturity-at-age.

Age	1	2	3	4	5	6	7	8	9	10+
Natural mortality	0.495	0.394	0.343	0.311	0.292	0.278	0.268	0.260	0.252	0.246
Maturity	0	0.5	0.5	1	1	1	1	1	1	1

Table 11.2.11. Plaice in Subarea 4 and Subdivision 20. Survey tuning indices.

BTS–Isis		Age							
Year	1	2	3	4	5	6	7	8	
1985	137	173.9	36.1	11	1.27	0.973	0.336	0.155	
1986	667	131.7	50.2	9.21	3.78	0.4	0.418	0.147	
1987	226	764.2	33.8	4.88	1.84	0.607	0.252	0.134	
1988	680	147	182.3	9.99	2.81	0.814	0.458	0.036	
1989	468	319.3	314.7	47.3	5.85	0.833	0.311	0.661	
1990	185	146.1	79.3	26.35	5.47	0.758	0.189	0.383	
1991	291	159.4	34	13.57	4.31	5.659	0.239	0.204	
1992	361	174.5	29.3	5.96	3.75	2.871	1.186	0.346	
1993	189	283.4	62.8	14.27	1.13	1.13	0.584	0.464	
1994	193	77.1	34.5	10.59	2.67	0.6	0.8	0.895	
1995	266	40.6	13.2	7.53	1.11	0.806	0.33	1.051	

BTS-IBTS Q3	1	2	3	4	5	6	7	8	9	10+
1996	26767	21493	6796	3410	3710	2488	998	660	363	548
1997	92288	36328	10637	3907	3129	2250	1143	992	393	454
1998	51215	74216	14337	6407	2472	1841	1119	884	487	501
1999	36895	25514	42807	6948	3227	1487	713	663	481	388
2000	44566	18578	15815	19175	2622	1312	692	610	234	413
2001	31390	23954	12471	8392	9540	1873	813	672	495	1009
2002	166583	26293	15546	9001	4899	3654	958	846	419	824
2003	36083	49923	13012	8092	3840	2511	2480	661	564	1144
2004	51815	19760	33331	7198	4265	2671	1817	2044	483	1371
2005	45282	42114	10977	18151	3091	3408	2032	942	1627	1731
2006	43724	26918	24336	7258	9006	2750	2789	1166	684	2416
2007	91108	38845	25507	21754	6088	8234	2250	2642	867	4118
2008	97547	81299	32036	21449	13954	4399	5457	1890	1832	3865
2009	64038	42250	48980	18233	11046	7300	3156	4756	1266	5034
2010	86639	40460	33659	30527	9952	6062	6569	2743	3059	6406
2011	131444	67102	42359	26519	17970	6448	4538	5261	1464	8174
2012	51599	91089	69839	35192	19466	12787	5919	4560	4641	8146
2013	65923	62823	73142	42430	18965	10976	8482	3710	2428	9479
2014	113601	78187	50562	41995	19163	8690	5890	4178	2559	8162
2015	35787	73963	61348	38900	28661	13883	6792	4451	3633	10640
2016	50290	43454	59295	39138	19713	13901	8549	4804	3419	10427
2017	93982	55426	28635	31965	18284	10271	6766	4666	2330	7878
2018	50179	70114	34643	19581	18336	10234	6885	4301	3392	8450
2019	189411	57089	41117	16131	9687	7731	5435	4157	2436	5886
2020	93084	97173	37398	26319	10129	6227	5156	4178	2503	5788
2021	89111	66867	68151	27644	16159	6840	4247	4084	2753	6538
2022	172142	46285	35368	29767	11066	8571	4966	3169	2830	7631
2023	89510	127025	47688	30699	27543	13246	8204	4899	2846	8709
2024	118500	85595	100441	33105	19088	18476	8897	6170	2661	7193

SNS1							SNS2						
Age							Age						
Year	1	2	3	4	5	6	Year	1	2	3	4	5	6
1970	9311	9732	3273	770	170	37.5	2000	22855	2493	891	983	17	2.0
1971	13538	28164	1415	101	50	23.6	2001	11511	2898	370	176	691	105.8
1972	13207	10780	4478	89	84	0.0	2002	30809	1103	265	65	69	30.7
1973	65643	5133	1578	461	15	5.7	2003	NA	NA	NA	NA	NA	NA
1974	15366	16509	1129	160	82	7.0	2004	18202	1350	1081	51	27	29.7
1975	11628	8168	9556	65	15	0.0	2005	10118	1819	142	366	8	19.0
1976	8537	2403	868	236	0	2.3	2006	12164	1571	385	52	54	0.0
1977	18537	3424	1737	590	213	0.0	2007	14175	2134	140	52	0	7.4
1978	14012	12678	345	135	45	13.6	2008	14706	2700	464	179	34	6.7
1979	21495	9829	1575	161	17	42.2	2009	14860	2019	492	38	20	0.0
1980	59174	12882	491	180	24	7.8	2010	11947	1812	529	55	10	0.0
1981	24756	18785	834	38	32	4.7	2011	18349	1143	308	75	60	28.0
1982	69993	8642	1261	88	8	8.7	2012	5893	2929	682	82	30	15.0
1983	33974	13909	249	71	6	1.3	2013	15395	3021	1638	428	89	31.1
1984	44965	10413	2467	42	0	0.0	2014	17313	2258	514	458	58	16.4
1985	28101	13848	1598	328	17	1.5	2015	16727	5040	1882	478	200	97.5
1986	93552	7580	1152	145	30	6.6	2016	10385	2434	1086	522	223	131.7
1987	33402	32991	1227	200	30	16.7	2017	15936	1716	1212	534	144	70.6
1988	36609	14421	13153	1350	88	12.1	2018	9465	5250	993	533	489	88
1989	34276	17810	4373	7126	289	113.6	2019	28309	1886	1533	338	196	62
1990	25037	7496	3160	816	422	48.8	2020	11393	3931	283	607	118	5
1991	57221	11247	1518	1077	128	74.4	2021	6014	1290	1135	249	230	143
1992	46798	13842	2268	613	176	52.0	2022	23731.8	1279.1	769.5	586	12.5	119.5
1993	22098	9686	1006	98	60	58.8	2023	10664.9	2171.1	721.2	224.3	330.6	85.5
1994	19188	4977	856	76	23	2.7	2024	32828	1812	1197	201	177	164
1995	24767	2796	381	97	38	0.0							
1996	23015	10268	1185	45	47	0.0							

SNS1							SNS2						
Age							Age						
Year	1	2	3	4	5	6	Year	1	2	3	4	5	6
1997	95901	4473	497	32	0	13.3							
1998	33666	30242	5014	50	10	0.0							
1999	32951	10272	13783	1058	17	0.0							

IBTS Q1	1	2	3	4	5	6	7	8+
2007	1862	4468	5278	6295	2241	1169	711	579
2008	1900	10141	7992	3684	2691	752	786	536
2009	2654	7127	13967	4567	2329	970	545	785
2010	891	4618	7733	7401	3539	1271	914	878
2011	1015	5466	6822	7207	5751	1965	1108	1292
2012	1731	11983	17223	8034	5608	3804	1619	2370
2013	1265	4937	10459	7170	3460	2019	1133	992
2014	2197	6846	9856	9371	5300	1956	1253	1419
2015	536	7414	10340	8819	6393	3037	1386	1616
2016	1190	3542	8484	7364	5245	2592	1573	2175
2017	1933	6153	4715	8038	5493	3488	1856	2635
2018	652	5361	6642	2690	3514	2104	1574	2893
2019	3670	3568	6605	3324	1983	1509	1433	2686
2020	1640	7264	4106	3527	1838	1352	1174	1890
2021	1044	6103	10523	4608	2859	1431	872	2199
2022	8753	6595	7067	10056	3018	1900	1228	2471
2023	2404	15089	10895	6807	7042	2655	1830	3060
2024	2618	7924	13971	6734	4253	3135	1614	2896

Table 11.3.1. Plaice in Subarea 4 and Subdivision 20. Stock summary table. Recruits are in thousands, weights in tonnes. Landings and discards since 2015 include part of the catches in Division 7.d in the first quarter. From 2016, discards include BMS landings from EU and UK fleets.

Year	Recruits (age 1)	SSB	Catch	Landings	Discards	Fbar2-6	Fbar hc2-6	Fbar dis2-3	Y/SSB
1957	1774476	813184	78442	70563	7881	0.109	0.090	0.046	0.09
1958	2579391	810578	88162	73354	14837	0.127	0.093	0.079	0.09
1959	3061077	803560	109180	79301	29863	0.145	0.093	0.096	0.10
1960	2709438	828816	117346	87541	29793	0.148	0.110	0.090	0.11
1961	2914968	833220	118369	85987	32489	0.148	0.100	0.114	0.10
1962	2101237	996347	125273	87474	37902	0.160	0.101	0.123	0.09
1963	2470739	898902	148439	107120	41257	0.182	0.115	0.151	0.12
1964	6258287	885396	147597	110542	37030	0.188	0.120	0.127	0.12
1965	2420223	782049	140210	97140	43081	0.169	0.119	0.117	0.12
1966	1900465	836660	166490	101833	64719	0.181	0.111	0.150	0.12
1967	1276416	960736	163517	108815	54548	0.183	0.103	0.142	0.11
1968	1313094	903237	139497	111534	27987	0.165	0.103	0.107	0.12
1969	1884167	802556	142811	121650	21169	0.177	0.129	0.084	0.15
1970	1740217	739080	159923	130342	29639	0.226	0.168	0.128	0.18
1971	1142069	678232	136964	113946	22994	0.206	0.156	0.113	0.17
1972	1009628	662033	142442	122843	19632	0.222	0.179	0.094	0.19
1973	3355622	543995	143711	130432	13353	0.250	0.220	0.071	0.24
1974	2590896	536249	157427	112539	44945	0.270	0.222	0.117	0.21
1975	1835726	542645	195349	108535	86699	0.307	0.201	0.241	0.20
1976	1505080	560470	166997	113673	53245	0.255	0.183	0.157	0.20
1977	2158422	561589	176623	119189	57500	0.290	0.188	0.185	0.21
1978	1929238	535428	159539	113977	45658	0.279	0.211	0.139	0.21
1979	1868078	511118	213145	145349	67935	0.383	0.278	0.202	0.28
1980	2705117	497701	171853	140763	31081	0.330	0.290	0.097	0.28
1981	2128664	467488	174141	141235	33030	0.331	0.290	0.097	0.30
1982	4605923	450284	205329	156155	49125	0.359	0.304	0.131	0.35
1983	3090475	544887	220538	145781	74482	0.347	0.280	0.157	0.27

Year	Recruits (age 1)	SSB	Catch	Landings	Discards	Fbar2-6	Fbar hc2-6	Fbar dis2-3	Y/SSB
1984	2937933	561816	236422	165767	70818	0.359	0.270	0.158	0.30
1985	3696933	599565	232241	171841	60547	0.341	0.282	0.144	0.29
1986	9483597	626335	308262	178881	129952	0.419	0.318	0.202	0.29
1987	4273482	750810	358749	168739	190533	0.454	0.327	0.310	0.22
1988	4249043	652264	324772	168557	156421	0.436	0.272	0.310	0.26
1989	2895315	708410	286704	178892	107793	0.379	0.244	0.260	0.25
1990	2557715	637808	240832	169455	71224	0.357	0.246	0.233	0.27
1991	2436142	589104	237986	157283	80932	0.393	0.264	0.253	0.27
1992	2127479	504042	193692	136728	57048	0.384	0.267	0.217	0.27
1993	1361437	455352	163524	128503	35018	0.353	0.279	0.153	0.28
1994	1174136	400112	145745	121925	23785	0.352	0.295	0.135	0.30
1995	2452752	382931	131248	109349	21827	0.365	0.311	0.130	0.29
1996	2760384	359320	143395	91387	52048	0.386	0.296	0.206	0.25
1997	5200563	387807	192969	92950	100149	0.437	0.296	0.348	0.24
1998	2124541	436684	183655	79811	103750	0.382	0.213	0.315	0.18
1999	2033982	409337	160881	89723	70978	0.353	0.206	0.252	0.22
2000	2509233	434519	135015	90752	44312	0.279	0.181	0.190	0.21
2001	1759141	486097	193368	92906	100313	0.373	0.171	0.311	0.19
2002	4933201	427957	134661	79496	55100	0.309	0.200	0.225	0.19
2003	1838648	503260	154063	74807	79274	0.293	0.181	0.217	0.15
2004	3498758	491858	127765	70500	57476	0.219	0.134	0.176	0.14
2005	2693434	555613	118819	62729	56264	0.202	0.105	0.174	0.11
2006	3162746	620290	132008	67356	64582	0.175	0.104	0.152	0.11
2007	4580133	643632	101787	58640	43070	0.140	0.078	0.122	0.09
2008	4062018	822726	105504	58113	47173	0.110	0.072	0.084	0.07
2009	3966141	960800	110990	62482	48541	0.100	0.062	0.080	0.07
2010	4412619	1114420	117481	70460	46968	0.099	0.060	0.075	0.06
2011	6109642	1098094	123596	76798	46802	0.102	0.056	0.076	0.07
2012	3754792	1141166	143123	82110	61093	0.108	0.059	0.089	0.07

Year	Recruits (age 1)	SSB	Catch	Landings	Discards	Fbar2-6	Fbar hc2-6	Fbar dis2-3	Y/SSB
2013	4432892	1183090	126602	86394	40182	0.102	0.062	0.080	0.07
2014	5403702	1256452	134504	81083	53325	0.110	0.054	0.092	0.06
2015	2717190	1111811	134882	85390	49448	0.116	0.061	0.100	0.08
2016	3055712	1063538	137274	92785	44478	0.133	0.067	0.111	0.09
2017	4642056	991148	114292	74941	39335	0.125	0.061	0.103	0.08
2018	2833913	825683	106398	57841	48402	0.134	0.058	0.114	0.07
2019	6312984	735901	86275	48780	37546	0.110	0.048	0.088	0.07
2020	3941222	756775	79134	40919	38293	0.096	0.041	0.078	0.05
2021	3524830	804449	74038	35952	38111	0.080	0.031	0.071	0.04
2022	8866432	816536	54452	25601	28866	0.059	0.023	0.052	0.03
2023	5118105	904330	56794	23154	33639	0.054	0.018	0.048	0.03
2024	7407716	1038434	44022	19499	24559	0.040	0.013	0.035	0.02
2025	4477139*	1230426							

* Recruitment in 2025 is the geometric mean of resampled recruitment estimates from 2015 to 2024.

Table 11.3.2. Plaice in Subarea 4 and Subdivision 20. Estimated parameters from SAM model.

	par	sd(par)	exp(par)	Low	High
logFpar_0	-8.774	0.157	0.000	0.000	0.000
logFpar_1	-8.653	0.132	0.000	0.000	0.000
logFpar_2	-9.381	0.132	0.000	0.000	0.000
logFpar_3	-10.139	0.133	0.000	0.000	0.000
logFpar_4	-10.869	0.136	0.000	0.000	0.000
logFpar_5	-11.138	0.141	0.000	0.000	0.000
logFpar_6	-11.389	0.197	0.000	0.000	0.000
logFpar_7	-11.123	0.207	0.000	0.000	0.000
logFpar_8	-3.646	0.076	0.026	0.022	0.030
logFpar_9	-3.411	0.069	0.033	0.029	0.038
logFpar_10	-3.254	0.075	0.039	0.033	0.045
logFpar_11	-3.220	0.083	0.040	0.034	0.047
logFpar_12	-3.327	0.101	0.036	0.029	0.044

	par	sd(par)	exp(par)	Low	High
logFpar_13	-7.827	0.124	0.000	0.000	0.001
logFpar_14	-5.905	0.089	0.003	0.002	0.003
logFpar_15	-5.049	0.090	0.006	0.005	0.008
logFpar_16	-5.055	0.116	0.006	0.005	0.008
logFpar_17	-5.541	0.132	0.004	0.003	0.005
logFpar_18	-4.063	0.087	0.017	0.014	0.020
logFpar_19	-4.429	0.100	0.012	0.010	0.015
logFpar_20	-5.568	0.146	0.004	0.003	0.005
logFpar_21	-6.963	0.171	0.001	0.001	0.001
logFpar_22	-7.940	0.179	0.000	0.000	0.001
logFpar_23	-8.576	0.197	0.000	0.000	0.000
logFpar_24	-5.251	0.104	0.005	0.004	0.006
logFpar_25	-6.543	0.117	0.001	0.001	0.002
logFpar_26	-7.252	0.139	0.001	0.001	0.001
logFpar_27	-7.854	0.190	0.000	0.000	0.001
logFpar_28	-8.450	0.245	0.000	0.000	0.000
logFpar_29	-8.709	0.255	0.000	0.000	0.000
logSdLogFsta_0	-1.907	0.191	0.149	0.101	0.218
logSdLogFsta_1	-1.852	0.126	0.157	0.122	0.202
logSdLogFsta_2	-1.853	0.095	0.157	0.130	0.190
logSdLogFsta_3	-1.851	0.125	0.157	0.122	0.202
logSdLogN_0	-0.691	0.097	0.501	0.412	0.609
logSdLogN_1	-2.402	0.097	0.091	0.075	0.110
logSdLogObs_0	-0.871	0.114	0.418	0.333	0.526
logSdLogObs_1	-2.241	0.089	0.106	0.089	0.127
logSdLogObs_2	-1.292	0.071	0.275	0.238	0.317
logSdLogObs_3	-0.683	0.224	0.505	0.323	0.790
logSdLogObs_4	-0.869	0.100	0.420	0.343	0.513
logSdLogObs_5	-0.516	0.160	0.597	0.434	0.821

	par	sd(par)	exp(par)	Low	High
logSdLogObs_6	-1.314	0.152	0.269	0.198	0.364
logSdLogObs_7	-1.729	0.074	0.177	0.153	0.206
logSdLogObs_8	-1.459	0.168	0.232	0.166	0.325
logSdLogObs_9	-1.329	0.172	0.265	0.188	0.373
logSdLogObs_10	-0.877	0.178	0.416	0.291	0.594
logSdLogObs_11	-0.816	0.169	0.442	0.316	0.619
logSdLogObs_12	-1.375	0.089	0.253	0.212	0.302
logSdLogObs_13	-1.213	0.135	0.297	0.227	0.390
logSdLogObs_14	-0.793	0.139	0.453	0.343	0.598
logSdLogObs_15	-0.634	0.137	0.530	0.403	0.697
logSdLogObs_16	-0.241	0.128	0.786	0.608	1.016
logSdLogObs_17	-0.087	0.083	0.917	0.777	1.082
logSdLogObs_18	-0.894	0.144	0.409	0.307	0.546
logSdLogObs_19	-0.742	0.139	0.476	0.361	0.628
logSdLogObs_20	-0.542	0.139	0.582	0.441	0.768
logSdLogObs_21	-0.169	0.136	0.844	0.643	1.108
logSdLogObs_22	0.097	0.115	1.102	0.875	1.388
transfIRARdist_0	0.389	0.255	1.476	0.887	2.457
transfIRARdist_1	0.152	0.215	1.164	0.757	1.789
transfIRARdist_2	-0.333	0.236	0.717	0.447	1.150
itrans_rho_0	1.723	0.182	5.603	3.892	8.064

Table 11.3.3. Plaice in Subarea 4 and Subdivision 20. Estimated Harvest (F) at age.

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	0.027	0.088	0.110	0.123	0.124	0.103	0.095	0.088	0.085	0.085
1958	0.031	0.102	0.129	0.143	0.143	0.120	0.109	0.100	0.095	0.095
1959	0.035	0.118	0.148	0.162	0.160	0.136	0.121	0.111	0.105	0.105
1960	0.036	0.121	0.151	0.164	0.164	0.142	0.125	0.113	0.108	0.108
1961	0.036	0.121	0.151	0.162	0.163	0.142	0.124	0.112	0.106	0.106

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1962	0.038	0.131	0.164	0.175	0.175	0.155	0.133	0.117	0.110	0.110
1963	0.042	0.147	0.186	0.199	0.199	0.179	0.153	0.134	0.125	0.125
1964	0.043	0.150	0.191	0.205	0.205	0.185	0.156	0.136	0.128	0.128
1965	0.040	0.136	0.172	0.184	0.184	0.169	0.145	0.127	0.120	0.120
1966	0.044	0.148	0.186	0.197	0.196	0.179	0.153	0.134	0.126	0.126
1967	0.047	0.152	0.188	0.199	0.198	0.181	0.154	0.133	0.125	0.125
1968	0.046	0.140	0.171	0.178	0.175	0.162	0.140	0.122	0.115	0.115
1969	0.051	0.151	0.181	0.190	0.187	0.177	0.154	0.135	0.128	0.128
1970	0.065	0.193	0.231	0.244	0.240	0.224	0.195	0.167	0.156	0.156
1971	0.061	0.173	0.206	0.221	0.221	0.207	0.184	0.162	0.153	0.153
1972	0.067	0.186	0.221	0.239	0.241	0.222	0.196	0.175	0.164	0.164
1973	0.077	0.210	0.251	0.273	0.274	0.244	0.214	0.191	0.178	0.178
1974	0.087	0.233	0.274	0.294	0.292	0.256	0.226	0.204	0.189	0.189
1975	0.102	0.273	0.315	0.332	0.328	0.287	0.254	0.229	0.211	0.211
1976	0.091	0.235	0.266	0.274	0.267	0.235	0.211	0.191	0.178	0.178
1977	0.103	0.268	0.305	0.314	0.304	0.261	0.231	0.205	0.188	0.188
1978	0.099	0.256	0.295	0.304	0.290	0.252	0.223	0.196	0.178	0.178
1979	0.127	0.345	0.408	0.423	0.398	0.343	0.298	0.258	0.230	0.230
1980	0.109	0.295	0.356	0.370	0.340	0.290	0.251	0.218	0.196	0.196
1981	0.106	0.291	0.354	0.374	0.344	0.292	0.254	0.221	0.200	0.200
1982	0.112	0.314	0.381	0.408	0.374	0.316	0.273	0.239	0.216	0.216
1983	0.108	0.304	0.361	0.395	0.365	0.310	0.267	0.234	0.212	0.212
1984	0.108	0.311	0.364	0.404	0.384	0.330	0.283	0.247	0.224	0.224
1985	0.101	0.292	0.338	0.380	0.371	0.326	0.280	0.243	0.221	0.221
1986	0.117	0.355	0.407	0.456	0.465	0.413	0.350	0.300	0.269	0.269
1987	0.120	0.378	0.436	0.487	0.511	0.458	0.390	0.328	0.292	0.292
1988	0.112	0.361	0.417	0.462	0.493	0.448	0.384	0.326	0.293	0.293
1989	0.096	0.313	0.365	0.400	0.428	0.389	0.335	0.285	0.257	0.257

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1990	0.087	0.287	0.341	0.376	0.407	0.371	0.319	0.268	0.241	0.241
1991	0.091	0.312	0.376	0.415	0.448	0.413	0.358	0.299	0.266	0.266
1992	0.084	0.293	0.363	0.408	0.445	0.413	0.363	0.303	0.269	0.269
1993	0.075	0.265	0.333	0.378	0.408	0.379	0.338	0.281	0.246	0.246
1994	0.074	0.270	0.341	0.382	0.397	0.369	0.332	0.269	0.231	0.231
1995	0.076	0.290	0.369	0.405	0.400	0.361	0.319	0.250	0.208	0.208
1996	0.083	0.331	0.413	0.435	0.402	0.348	0.298	0.232	0.190	0.190
1997	0.094	0.390	0.490	0.504	0.440	0.362	0.289	0.219	0.175	0.175
1998	0.082	0.336	0.429	0.460	0.384	0.303	0.227	0.167	0.131	0.131
1999	0.073	0.295	0.372	0.428	0.373	0.299	0.214	0.152	0.114	0.114
2000	0.060	0.239	0.289	0.335	0.298	0.236	0.165	0.113	0.081	0.081
2001	0.083	0.346	0.406	0.441	0.384	0.286	0.195	0.126	0.085	0.085
2002	0.072	0.290	0.326	0.350	0.321	0.256	0.182	0.114	0.073	0.073
2003	0.074	0.292	0.313	0.326	0.296	0.239	0.169	0.103	0.063	0.063
2004	0.064	0.241	0.244	0.239	0.207	0.163	0.112	0.070	0.042	0.042
2005	0.060	0.224	0.224	0.217	0.192	0.151	0.104	0.068	0.041	0.041
2006	0.056	0.203	0.200	0.187	0.162	0.126	0.088	0.061	0.038	0.038
2007	0.046	0.159	0.156	0.148	0.131	0.104	0.074	0.053	0.034	0.034
2008	0.037	0.123	0.120	0.117	0.105	0.085	0.063	0.046	0.029	0.029
2009	0.034	0.109	0.108	0.108	0.096	0.078	0.057	0.044	0.029	0.029
2010	0.033	0.103	0.105	0.109	0.098	0.080	0.058	0.045	0.029	0.029
2011	0.032	0.097	0.106	0.115	0.106	0.086	0.062	0.049	0.032	0.032
2012	0.034	0.104	0.115	0.123	0.111	0.089	0.062	0.048	0.031	0.031
2013	0.032	0.095	0.108	0.116	0.107	0.087	0.062	0.047	0.029	0.029
2014	0.034	0.101	0.116	0.124	0.114	0.091	0.066	0.050	0.030	0.030
2015	0.036	0.108	0.124	0.132	0.121	0.097	0.071	0.054	0.031	0.031
2016	0.038	0.117	0.139	0.153	0.142	0.114	0.084	0.063	0.035	0.035
2017	0.036	0.110	0.133	0.146	0.133	0.104	0.077	0.057	0.032	0.032

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2018	0.038	0.116	0.144	0.158	0.141	0.110	0.083	0.062	0.034	0.034
2019	0.032	0.094	0.118	0.131	0.116	0.092	0.073	0.056	0.031	0.031
2020	0.028	0.080	0.102	0.115	0.103	0.082	0.070	0.055	0.031	0.031
2021	0.025	0.070	0.086	0.095	0.085	0.065	0.056	0.045	0.025	0.025
2022	0.020	0.054	0.063	0.068	0.061	0.048	0.042	0.034	0.019	0.019
2023	0.019	0.050	0.056	0.061	0.056	0.044	0.039	0.032	0.018	0.018
2024	0.014	0.037	0.041	0.045	0.042	0.034	0.031	0.026	0.015	0.015

Table 11.3.4. Plaice in Subarea 4 and Subdivision 20. Estimated stock numbers (thousands).

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1957	1774476	732593	882539	506999	341314	126448	153465	124335	76506	165452
1958	2579391	1047240	457181	575044	328692	217533	92015	106401	87032	173217
1959	3061077	1530185	652867	286698	364524	211654	143855	65206	74341	184119
1960	2709438	1791334	905950	404383	177763	232755	139289	97414	46064	180536
1961	2914968	1585049	1046325	549169	255927	115198	149125	94223	67136	159258
1962	2101237	1742784	954854	629883	345382	160820	78589	100225	64258	158261
1963	2470739	1224588	1024197	581021	385622	215979	103569	53995	69214	158619
1964	6258287	1400552	703500	588443	358390	244074	131598	66805	36732	160721
1965	2420223	4100899	769084	404851	330604	216192	154190	86024	44733	139862
1966	1900465	1420803	2698309	440053	247710	204448	133494	101845	58694	128690
1967	1276416	1141875	811642	1696232	256493	151645	133283	84199	68742	127881
1968	1313094	761233	660957	469107	1076689	145197	93124	90861	55361	137290
1969	1884167	740157	461285	394865	283751	717214	90971	60420	63585	135860
1970	1740217	1100211	425519	283935	243311	185750	457390	59681	39897	136210
1971	1142069	1029778	588475	228623	161742	146036	114744	274193	40146	120018
1972	1009628	655368	600227	333854	133635	97345	92345	73036	174255	107906
1973	3355622	546897	359843	353332	196977	78425	57975	58810	46882	183789
1974	2590896	1977146	294481	198117	192829	108126	47287	36510	37386	149076

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
1975	1835726	1487572	1114641	153741	107580	103194	64096	29657	23157	118885
1976	1505080	1002777	772266	592869	78525	56331	56260	38615	18372	87963
1977	2158422	810525	545563	431374	351408	47503	33537	33614	24691	67711
1978	1929238	1184597	410234	288895	237640	183965	29990	20677	20555	58625
1979	1868078	1070556	617089	221351	161738	141866	105918	18755	13308	51621
1980	2705117	969274	492828	292027	106710	83042	74469	58680	11302	40270
1981	2128664	1473775	491485	227468	145434	58805	48259	44553	36117	33775
1982	4605923	1151682	745436	244586	110377	75022	35194	28792	27483	45367
1983	3090475	2561654	570986	345570	117555	56356	41464	21105	17520	46150
1984	2937933	1661392	1301446	293810	156977	62108	31795	25014	13147	40272
1985	3696933	1544624	805054	659598	148670	74565	34259	18780	15268	33883
1986	9483597	2069988	775768	427796	316193	83786	39856	19851	11629	31614
1987	4273482	5396787	984186	369471	212914	142526	43393	21703	11403	26525
1988	4249043	2340475	2509464	464620	172745	96649	69074	21908	12499	23229
1989	2895315	2337295	1111254	1173307	221404	78323	46662	36676	12198	21718
1990	2557715	1562947	1142501	550670	564916	108911	39446	25762	21401	20862
1991	2436142	1450676	813104	575143	270823	265734	57146	22558	15415	26576
1992	2127479	1300657	700863	401641	281700	128937	119780	30803	13230	26103
1993	1361437	1152897	642598	344937	202809	130590	61229	59395	17602	23891
1994	1174136	735575	585377	325029	176339	101398	66107	32668	32824	24966
1995	2452752	625426	373817	293525	162739	91585	54965	33882	18448	33197
1996	2760384	1415361	329286	185777	141646	81439	49720	29435	18974	30964
1997	5200563	1695421	698285	164727	98331	71264	43677	28808	17579	30222
1998	2124541	3058000	755502	299253	85907	46807	35322	25185	17495	29507
1999	2033982	1124542	1589866	312239	125944	50198	25757	20787	16233	30203
2000	2509233	1122731	569520	856724	127098	61604	26456	16381	12989	30518
2001	1759141	1424055	658564	362323	466607	65800	34196	17360	11353	30352
2002	4933201	983620	631527	308129	186236	221847	38932	22313	11878	28105

Year	Age									
	1	2	3	4	5	6	7	8	9	10+
2003	1838648	2672199	489667	322626	169995	102476	136470	23574	15110	26863
2004	3498758	997913	1325312	256396	177441	101497	59387	85918	15845	28901
2005	2693434	2051043	487837	714651	146280	125278	68783	40554	62563	32652
2006	3162746	1548132	1129206	295841	423248	89831	87875	48198	29670	73721
2007	4580133	1822276	872907	658689	200029	292791	63145	65986	36587	80939
2008	4062018	2818705	996409	545050	423374	125980	227911	46448	49087	93405
2009	3966141	2289388	1640014	594677	346420	283717	86995	165139	33779	123657
2010	4412619	2256715	1373910	1035880	374576	218556	218218	64525	124079	122683
2011	6109642	2613831	1357519	943339	689963	246315	147444	165365	47117	207022
2012	3754792	3736827	1860784	906945	612209	442183	166237	108640	123584	186382
2013	4432892	2368350	2196380	1156382	553366	388156	283987	117403	79455	212863
2014	5403702	2715085	1593381	1337622	729479	336271	245194	187616	86458	204497
2015	2717190	3116175	1752517	1056198	801070	452157	225565	171752	132223	209017
2016	3055712	1621949	1713936	1100694	681145	474662	292740	155891	124059	233906
2017	4642056	1837792	948109	1010491	654244	406193	278192	192898	106477	236015
2018	2833913	2796450	1084080	570981	589961	385143	255554	182091	137136	233260
2019	6312984	1699463	1609127	602369	336817	340981	238975	173223	125223	249152
2020	3941222	3561354	1050020	978686	367022	211431	232158	167111	124296	259824
2021	3524830	2258281	2248905	714806	613948	247766	142998	159598	121647	265507
2022	8866432	2049043	1373575	1405448	454871	385332	185894	110020	114383	273352
2023	5118105	5332726	1342270	909245	990369	351990	285173	145780	85989	276965
2024	7407716	2960664	3179930	932325	620256	668121	264957	212945	113026	267513

Table 11.5.1. Plaice in Subarea 4 and Subdivision 20. Annual catch scenarios. All weights are in tonnes.

Basis	ple.27.420 plaice stock			Plaice in Subarea 4 and Subdivision 20										
	Total catch (2026)	Projected landings (2026)	Projected discards* (2026)	F _{total} (ages 2–6)** (2026)	F _{projected} landings (ages 2–6) (2026)	F _{projected} discards (ages 2–3) (2026)	SSB (2027)	% SSB change***	% advice change^	Total catch^^ (2026)	Projected landing (2026)	Projected discards* (2026)	% change in projected catches ^^^	% TAC change#
ICES advice basis														
MSY approach: F _{MSY}	164129	85387	78742	0.123	0.043	0.080	1348073	2.6	-7.3	163951	85343	78607	-7.2	-7.2
Other scenarios														
F _{MSY upper}	186721	97126	89595	0.141	0.049	0.092	1327669	1.10	5.5	186543	97082	89460	5.6	5.6
F _{MSY lower}	127001	66047	60954	0.094	0.033	0.061	1381109	5.2	-28	126823	66003	60819	-28	-28
F = 0 ^{##}	0	0	0	0	0	0	1495494	13.9	-100	0	0	0	-100	-100
F _{pa}	186721	97126	89595	0.141	0.049	0.092	1327669	1.10	5.5	186543	97082	89460	5.6	5.6
SSB (2027) = B _{jim}	1285312	677246	608066	1.75	0.61	1.14	391695	-70	630	1285134	677202	607931	630	630
SSB (2027) = B _{pa} = MSY B _{trigger}	1094972	574576	520396	1.28	0.44	0.84	544290	-59	520	1094794	574532	520261	520	520
F = F ₂₀₂₅	54965	28565	26400	0.04	0.014	0.026	1445596	10.1	-69	54787	28521	26265	-69	-69

* Including below minimum size (BMS) landings, assuming recent discard rates (i.e., 2022–2024).

** F_{projected} landings and F_{projected} discards do not add up to the F_{total}, as they are calculated using different ages.

*** SSB 2027 relative to SSB 2026.

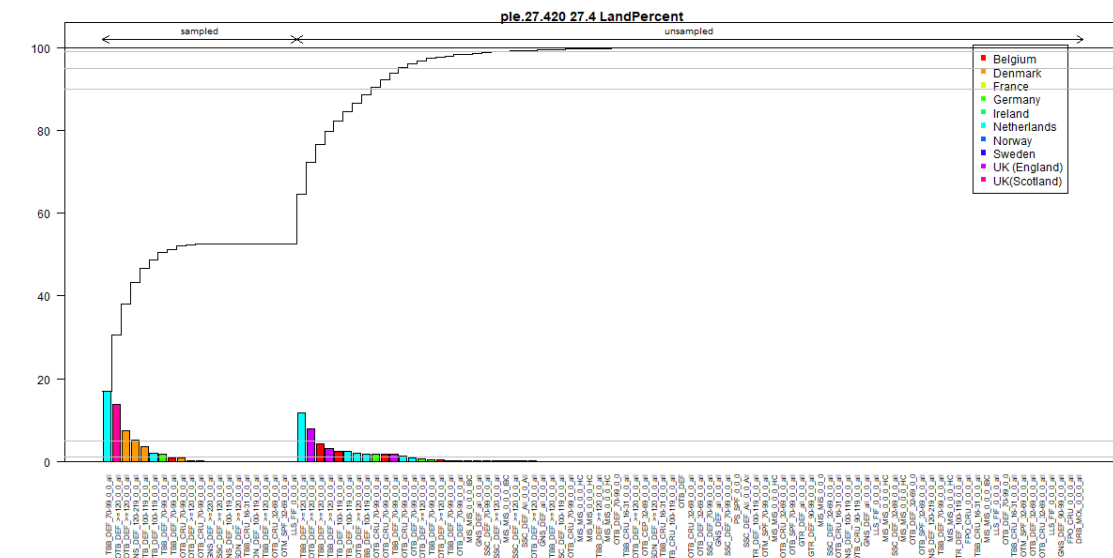
^ Projected catch of plaice stock in 2026 relative to the advised value for 2025, adding the proportion caught in Division 7.d (176 593 + 395 = 176 988 tonnes).

^^ Projected catches for plaice caught in Subarea 4 and Subdivision 20 in 2026 are calculated as the total catch advice for the stock (164 129 tonnes) reduced by the projected catch of the ple.27.420 stock to be taken in Division 7.d when fishing at catch advice level (44 tonnes landing + 135 tonnes discards).

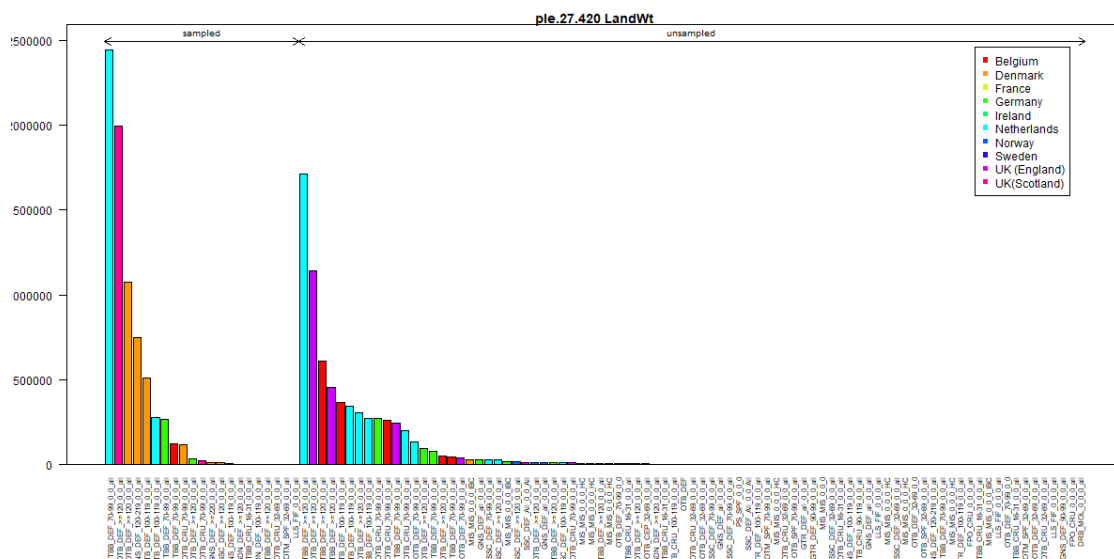
^^^ Projected catch of plaice caught in Subarea 4 and Subdivision 20 in 2026 relative to the corresponding value in 2025 (176 593 tonnes).

Projected catch of plaice caught in Subarea 4 and Subdivision 20 in 2026 relative to the 2025 TAC (176 593 tonnes) for Subarea 4 and Subdivision 20.

This scenario implies zero fishing in Division 7.d in the first quarter.



(a)



(b)

Figure 11.2.1. Plaice in Subarea 4 and Subdivision 20. Summary of data upload in InterCatch for Subarea 4: (a) Percentage of landings. Sampled and unsampled refers to availability of age-composition information. (b) Percentage of landings provided with discards, by country by métier.

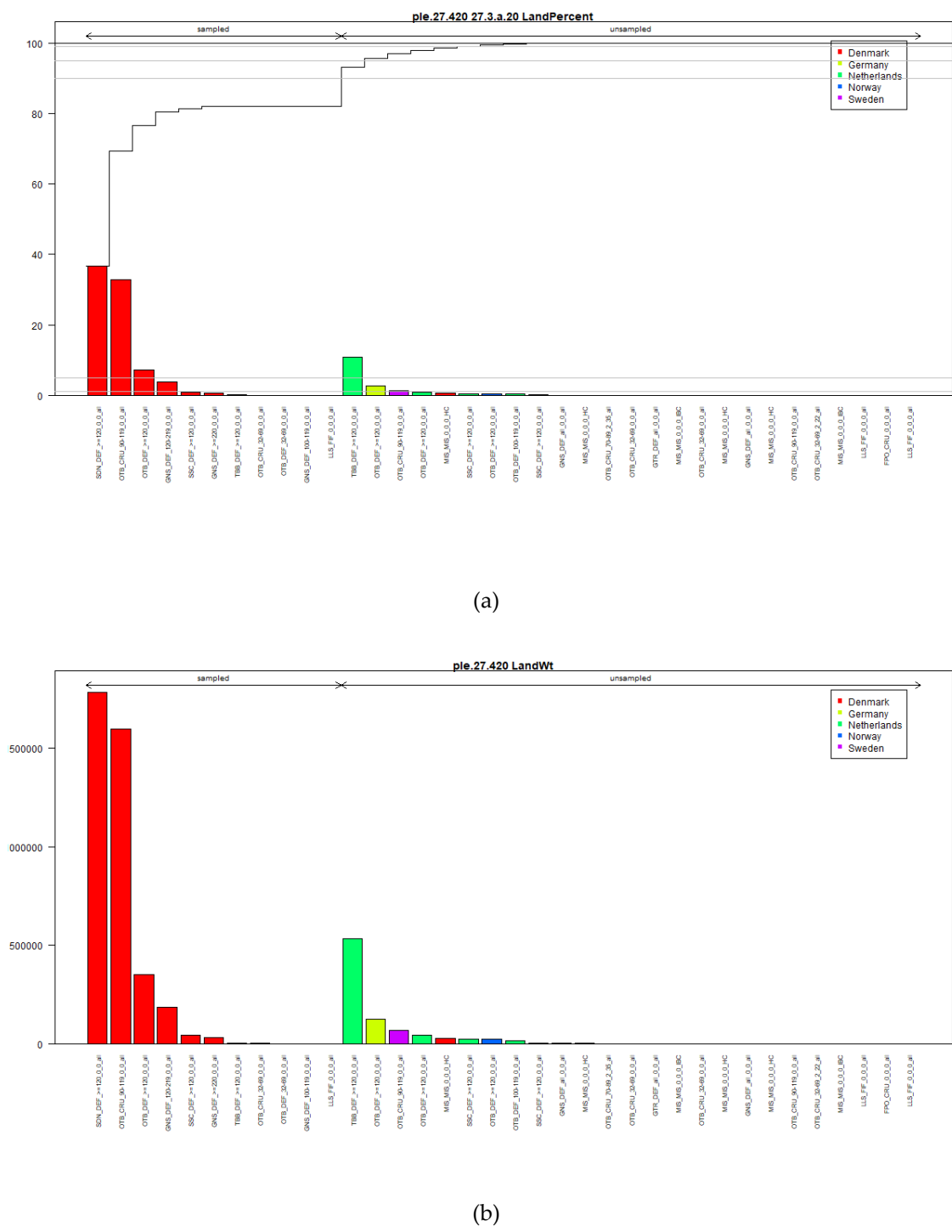


Figure 11.2.2. Plaice in Subarea 4 and Subdivision 20. Summary of data upload in InterCatch for Subdivision 20: (a) Percentage of landings. Sampled and unsampled refers to availability of age-composition information. (b) Percentage of landings provided with discards, by country by métier.

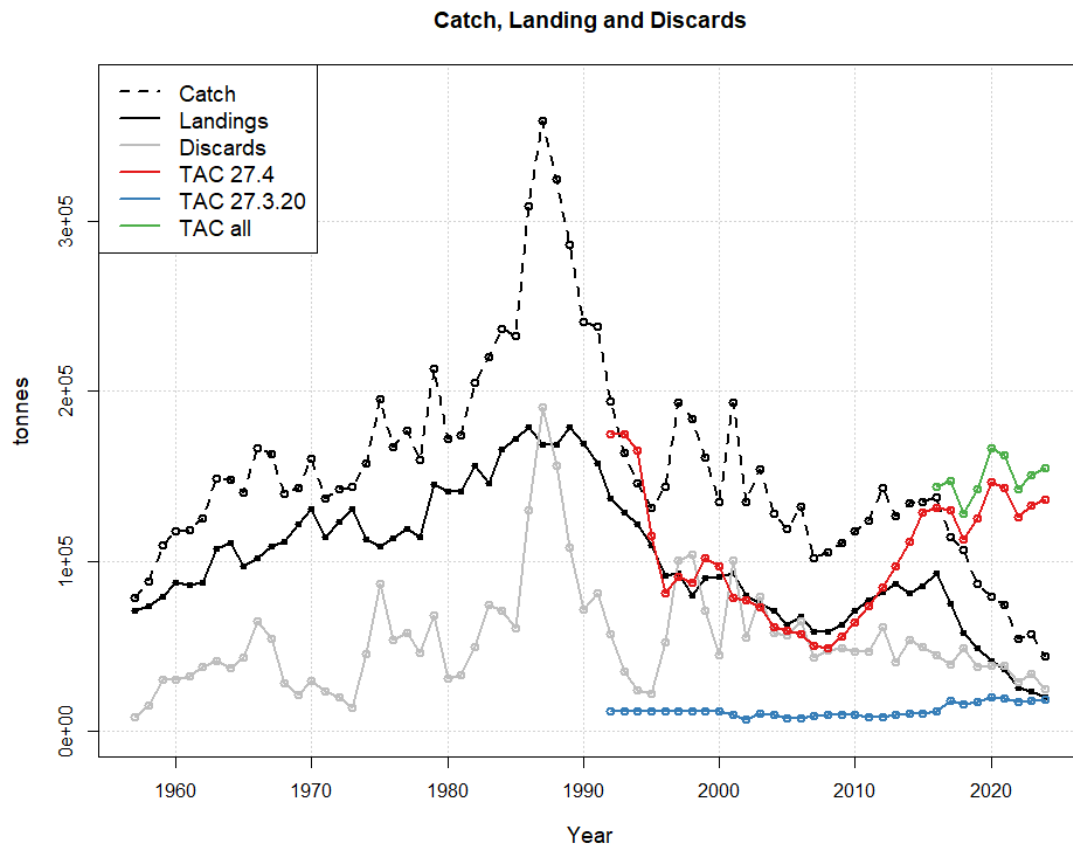


Figure 11.2.3. Plaice in Subarea 4 and Subdivision 20. Time series of catch (dashed line), landings (solid black line) and discards (grey line) estimates. TAC for Subarea 4 (red), Subdivision 20 (blue) and combined area (green) are also plotted. Discards before 2000 were reconstructed using a model-based method. Landing TAC was given before 2019 and catch TAC was given since 2019.

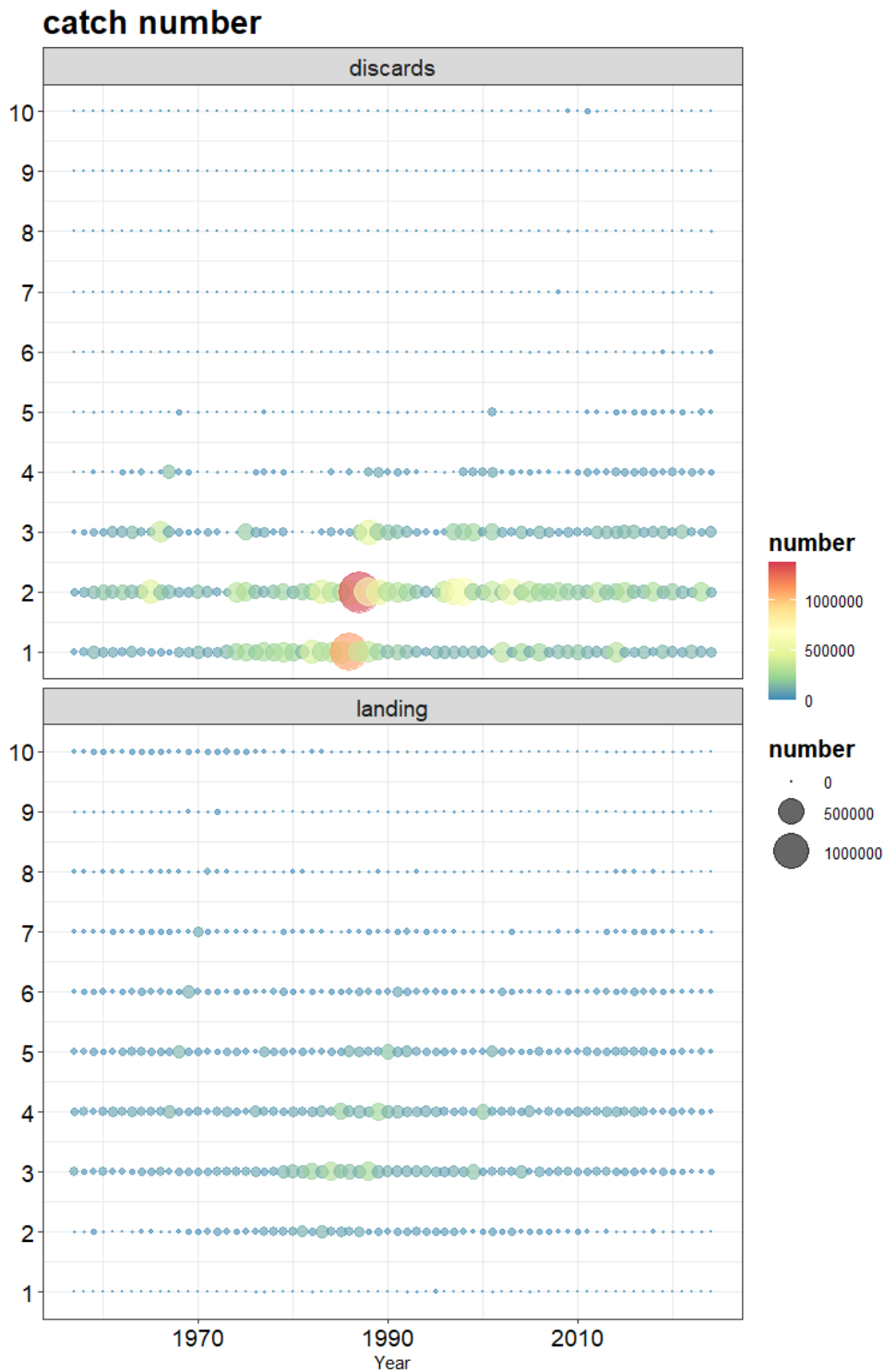


Figure 11.2.4. Plaice in Subarea 4 and Subdivision 20. Discards numbers-at-age (top) and landings numbers-at-age (bottom). Discards before 2000 were reconstructed using a model-based method.

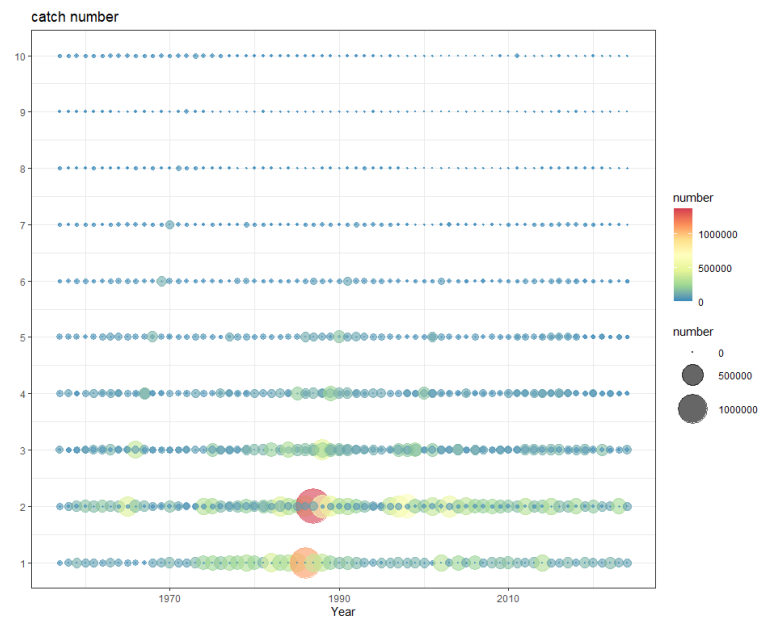


Figure 11.2.5. Plaice in Subarea 4 and Subdivision 20. Catch numbers-at-age: Discards before 2000 were reconstructed using a model-based method.

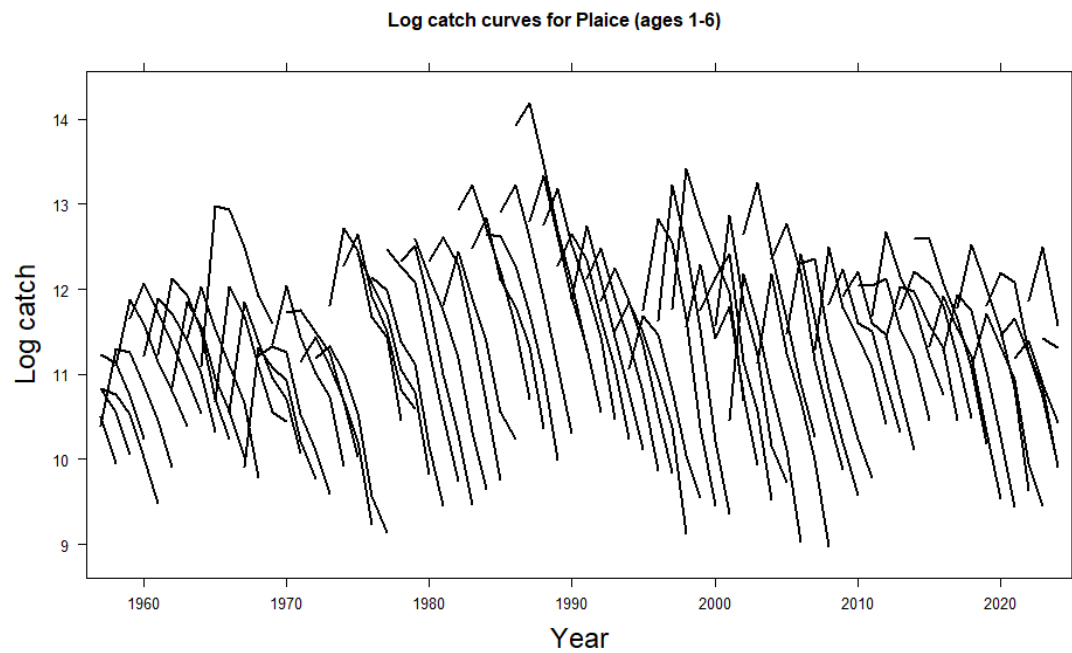


Figure 11.2.6. Plaice in Subarea 4 and Subdivision 20. Catch curves for catches in age 1–6.

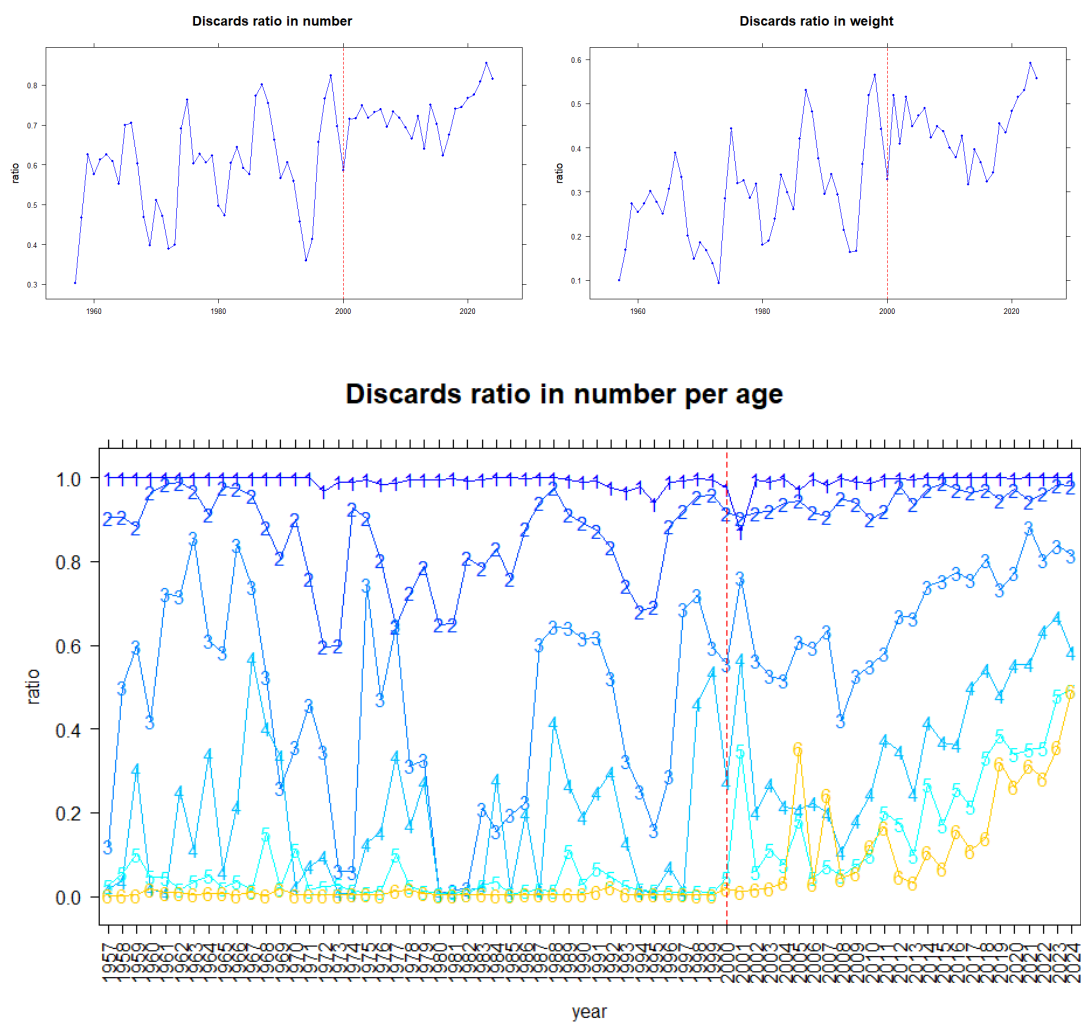


Figure 11.2.7. Plaice in Subarea 4 and Subdivision 20. Discards ratio. Discards before 2000 were reconstructed using a model-based method.

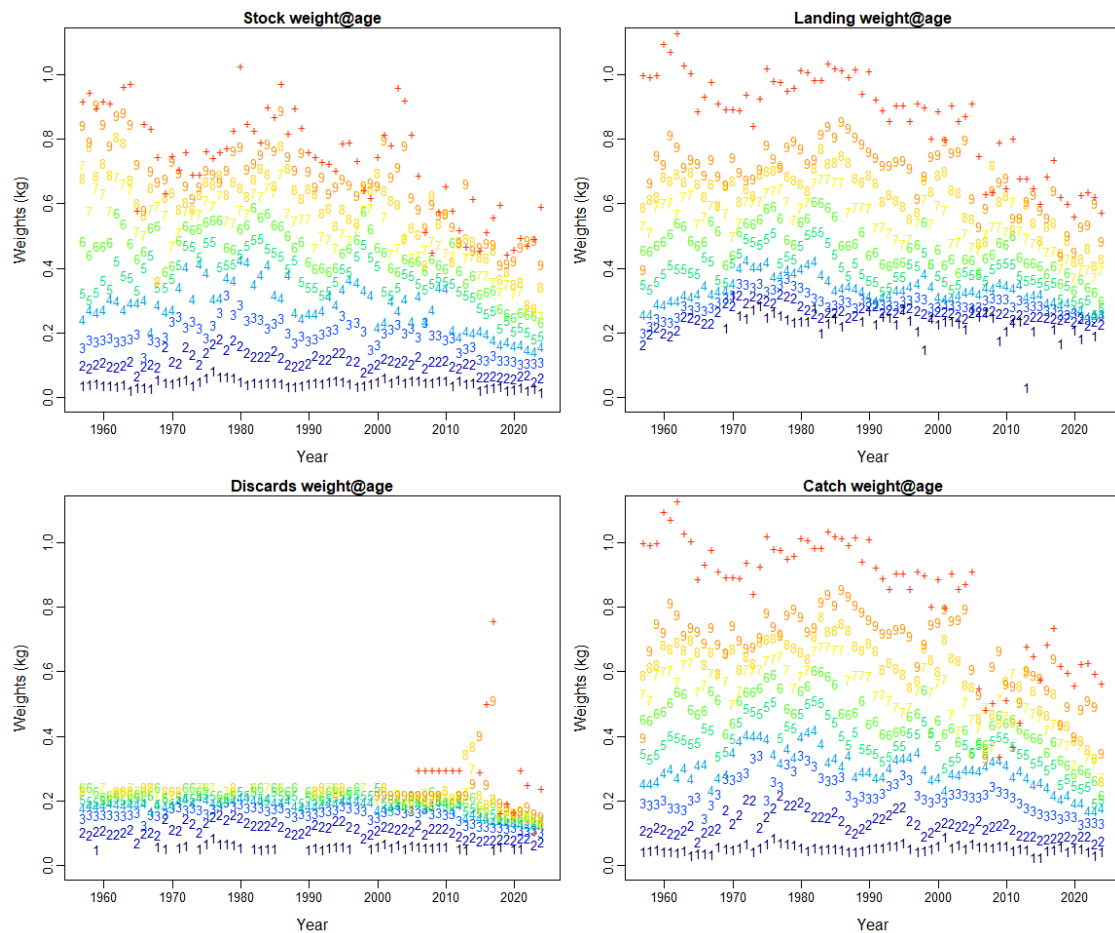


Figure 11.2.8. Plaice in Subarea 4 and Subdivision 20. Stock weight-at-age (top left), landings weight-at-age (top right), discards weight-at-age (bottom left) and catch weight-at-age (bottom right).

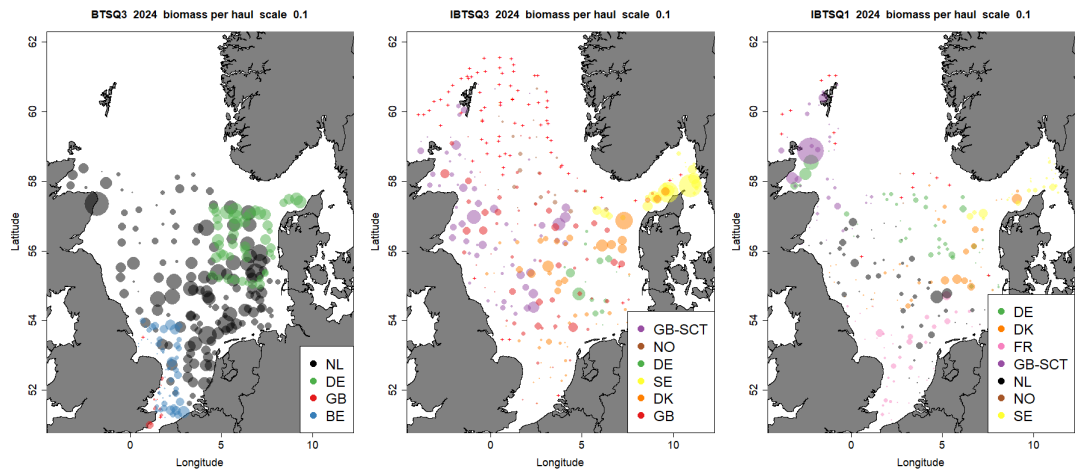
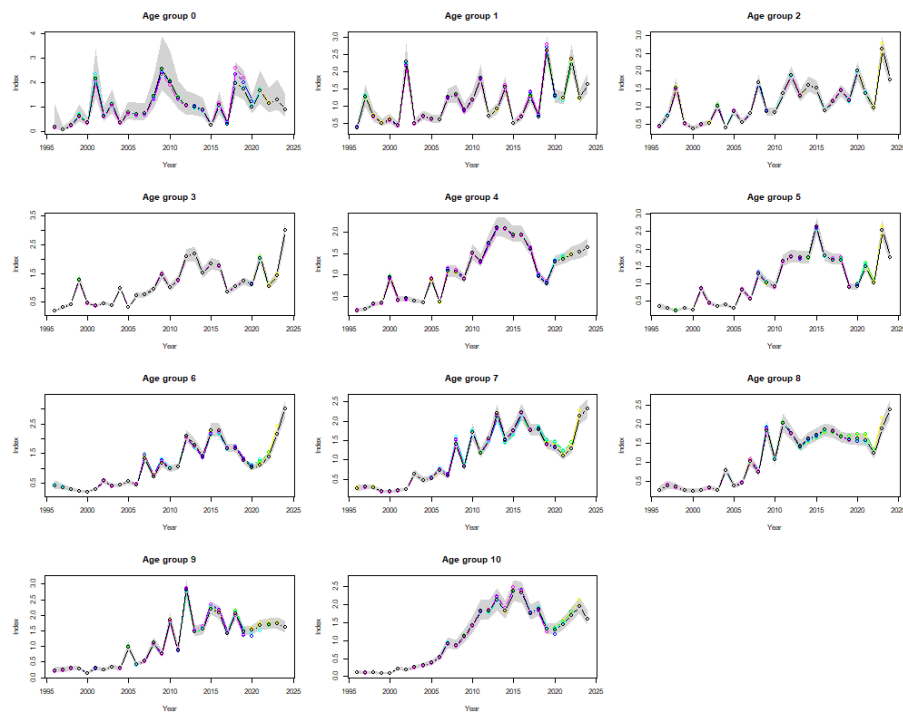


Figure 11.2.9. Plaice in Subarea 4 and Subdivision 20. Spatial distribution of biomass per haul for BTS-Q3, IBTS-Q3 and IBTS-Q1 surveys in 2024. The age length key was estimated annually using GAM including spatial effects. Samples in grey area were excluded due to low coverage.

(a) BTS-IBTS Q3



(b) IBTS Q1

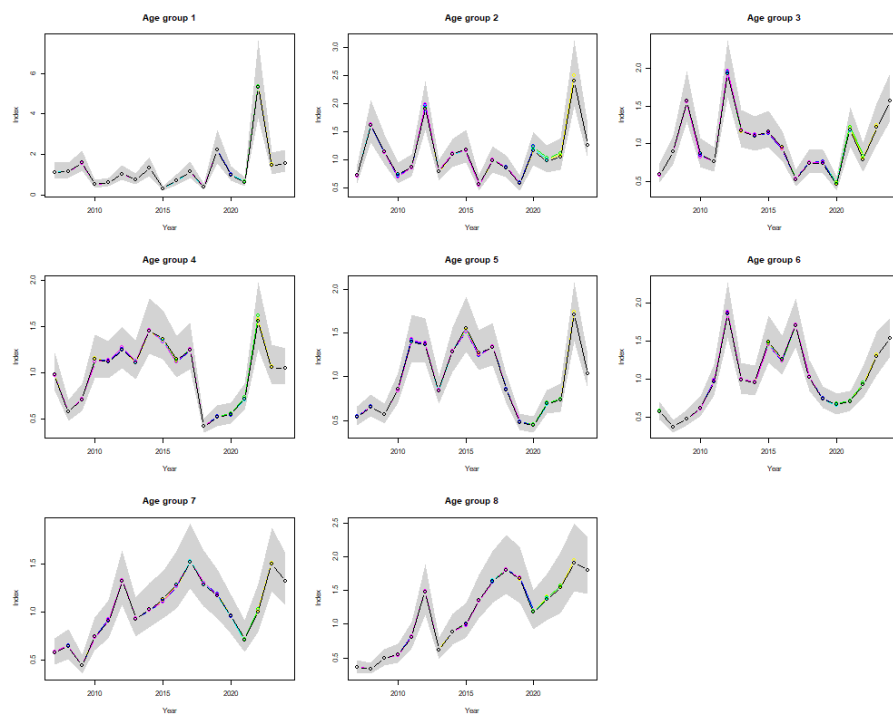


Figure 11.2.10. Plaice in Subarea 4 and Subdivision 20. Retrospective analysis of (a) BTS-IBTS Q3 and (b) IBTS Q1. All Mohn's rho values are low (<0.2).

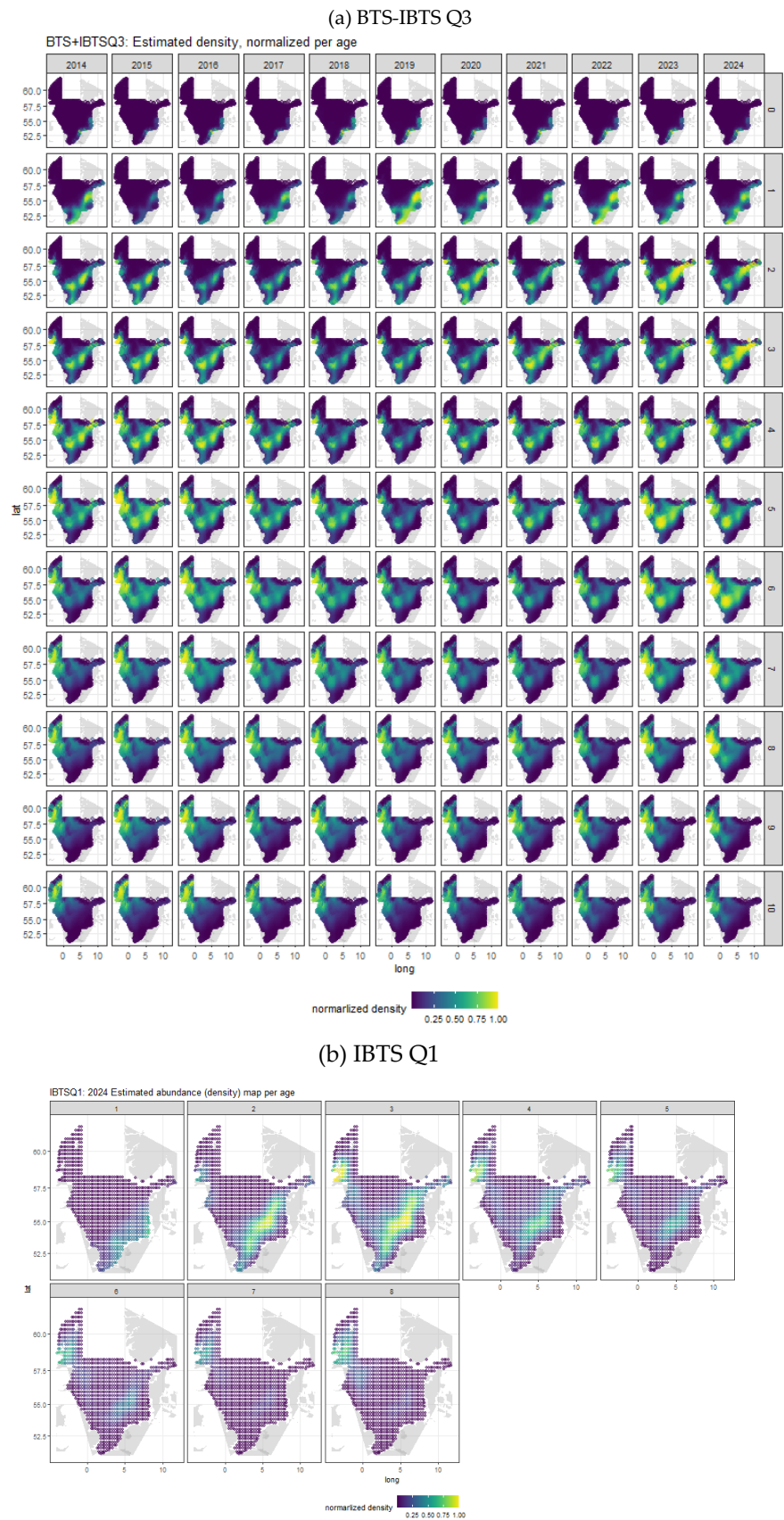


Figure 11.2.11. Plalice in Subarea 4 and Subdivision 20. The abundance map per age for (a) BTS-IBTS Q3 (2014-2024), (b) IBTS-Q1, estimated using delta-GAM method. BTS-IBTS Q3 includes age 0-10+. IBTS Q1 includes age 1-8+. Abundance decreasing from yellow to purple colour.

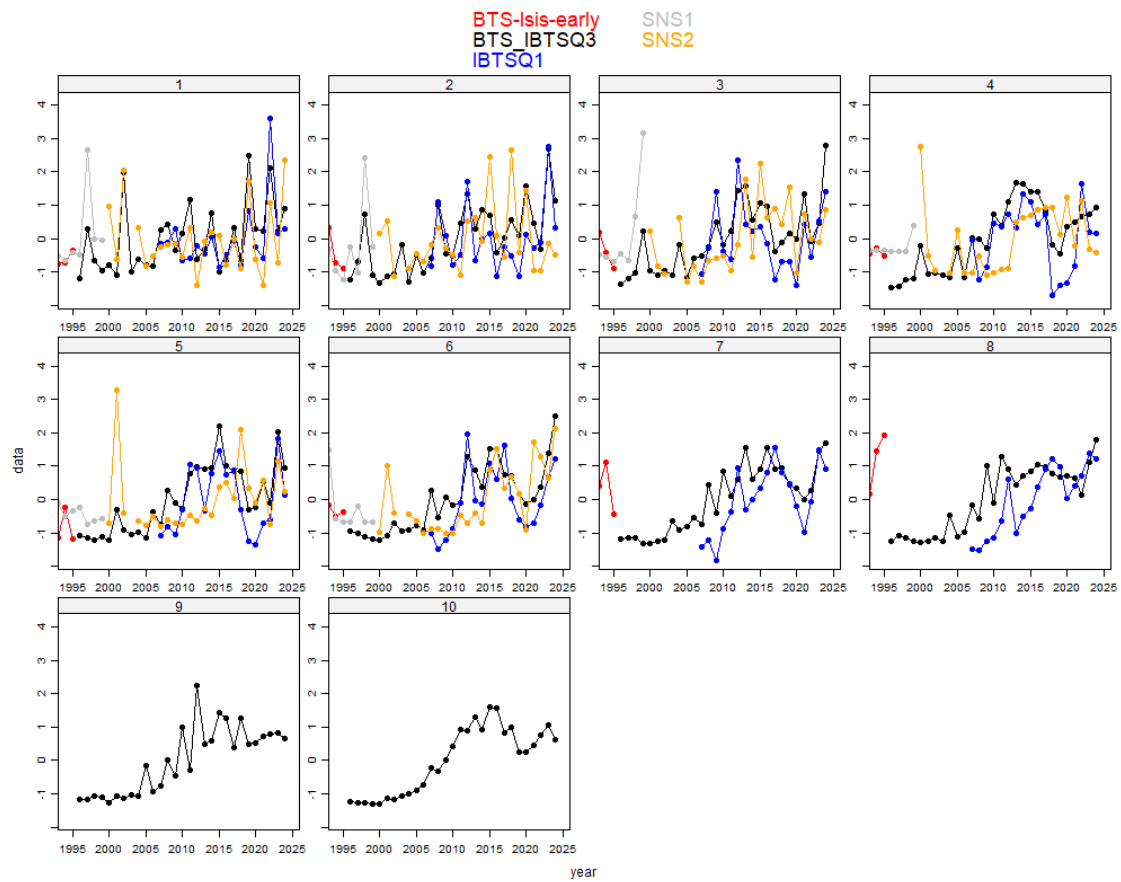


Figure 11.2.12. Plaice in Subarea 4 and Subdivision 20. Standardized survey tuning indices used for tuning stock assessment model: BTS-IBTS Q3 combined (1996–2024, black), BTS-Isis-early (1985–1995, red), IBTS Q1 (2007–2024, blue), SNS-1 (1970–1999, gray), and SNS-2 (2000–2024, yellow). Note: only ages used in the assessment are presented. Time series are presented only from 1993.

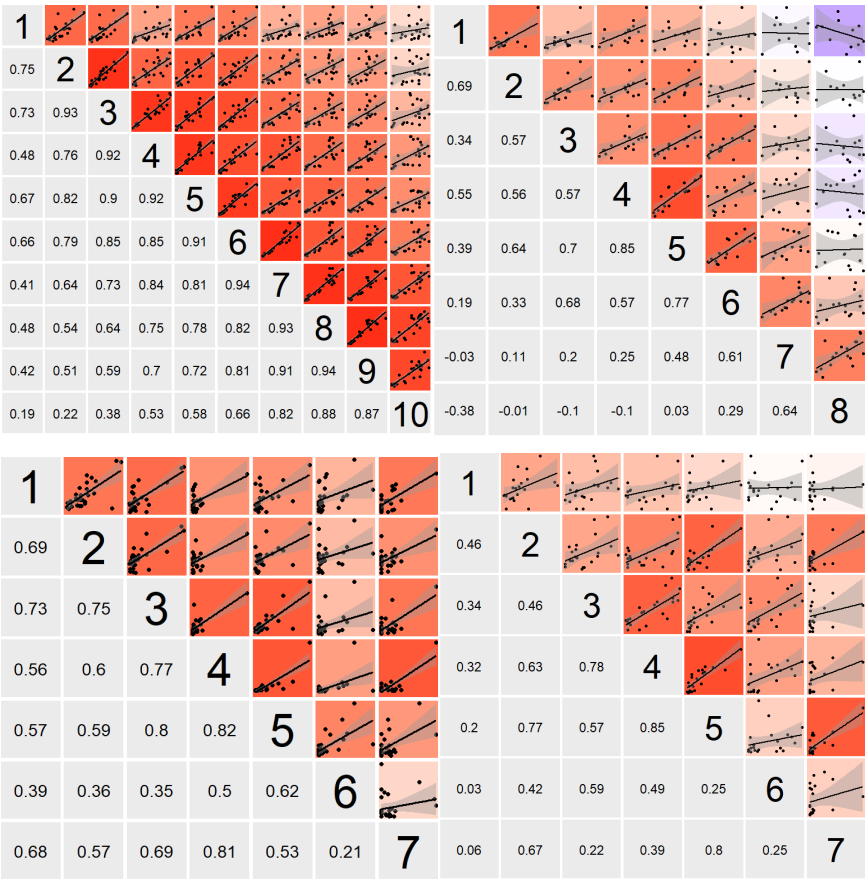


Figure 11.2.13. Plaice in Subarea 4 and Subdivision 20. Internal consistency plot for surveys. top left: BTS-IBTSQ3; top right: IBTS-Q1, bottom left: SNS1 1970-1999; bottom right: SNS2 2000-now

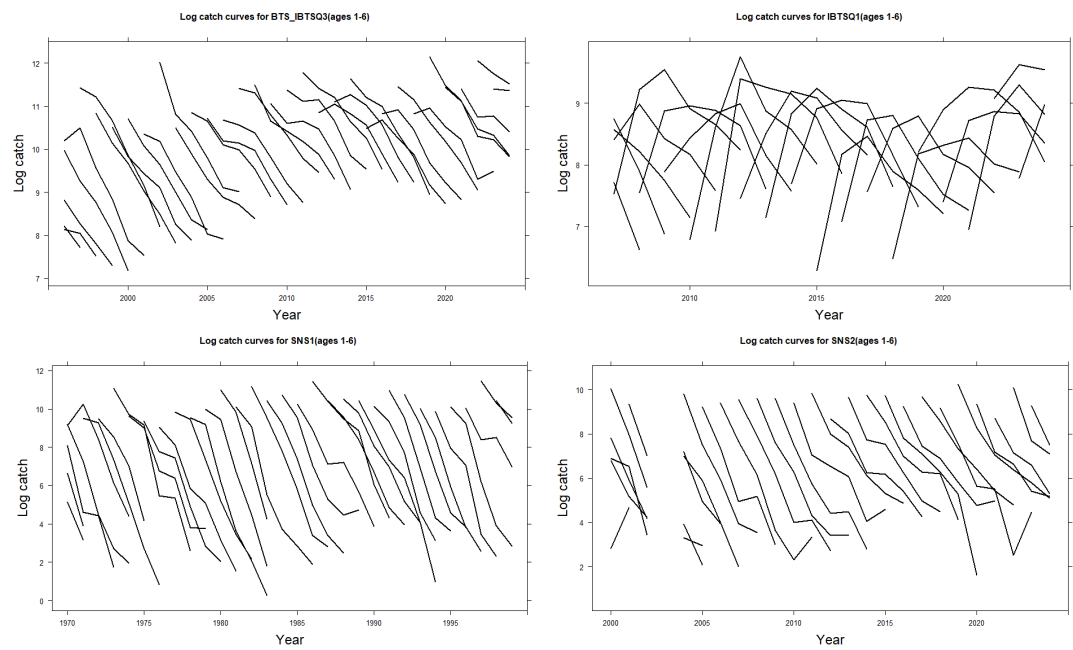


Figure 11.2.14. Plaice in Subarea 4 and Subdivision 20. Catch curves for Surveys in age 1–6.

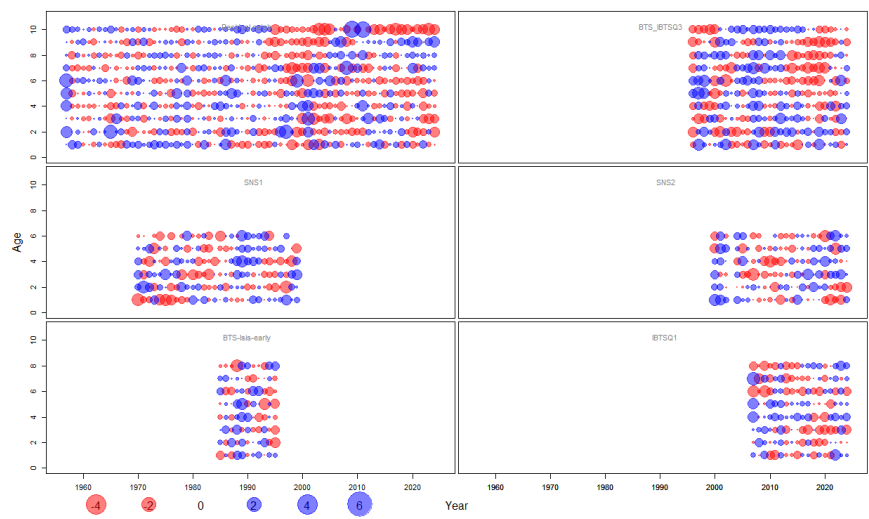


Figure 11.3.1. Plaice in Subarea 4 and Subdivision 20. Standardized one step ahead observation residuals for catch and survey indices. Positive values are in blue and negative values are in red.

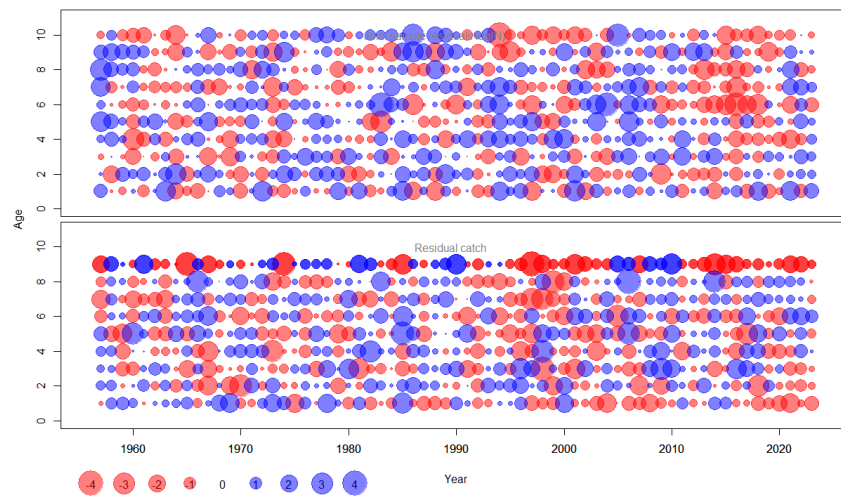


Figure 11.3.2. Plaice in Subarea 4 and Subdivision 20. Process error for logN and logF. Positive values are in blue and negative values are in red.

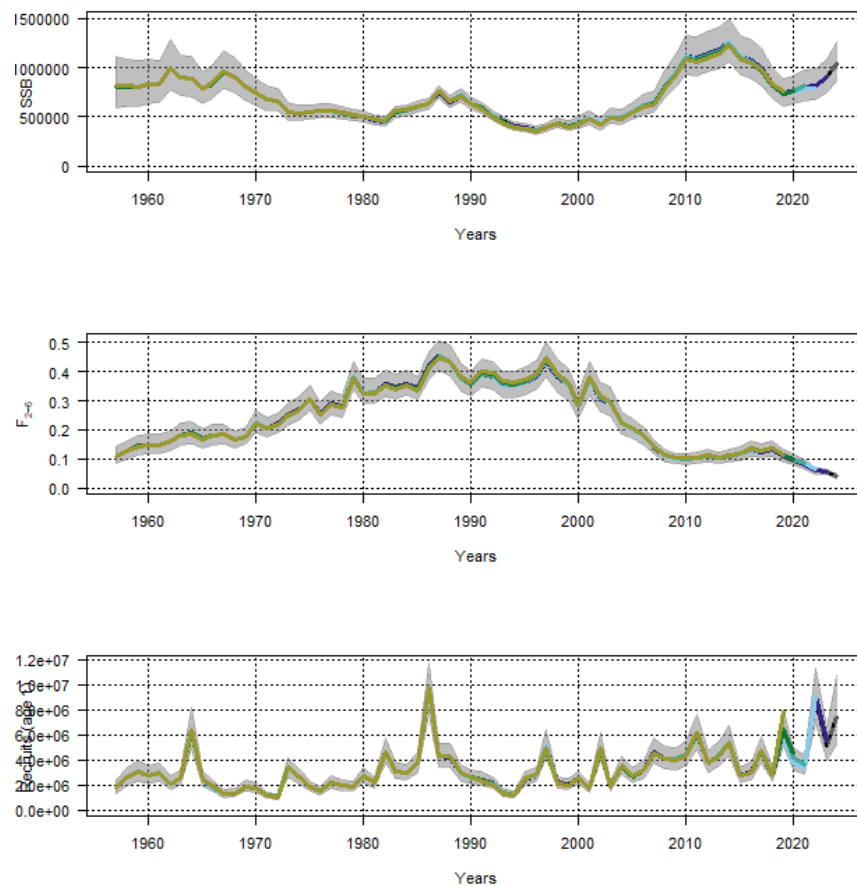


Figure 11.3.3. Plaice in Subarea 4 and Subdivision 20. Retrospective analysis for SSB (Mohn's $\rho=0.0046$), recruitment (Mohn's $\rho=0.103$) and F (Mohn's $\rho=0.047$).

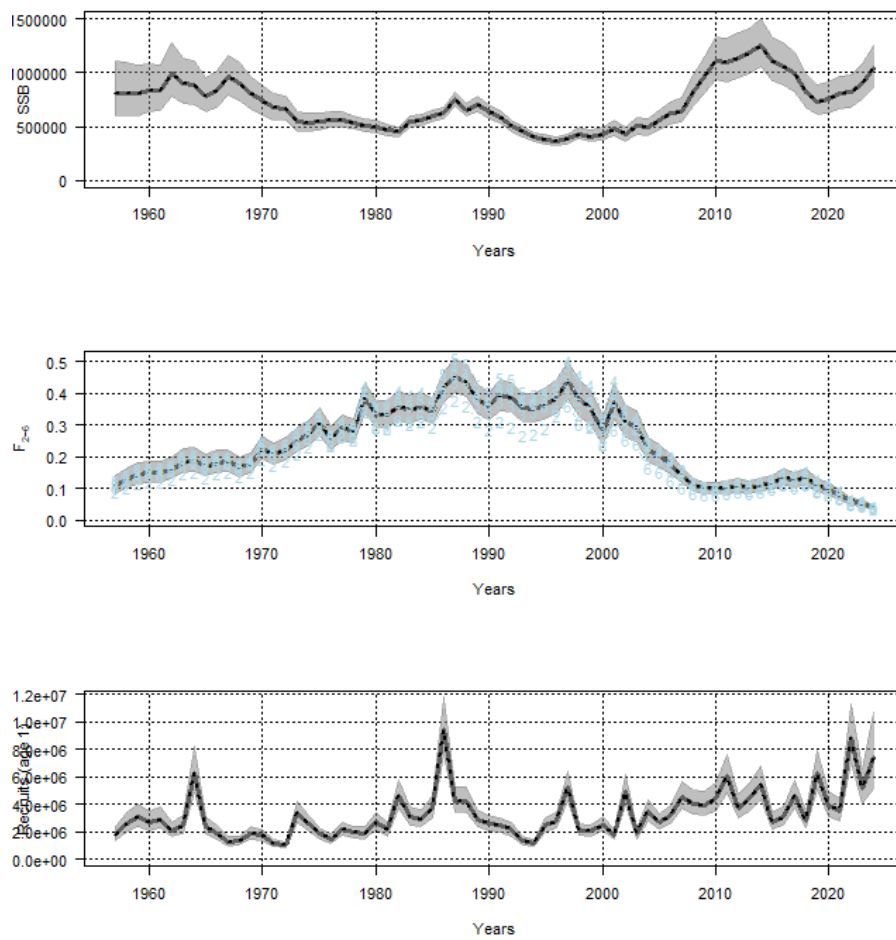


Figure 11.3.4. Plaice in Subarea 4 and Subdivision 20. Stock assessment output for ple.27.420. SSB (top), fishing mortality (middle), recruitment (bottom) estimates of the assessment.

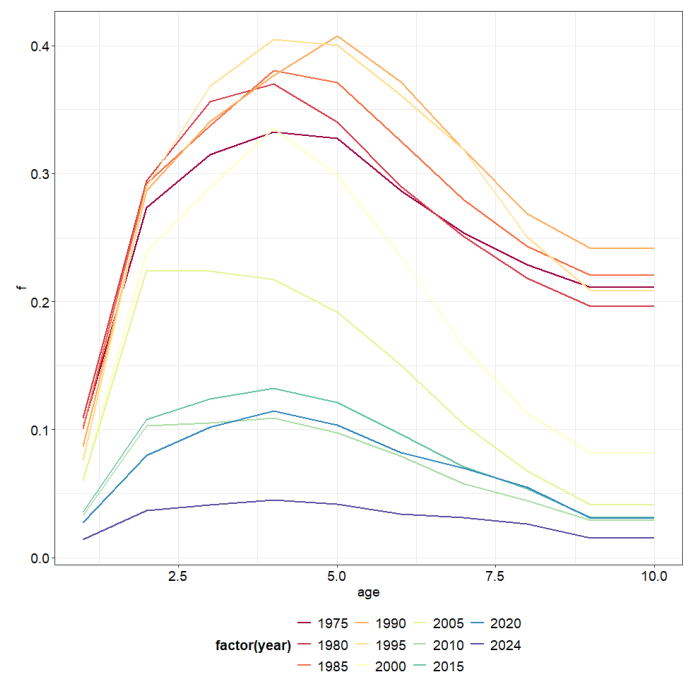


Figure 11.3.5. Plaise in Subarea 4 and Subdivision 20. Estimated fishing mortality by age since 1975.

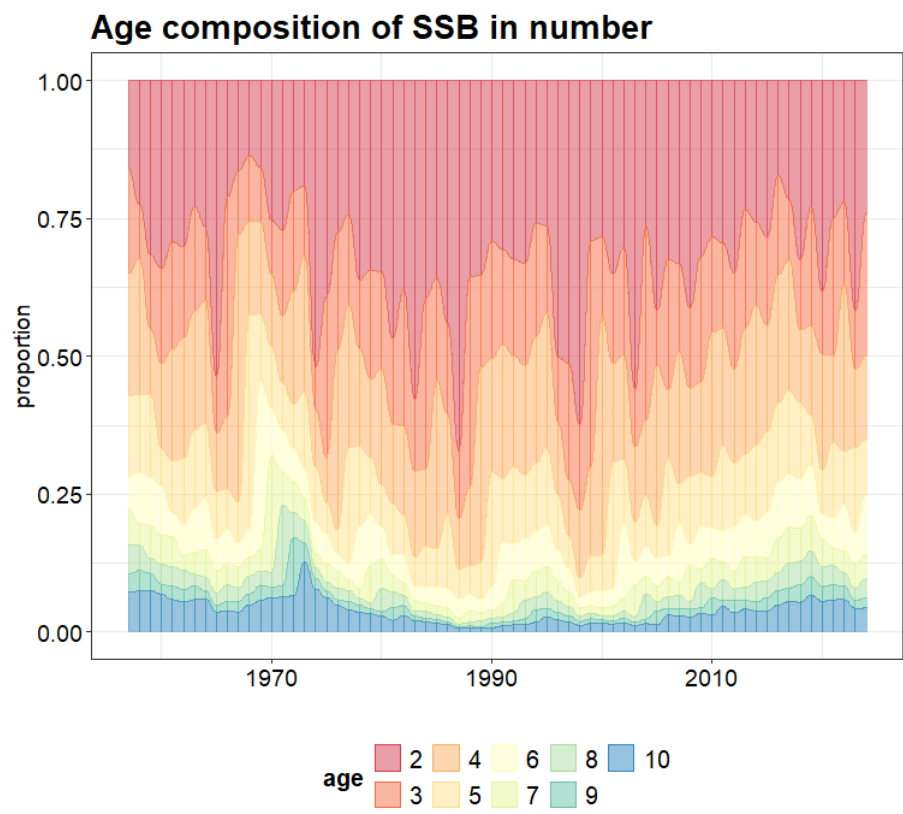


Figure 11.3.6. Plaice in Subarea 4 and Subdivision 20. Age compositions in the estimated SSB.

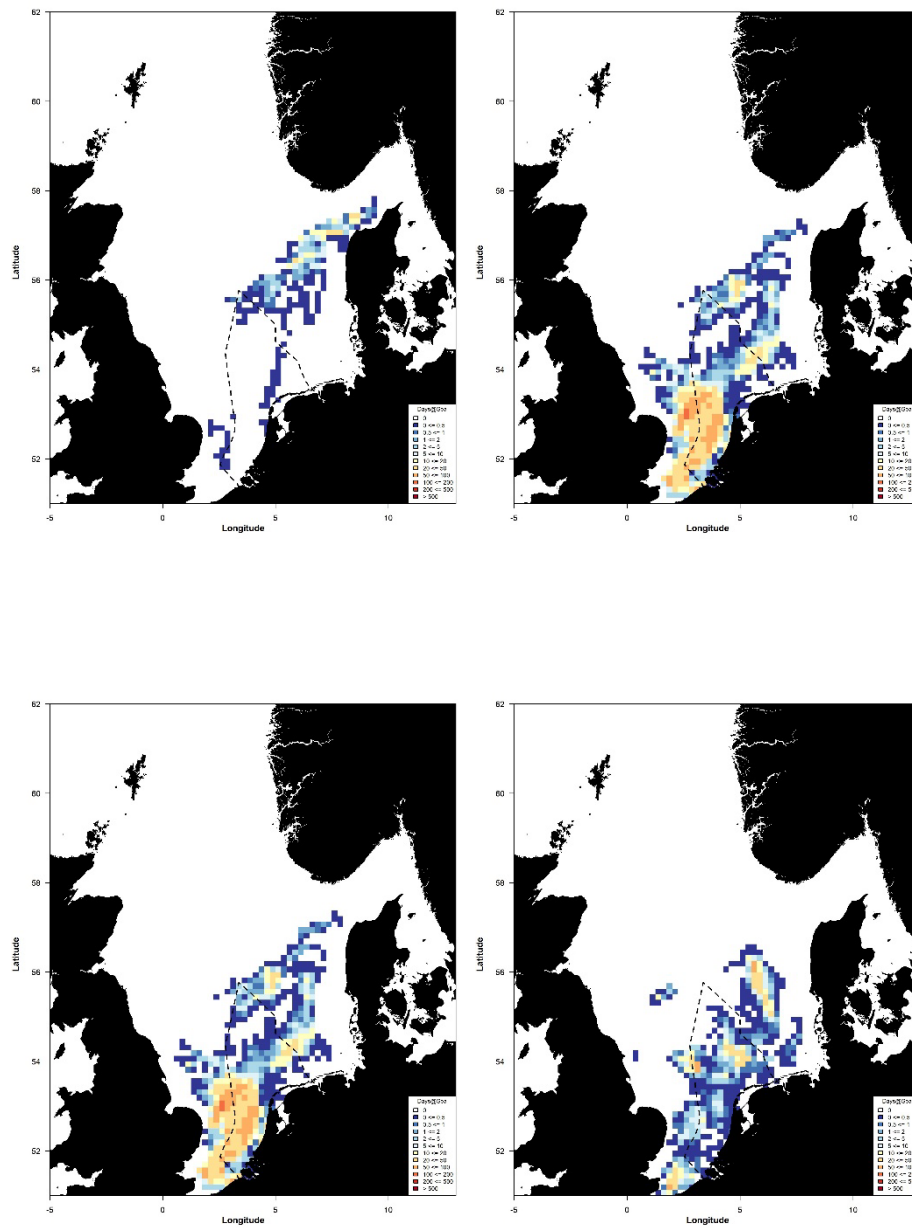


Figure 11.3.7. Plaice in Subarea 4 and Subdivision 20. Spatial distribution of fishing effort (hours fished) from The Netherlands in 2023: top left: BT1 fleet; top right: BT2 fleet; bottom left: TR1 fleet; bottom right: TR2 fleet. BT2 and TR1 are the two major gears in catching plaice in North Sea.

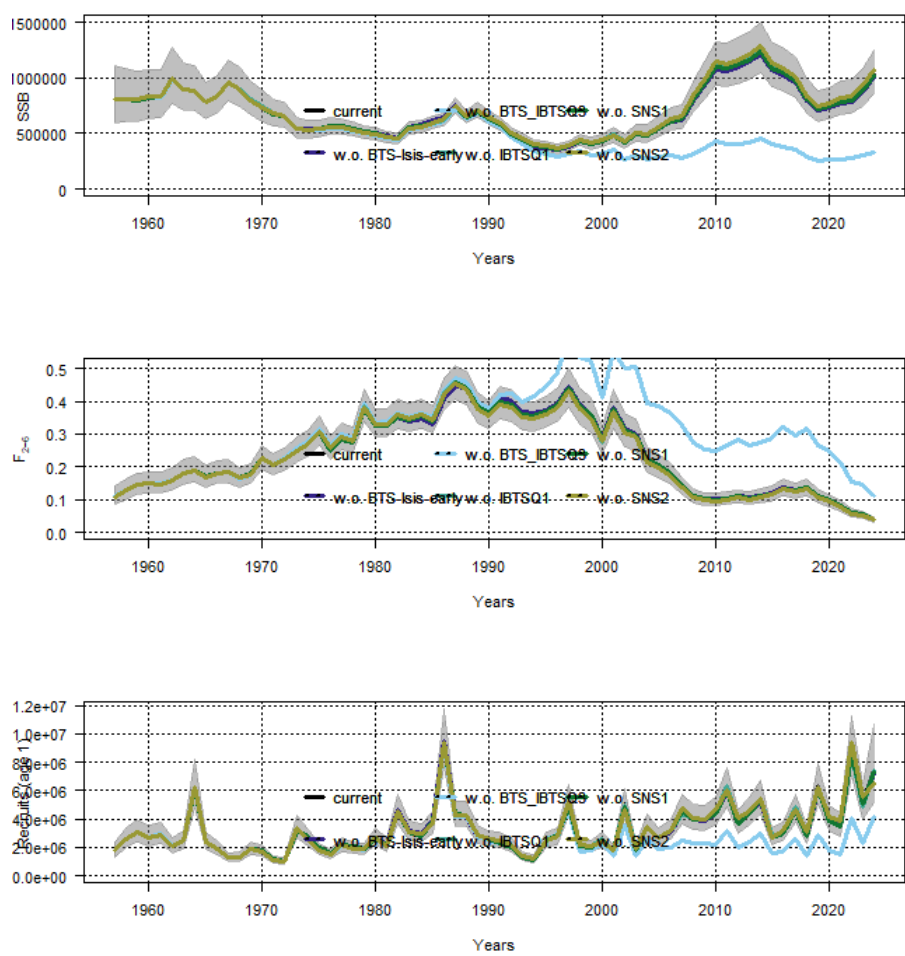


Figure 11.3.8. Plaice in Subarea 4 and Subdivision 20. Leave one out runs.

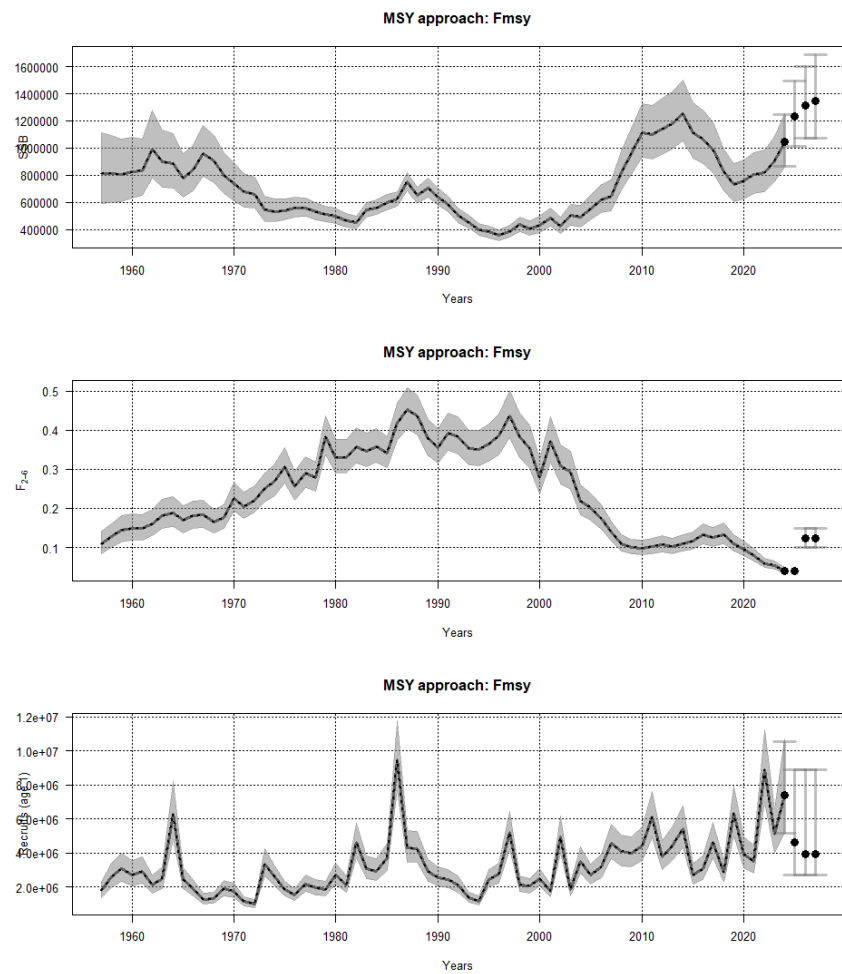


Figure 11.3.9. Plaine in Subarea 4 and Subdivision 20. Stock short term forecast under F_{MSY} option.



Figure 11.3.10. Plaice in Subarea 4 and Subdivision 20. Comparison between the assessment carried out during WGNSSK 2025 and previous years WGNSSK assessments.

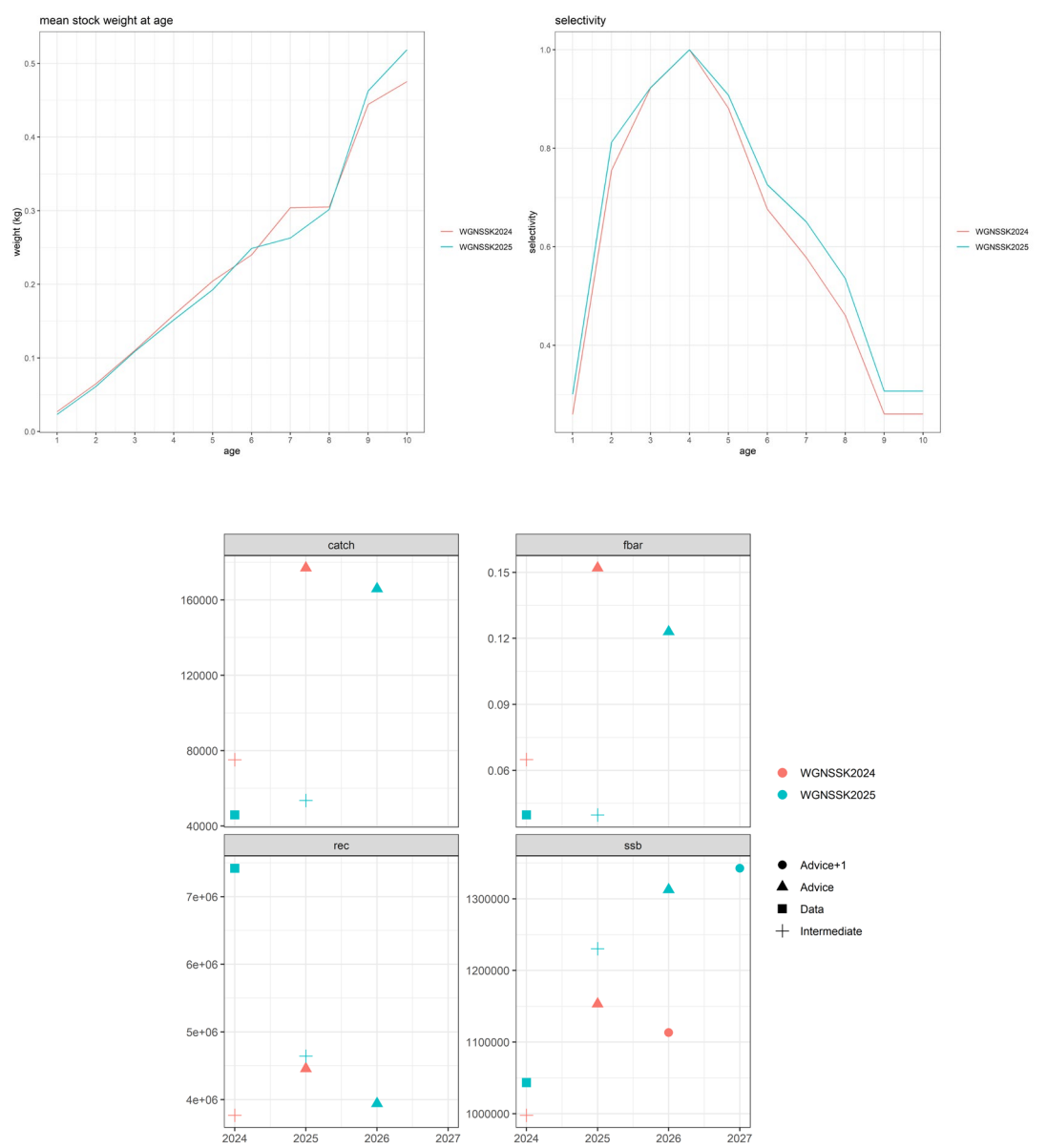


Figure 11.3.11. Plaine in Subarea 4 and Subdivision 20. Forecast assumptions comparison between the assessment carried out during WGNSSK 2024 and WGNSSK 2025. Top left: stock mean weight; top right: selectivity; bottom: catch, Fbar, R, SSB.

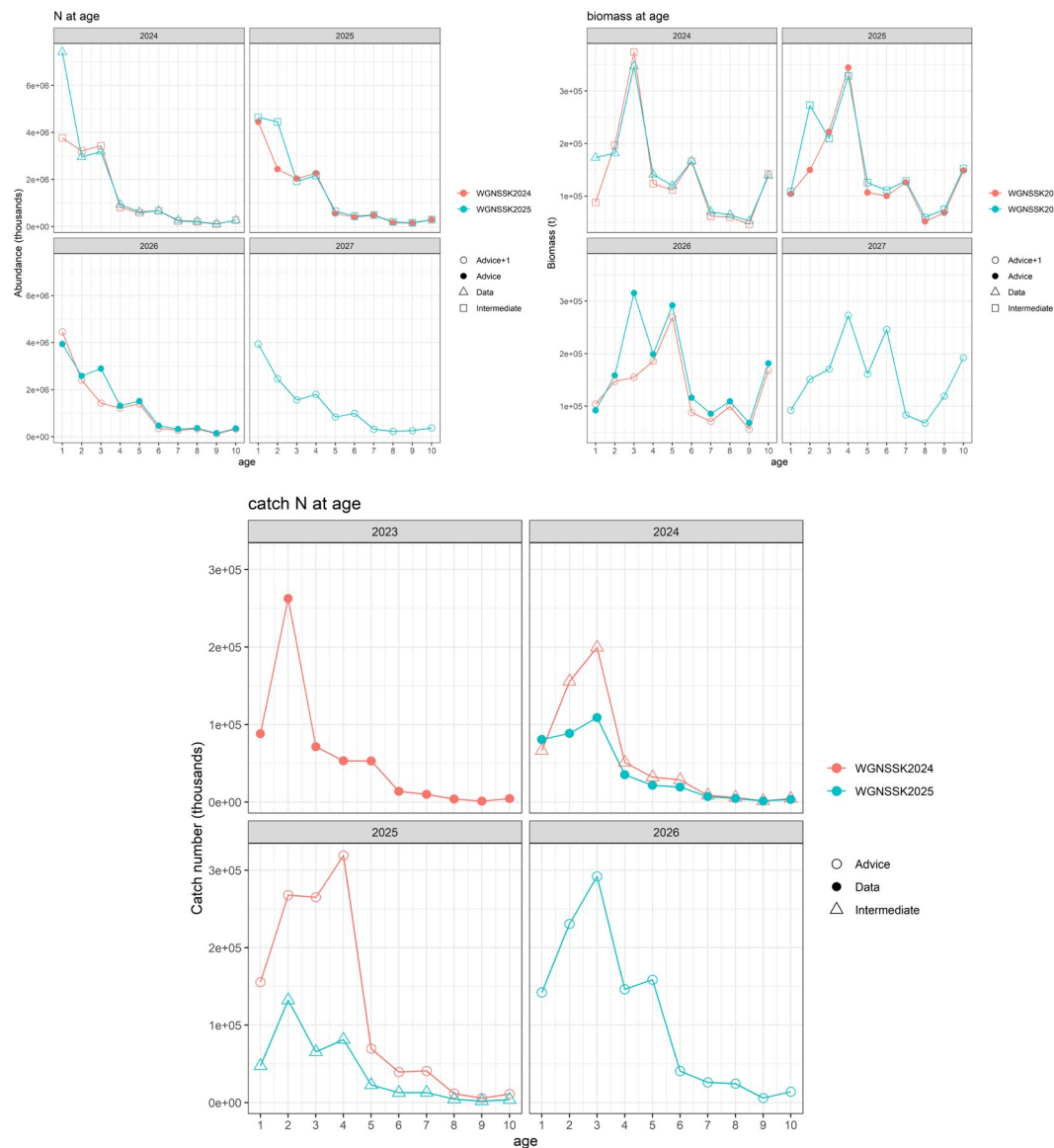


Figure 11.3.12. Plaice in Subarea 4 and Subdivision 20. Comparison between the assessment and forecast carried out during WGNSSK 2024 and WGNSSK 2025. Top left: numbers at age; top right: biomass at age; bottom: catch number at age.

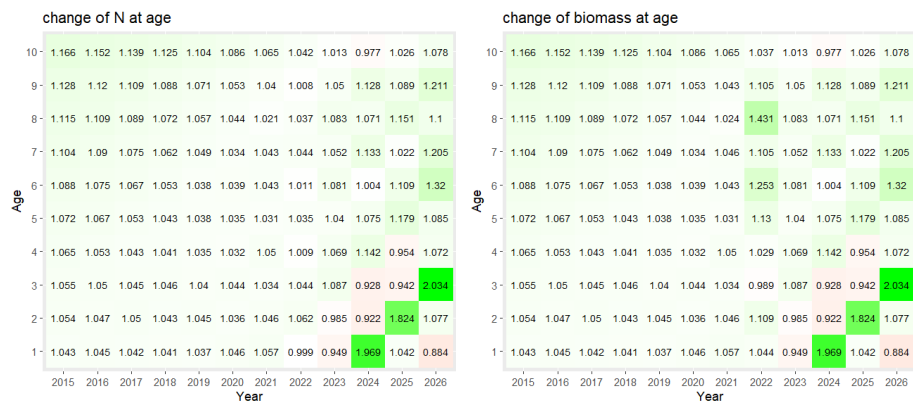


Figure 11.3.13. Plaice in Subarea 4 and Subdivision 20. Comparison between the assessment and forecast carried out during WGNSSK 2024 and WGNSSK 2025. Left: ratio of N-at-age from assessment and forecast at WGNSSK 2025 / N-at-age from assessment and forecast at WGNSSK 2024; Right: ratio of biomass-at-age from assessment and forecast at WGNSSK 2025 / biomass-at-age from assessment and forecast at WGNSSK 2024.

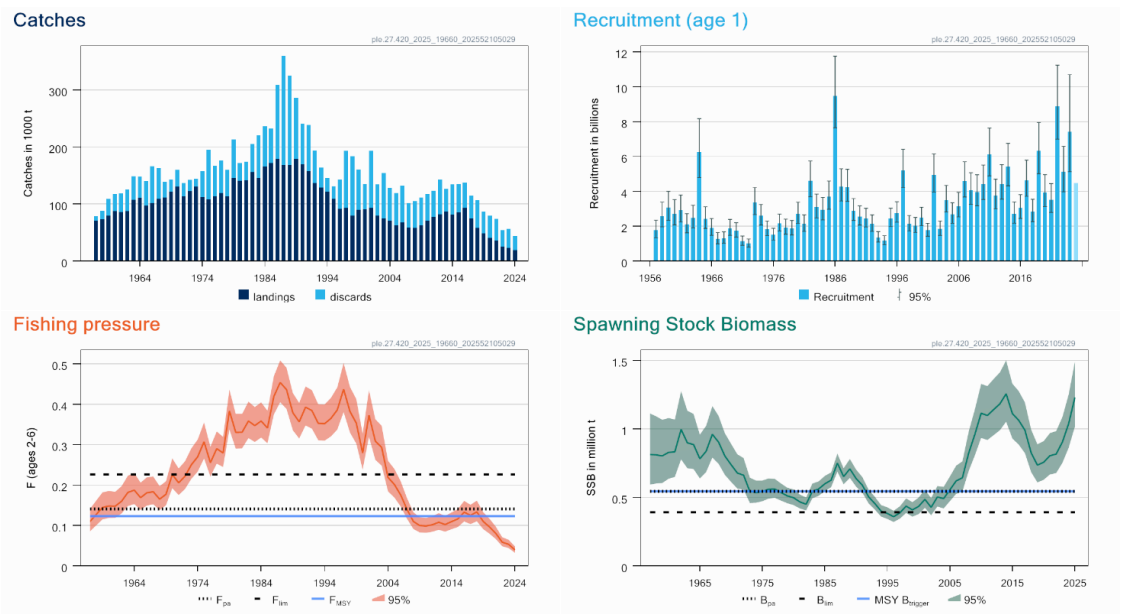


Figure 11.4.1. Plaice in Subarea 4 and Subdivision 20. Stock development over time.

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